Banks as Firms:
The Macroeconomics of Financial Firm Dynamics*

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Abstract

This paper studies the firm dynamics of banks and their role in shaping aggregate and regional business cycles. I develop a framework in which banks are heterogeneous firms that endogenously enter and exit oligopolistic regional loan markets, competing on screening ability, funding costs, loan appeal, and regulatory costs. The model delivers sharp predictions linking loan rates, market shares, and entry decisions to bank fundamentals and local frictions. Guided by theory, I estimate bank-level fundamentals and region-level loan market frictions using a structurally identified dynamic state-space model. Screening ability emerges as the primary driver of bank growth and market share, while loan market frictions explain over half of aggregate loan spreads. Embedding the estimated bank dynamics in a quantitative multi-region New Keynesian DSGE model, I show that banking shocks account for roughly one-third of output fluctuations and generate substantial regional heterogeneity. Bank dynamics drive the uneven amplification and spatial propagation of shocks through novel credit elasticity and banking network channels, leading to the asymmetric regional transmission of national monetary policy.

JEL Classifications: G21; E44; E52; D22

Keywords: Bank Heterogeneity; Firm Dynamics; Credit Allocation; Monetary Policy Transmission; Regional Business Cycles; Financial Frictions

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Banks are unique firms that intermediate credit. These intermediaries raise short-term funds from savers to make long-term loans to entrepreneurs and households, making them a key transmission channel for monetary policy. However, banks are also like any other firm: they enter and exit regional product markets, compete for market share, and set prices based on supply and demand.

Despite this fact, banks are not treated as firms in most macroeconomic analysis. Instead, banks are modeled as pass-through entities that transmit aggregate shocks through reduced-form loan spreads or as perfectly competitive intermediaries subject to balance sheet constraints. This abstraction reflects two limitations. Conceptually, the core economic function of banks —maturity transformation— does not map cleanly into a standard production technology framework, encouraging reduced-form representations in which intermediation appears as a wedge rather than as the outcome of firm-level choices. Practically, the absence of a well-defined notion of bank productivity has hindered the construction of firm-level measures of fundamentals that could discipline models of lending heterogeneity, competition, and growth.

This paper studies banks as firms and asks two questions: what forces drive the firm dynamics of banks, and how do those dynamics shape aggregate and regional business cycles? To answer these questions, I develop a theoretical framework in which banks compete for market share and grow based on their screening ability, capital funding costs, and non-pecuniary loan appeal. I then estimate these bank-level fundamentals, along with regional credit market frictions, and embed them in a quantitative model to assess how bank heterogeneity affects credit allocation, business cycle fluctuations, and the transmission of monetary policy across the United States.

With the measurements and model in hand, I document three findings. First, banks' screening ability is the primary driver of bank-level firm dynamics in lending growth, market share, and loan spreads. Second, changes in bank fundamentals and loan market frictions are a source of aggregate business cycle fluctuations, explaining 31 percent of medium run variation in output growth, while heterogeneity in bank dynamics generates substantial heterogeneity in non-financial activity, accounting for 45 percent of the regional dispersion in output growth. Third, heterogeneity in bank fundamentals leads to the asymmetric regional transmission of national monetary policy. Banks with weaker fundamentals are more sensitive to policy changes, and, combined with endogenous differences in geographic lending footprints, this sensitivity produces heterogeneous spatial propagation of national shocks. Consequentially, aggregate banking sector conditions are insufficient to assess the transmission and effectiveness of monetary policy during geographically concentrated crises such as the Global Financial Crisis or the COVID-19 recession.

Extended Summary. I first formalize the forces that shape bank dynamics in a novel theoretical framework. The economic setting begins with a regional New Keynesian DSGE model but with a financial sector made up of heterogeneous commercial banks. Banks are ex ante identical, born in a headquarters region, and endogenously decide to enter (or exit) other oligopolistic regional lending markets. When a bank enters a region, it competes based on three fundamentals, its screening ability, capital costs, and loan appeal. A bank's *screening ability* is its ability to assess whether a loan applicant will default or not if given a loan, and is analogous to the bank's lending productivity. A bank's *capital cost* is the premium the bank pays for capital funding relative to the federal funds rate. The bank's *loan appeal* is analogous to a non-financial firm's product appeal and captures firm-specific demand generated by the non-pecuniary reasons a borrower would want to work with a specific bank (for example, the ease of loan covenants, distribution of bank branches, or advertising). Banks also pay a regulatory cost based on their risk-adjusted leverage ratio (introducing dynamics similar to a risk-averse investor pricing loans), and face region-specific loan market frictions which may represent search and matching costs that cause sticky borrower-lender relationships.

The bank-level fundamentals, local frictions, and regulatory costs all influence a bank's regional entry decisions, market share, and loan rates. I analytically characterize these relationships with three results. First, loan rates reflect funding costs, expected default risk, regulatory distortions, and markups arising from both market frictions and market power. Second, a bank's market share increases as its fundamentals strengthen relative to its regional competitors. Third, banks enter a region when their expected profits exceed the fixed costs of operating in the region. As a result, several forces can push banks towards entering new credit markets: more productive banks expand geographically to capture oligopolistic rents, banks are attracted to larger markets with greater loan demand, remain close to headquarters, and move to regions with high frictions that protect profits.

The analytical model then guides the empirical estimation of bank-level fundamentals and market frictions. I use a theoretically disciplined dynamic state-space model to jointly estimate the bank-level fundamentals (screening ability, cost of capital, regional loan appeal), region-level fundamentals (firm exit rates, and non-financial productivity), loan market frictions, and regulatory costs. The model is estimated at the bank-level using balance sheet and income statements from regulatory U.S. Call Report filings, geographic footprints from the FDIC Summary of Deposits, and state-level non-financial conditions from the U.S. Census Bureau.

The estimated fundamentals and frictions have evolved over time. First, the average screening ability is broadly stable from 1995 to 2020, but with a large decline during the global financial crisis (GFC). However, the evolution of the distribution is not symmetric. The aggregate deterioration

during the crisis was preceded by the lower tail of banks, those with relatively low screening ability, who showed signs of lending productivity falling at least two years before the crisis. In contrast, other fundamentals have converged. Capital cost premia have declined secularly and compressed from a dispersion exceeding seven percentage points in 1995 to less than one percentage point by 2020. Median region-specific loan appeal is stable, but the mean has fallen as early high-appeal outliers lost their relative advantage. Over the same period, loan market frictions increased, raising the implied loan rate premium from 2.5 percentage points in 1995 to 3.07 percentage points in 2019.

Having measured the fundamentals and frictions that theoretically drive bank-level outcomes, I next turn to answering the first question: what forces shape the firm dynamics of banks?

I rely on two exercises. First, I conduct an empirical accounting decomposition of observed bank-level net interest margins (the spread between capital costs and loan rates) and regional loan market shares. By appealing directly to the data, the empirical decomposition will characterize the full distribution of realized bank-level outcomes, including the complete size distribution of banks which has been shaped by historical regulations and crisis interventions. However, the accounting decomposition is an analysis of variance and does not necessarily measure the causal relationship between bank-level fundamentals, market frictions, and banking dynamics. Therefore, I additionally calibrate a quantitative version of the model presented in Section 2 to estimate a structural variance decomposition of bank-level outcomes. This second exercise measures causal relationships while abstracting from contaminating historical policies.

A bank's market share is primarily determined by its screening ability. Across banks, differences in screening ability account for the majority of variation in market shares in both the empirical and structural decompositions. Within banks, fluctuations in screening ability account for the majority of changes in market share over the bank's life. Market share dynamics therefore reflect differences in screening ability, rather than differences in funding costs, borrower quality, or oligopolistic rents.

Meanwhile, a bank's loan spread is jointly determined by its screening ability and loan market frictions. Loan market frictions account for 51.4 percent of the observable historical loan spreads in the U.S, while screening ability accounts for approximately 34 percent. However, market frictions are relatively uniform across banks, and the cross-sectional variation in loan spreads is primarily determined by differences in banks' screening ability. The empirical and structural variance decompositions are unified in this conclusion, where screening ability accounts for approximately 52.5 and 75.1 percent of across-bank variation in loan spreads across the two exercises respectively. Taken together, these results imply that screening ability is central not only to competitive outcomes, but

also to loan pricing and the evolution of bank-level profits.

I next turn to answering the second question: how do bank dynamics shape aggregate and regional business cycles? To answer this question, I continue to rely on the quantitative model to isolate the causal relationship between bank dynamics and fluctuations in economic activity.

Shocks originating in the banking sector are a direct source of economic fluctuations. Aggregate shocks to bank fundamentals and loan market frictions induce changes in loan rates and in turn account for 30.7 percent of aggregate fluctuations in output growth over the medium run. Meanwhile, banking sector shocks may also generate regional dispersion in business cycle fluctuations by creating differences in local loan rates. Regionally concentrated banking shocks account for 45.2 percent of a region's output growth relative to aggregate business cycle fluctuations.

Simultaneously, bank dynamics are an indirect source of regional heterogeneity through their effect on the asymmetric transmission and propagation of shocks. Bank dynamics create regional dispersion through two channels: the credit elasticity and credit exposure channels.

First, the level of local banking sector fundamentals and market frictions mediate how regional economies respond to shocks through the *credit elasticity channel*. Banks with weaker fundamentals are more sensitive to shocks and charge higher loan rates, holding non-financial conditions fixed. Meanwhile, regions produce with decreasing returns to scale, and because external finance is a necessary input for production while loan demand is downward sloping, the real effects of shocks depend on baseline loan rates. In regions with weak banking fundamentals, high loan rates place the economy low on the credit demand schedule and production curve, so marginal changes in rates generate relatively large changes in borrowing and output.

Second, the banking network asymmetrically propagates shocks through the *credit exposure channel*. A bank's loan rates are in part determined by its net worth. Therefore, banks operating in multiple regions will adjust loan rates across all markets in response to aggregate or region-specific shocks that impair profits. Since regions are exposed to multi-region banks, they are exposed to the downstream effects of different region-specific shocks or responses to aggregate fluctuations (due to the credit elasticity channel, for example). Then as regions are differently exposed to the banking network, they are asymmetrically exposed to the propagation of regional and national shocks.

I lastly quantify the effects of the credit elasticity and exposure channels through their effects on the regional transmission of national monetary policy shocks. To do so, I simulate a 25 basis

point tightening shock simultaneously impacting 50 regions. I first find that the credit elasticity channel generates substantial and economically meaningful differences in the regional response to the common policy shock. Regions with weaker fundamentals had a larger response to the policy shock, with loan growth falling 7.5 percent more than in regions with strong bank fundamentals. I then find that the credit exposure channel leads to a persistent and asymmetric propagation of the policy shock. The multi-region banking network leads to a 3.92 percent increase in the medium run regional dispersion of loan growth.

Roadmap. Section 2 presents the theoretical framework, Section 3 analytically characterizes bank dynamics and their role in the macroeconomy, Section 4 measures the economic fundamentals, market frictions, and regulatory distortions, Section 5 examines which forces drive bank dynamics, Sections 6 and 7 in turn put these forces in a macro context, and Section 8 concludes. The remainder of this section highlights the contributions of this work and reviews the related literature.

Literature review. This paper's primary contribution is to bridge the two important but largely separate literatures on firm dynamics and the macroeconomics of banking and financial frictions.

Macroeconomic models have evolved in their treatment of financial intermediation. Early frameworks abstracted from intermediation, consistent with the frictionless benchmark of Modigliani and Miller (1958). Subsequent work introduced financial intermediaries as a source of amplification of non-financial shocks through the so-called financial accelerator mechanism, formalized in Bernanke et al. (1999) and Kiyotaki and Moore (1997). Following the GFC, banks became explicit economic agents and potential sources of aggregate fluctuations. Seminal contributions such as Gertler and Kiyotaki (2010) and Brunnermeier and Sannikov (2014) incorporated balance-sheet constraints and intermediary net worth into dynamic general equilibrium frameworks, while others, including Jermann and Quadrini (2012) and Ajello (2016), identified intermediation shocks as distinct drivers of aggregate business cycles. More recent work emphasizes the role of cross-bank heterogeneity in determining macroeconomic outcomes, as in Jamilov and Monacelli (2025) and Coimbra and Rey (2024). Despite this progress, models with banks still typically abstract from endogenous entry and exit, regional expansion, competitive interactions, and firm-level growth. Key exceptions include Corbae and D'Erasmo (2021, 2025), which study banking *sector* dynamics and incorporate oligopolistic competition and regional dispersion among banks. However, these studies focus on sector-wide dynamics and take the initial size distribution (abstracted to a dominant-fringe paradigm) and regional distribution of banks as given and fixed.

In contrast, I model banks as heterogeneous firms, building on foundational work in firm dynamics

Hopenhayn (1992); Hopenhayn and Rogerson (1993); Melitz (2003) and its quantitative synthesis Hottman et al. (2016); Eslava et al. (2024). Bank market shares, geographic expansion, and competitive interactions evolve endogenously, linking financial firm dynamics to regional and aggregate fluctuations and to the transmission of monetary policy.

This paper's second contribution is to construct bank-level measures of screening ability and loan appeal. I am, to the best of my knowledge, the first to structurally model and estimate loan appeal, an object analogous to product appeal in the firm dynamics literature (see for example Foster et al. (2008), Hottman et al. (2016), or Eslava et al. (2024)). The estimation of screening ability builds on a large micro-oriented literature studying banking competition under asymmetric screening technologies, whose early work analyzed credit market competition under symmetric screening technologies Broecker (1990), and whose subsequent studies allow for asymmetric screening ability across banks Hauswald and Marquez (2003). Other contributions endogenize screening ability through costly information acquisition Hauswald and Marquez (2006); Karapetyan and Stacescu (2014), introduce fixed costs of screening Yannelis and Zhang (2023), or model screening as the outcome of multiple information inputs Blickle et al. (2024); He et al. (2024). Despite their insights into the role of informational frictions in credit markets, these studies are primarily theoretical and do not attempt to measure bank-level screening ability in the data.

A related macro-finance literature interprets monitoring and screening costs as financial-sector productivity, but focuses primarily on aggregate measures. For example, Philippon (2015, 2022) document the long-run stability of U.S. financial intermediation costs, building on earlier efforts to measure financial-sector value added Wang et al. (2008); Haldane et al. (2010); Basu et al. (2011), with complementary European evidence in Bazot (2018, 2024). These approaches are informative about long-run trends but abstract from cross-bank heterogeneity and leave bank-level productivity unmeasured. My framework bridges these literatures by structurally identifying and estimating screening ability and loan appeal at the bank level.

This paper's third contribution is to the literature on the regional effects of credit supply shocks and monetary policy. An empirical literature documents that shocks to, or mediated by, banks generate significant and uneven regional effects Drechsler et al. (2017); Huber (2018); Gilje (2019); Greenstone et al. (2020); Davis and Haltiwanger (2024). These studies isolate credit supply shocks but abstract from non-financial shocks, leaving open questions about joint drivers of regional cycles. A related quantitative literature examines regional heterogeneity in monetary policy transmission, showing that common policy shocks generate differential regional responses due to variation in household balance sheets or industrial composition Beraja et al. (2019); Jones et al. (2022); Hazell

et al. (2022). However, this literature has abstracted from regional heterogeneity in the banking sector, despite the bank lending channel being a key transmission mechanism of monetary policy shocks. My paper complements these works by identifying and quantifying the extent to which the credit elasticity and exposure channels lead to the asymmetric regional transmission of monetary policy through heterogeneity in regional banking sectors.

2 Theoretical Framework of Bank Firm Dynamics

I first develop a novel theoretical framework of bank firm dynamics. The following section will describe the economic environment and agents before Section 3 analytically characterizes the firm dynamics of banks and how they shape aggregate and regional economic outcomes.

2.1 Summary of the economic setting

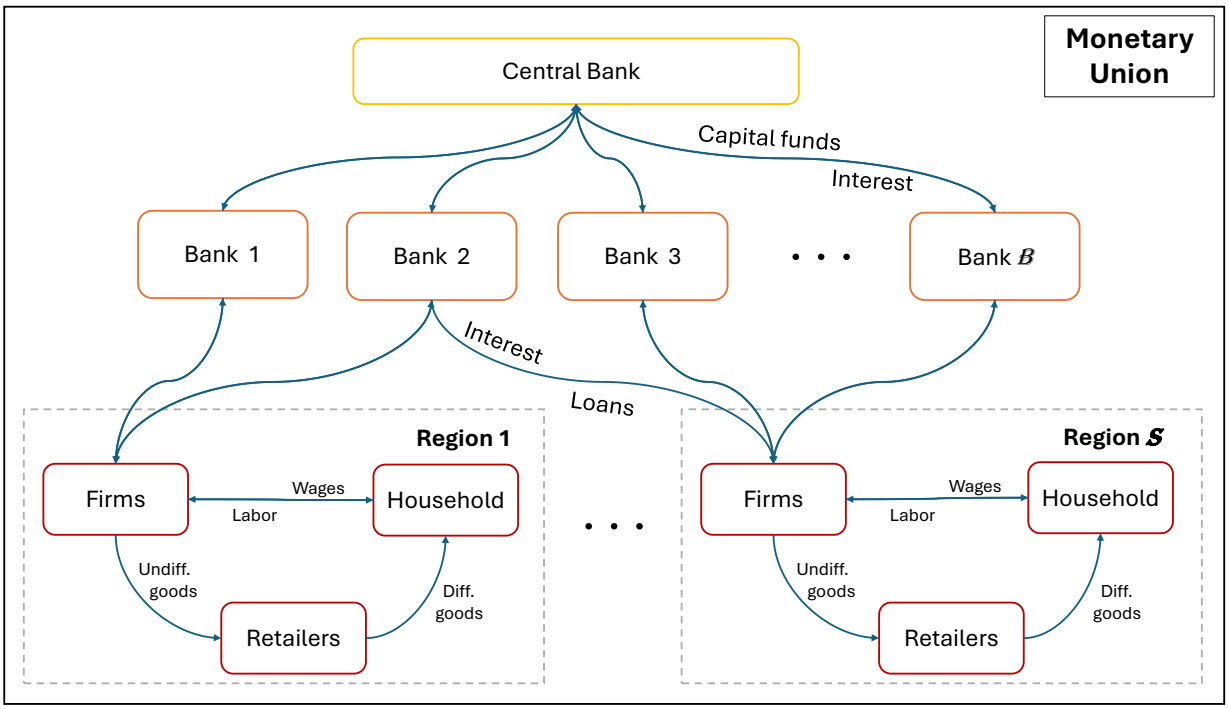
The model is a regional New Keynesian DSGE. Time is discrete and runs to infinity. The economy is a set of independent islands that are home to immobile labor (households) and idiosyncratic production technology (firms and retailers), but are spanned by a financial sector that owns capital and lends to regional industries to enable production. The structure and flow of resources in the economy are summarized by Figure 1.

Banks decide each period which oligopolistic regional lending markets (islands) to enter subject to a fixed cost of operating that increases with market distance from the bank's headquarters. After deciding which markets to serve in a given period, banks pay capital and regulatory costs to borrower funds from the Central Bank and issue risky loans with an imperfect screening technology.

A regional market is home to a continuum of households that save in local bond markets and supply labor to local firms. The market also has a continuum of firms that use bank loans as working capital to produce a homogeneous good that is sold to retailers in a perfectly competitive goods market. However, firms are subject to an exogenous exit shocks that result in them defaulting on their loans and ceasing production. Retailers differentiate the intermediate goods and sell the new variety of consumption goods to households, setting prices subject to a standard menu cost nominal rigidity. A central bank conducts monetary policy to stabilize aggregate inflation and output across islands in the monetary union.

The financial block applies a firm dynamics modeling approach to the financial sector of heterogeneous banks. The imperfectly competitive regional loan markets and loan "exporting" dynamics

Figure 1: Map of the economy



positions my treatment of banks closest to the Melitz (2003) tradition of firm dynamics models, although extensions to oligopolistic competition and the inclusion of firm-specific appeal brings this model in line with modern firm dynamic frameworks, such as Hottman et al. (2016) and Eslava et al. (2024). The non-financial block of the model combines the heterogeneous firm structure of Ottonello and Winberry (2020) and regional monetary union setting similar to Nakamura and Steinsson (2014), Beraja et al. (2019), and Jones et al. (2022). An extended summary follows before moving to describe in detail the key agents in the economy, banks.

2.2 Regional economies and the monetary authority

I first describe the regional economies, in particular the production sector that demands loans from banks and the monetary authority that supplies banks with funds to lend.

Individual firms and exit rates. There is a continuum of individual firms in each region. Each firm applies to a bank for a loan to fund its labor bill before starting production. If a firm is denied a loan then it cannot produce that period.[1] However, each firm potentially receives an exogenous

[1]Because firms only match with and submit loan applications to a single bank each period, and there is no explicit

exit shock. If a firm is forced to exit, then it does not produce that period and defaults on its loan.

The probability of surviving (i.e. not receiving an exit shock) is time-varying and region specific, with law of motion:

$$q_{s,t} = \rho^q q_{s,t-1} + \epsilon^q_{s,t}$$

where ρ^q is the regional persistence of exit rates and $\epsilon^q_{s,t}$ is the stochastic exit shock, which may be correlated across islands.

The region's production sector and loan demand. The decisions and outcomes of the individual firms can be studied as a single regionally representative firm. The representative firm aggregates bank-specific loans into a composite credit product $L_{s,t}$ using CES aggregator technology:

$$L_{st} = \left(\int_{i \in \mathcal{B}_{s,t}} \eta_{ist}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}} \, d_s \right)^{\frac{\sigma_{s,t}}{\sigma_{s,t}-1}}, \qquad \prod_{\mathcal{B}_{s,t}} \eta_{i,s,t}^{\frac{1}{||\mathcal{B}_{s,t}||}} = 1, \tag{1}$$

where $\sigma_{s,t} > 1$ is the elasticity of substitution across bank loans, and $\eta_{ist} \geq 0$ is the firm's idiosyncratic preference weight for bank i. The bank's product appeal $\eta_{i,t}$ may subsume aggregate and bank-specific demand shocks, as well as account for non-pecuniary differences that create persistent differences in bank-specific demand (such as the strictness of loan terms and branch locations).[2] Simultaneously, the elasticity of loan substitution may be considered a stand-in for local frictions that make substituting across bank loans more difficult for borrower. These frictions may include, for example, search and matching costs and transportation costs that generate sticky borrower-lender relationships (see Freixas and Rochet (2008) for a textbook treatment of these frictions).

The firm uses the composite loan $L_{s,t}$ to hire labor as a composite input to produce output with standard decreasing returns to scale technology and time-varying total factor productivity:

$$Y_{s,t} = A_{s,t} \cdot M^{\alpha}_{s,t}, \quad \text{with } \alpha \in (0,1),$$

$$a_{s,t} = \rho^a a_{s,t-1} + \epsilon^a_{s,t} \qquad a_{s,t} = \log A_{s,t}$$

where $A_{s,t}$ is region-specific productivity, $M_{s,t}$ is region-specific employment, and $\epsilon^a_{s,t}$ is the firm's productivity shock.

persistence in borrower-lender relationships, there is no multiple-equilibria or "winners curse" present in this economy. These outcomes can arise in the presence of sticky borrower-lender relationships and asymmetric screening ability, as analyzed by works recently reviewed by Blickle et al. (2024).

[2] See Hottman et al. (2016) or Eslava et al. (2024) for a more thorough description of firm-specific appeal in a similar oligopolistic competition and CES demand setting, but in a non-financial firm application.

Combining the loan aggregation and output production technology, the firm's problem is to maximize profits by choosing labor inputs N_t, taking wages as given, while minimizing financing costs by choosing loans $\{L_{ist}\}_{i\in\mathcal{B}_{s,t}}$, taking interest rates $\{R_{i,s,t}\}_{i\in\mathcal{B}_{s,t}}$ as given:

$$\max_{M_{s,t},\ \{L_{is,t}\}_{i\in\mathcal{B}_{s,t}}} \quad A_{s,t}M_{s,t}^{\alpha} - R_{s,t}L_{s,t} \quad \text{s.t} \tag{2}$$

$$L_{s,t} \geq W_{s,t}M_{s,t} \qquad L_{s,t} = \left(\sum_{i\in\mathcal{B}_{s,t}} \eta_{i,s,t}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}}\right)^{\frac{\alpha\sigma_{s,t}}{\sigma_{s,t}-1}}$$

where $L_{s,t} \geq W_{s,t}M_{s,t}$ is the condition that the Firm must pay its labor bill with a working capital loan before production can begin.

Retailers and households. The remaining regional agents, retailers and households, are standard and I leave their formal problems and discussion of non-financial regional outcomes to Appendix A.

Central Bank. The Central Bank sets the nominal interest rate following an inertial Taylor Rule:

$$I_t = I_{t-1}^{\rho_i}\left(\hat{Y}_t^{\psi_y}\hat{\Pi}_t^{\psi_\pi}\right)^{(1-\rho_i)} V_t \tag{3}$$

where $\hat{Y}_t$ is the aggregate output gap (with respect to steady state Y_t) and $\hat{\Pi}_t$ is the inflation gap (with respect to an exogenous determined inflation target), and V_t is the monetary policy shock that follows the standard log AR(1) process:

$$\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t^{\nu} \qquad \nu_t = \log V_t.$$

The Central Bank interacts with the economy primarily through lending capital funds directly to banks at the nominal policy rate. I assume the Central Bank is a deep-pocketed investor and can lend as much capital as is demanded by banks at the policy rate it sets.

2.3 Banks

Banks operate across a set of oligopolistic (Cournot) competitive regional loan markets, $s \in S$, with the goal of maximizing their expected present discounted franchise value (i.e. net worth). Banks compete in these loan markets by issuing uncollateralized, one period, risky loans. Loans are risky because borrowers will default with an exogenous time-varying probability. When loans default the entire principal is lost.

2.3.1 The bank's problem

Bank i's formal dynamic programming problem is:

$$\max_{\{L_{i,s,t}\}_{s\in S}} \mathbf{V}(N_{i,t}, \mathcal{M}_t, \Omega_t) = \mathbb{E}_t \Pi_{i,t} + (1-m)\beta \mathbb{E}_t \mathbf{V}(N_{i,t+1}, \mathcal{M}_{t+1}, \Omega_{t+1}) \quad \text{s.t.} \tag{4}$$

$$\Pi_{i,t} = \sum_{s\in S} \left\{ R^*_{i,s,t} L_{i,s,t} - f(d_{i,s}) \right\} - (L_{i,t} - N_{i,t}) C_{i,t} - \Gamma(\sigma^R \frac{L_{i,t}}{N_{i,t}}) \qquad \text{(Profit function)}$$

$$R^*_{i,s,t} = (1 - \lambda_{i,s,t}) R_{i,s,t} \qquad \text{(Realized gross return)}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \qquad \text{(Net worth LoM)}$$

where the bank maximizes profits, and in turn net worth, by issuing a portfolio of region specific loans $\{L_{i,s,t}\}_{s\in S}$. The region-specific gross realized portfolio return on loans, $R^*_{i,s,t}$, and total profits $\Pi_{i,t}$ realized at end of period t after $\lambda_{i,s,t}$ percent of borrowers default. The bank's net worth in period t is denoted $N_{i,t}$ and is simply the previous period's net worth plus realized gains or losses from the current period's loan portfolio. Future net worth is discounted at rate $\beta < 1$. Banks exogenously exit with probability m at the end of the period after profits are realized.[3]. Banks take as given the cross-sectional distribution of banks' balance sheets and fundamentals, $\mathcal{M}_t$, (which includes their own states) and the distribution of region-specific fundamentals, Ω_t.

2.3.2 The bank's costs

Banks issue loans after paying capital, regulatory, and trade costs.

Capital costs. Banks can use their net worth, $N_{i,t}$, to fund loans and can borrow additional capital from the Central Bank. The cost of capital funding for banks will be the federal funds rate and a multiplicative markup, so that $C_{i,t} = G_{i,t} I_t$, where $C_{i,t}$ is the banks marginal cost of capital, $G_{i,t}$ is the banks funding technology (or capital cost premium), and I_t is the nominal federal funds rate. Capital funds are fungible and can be used in any market to make loans.

The assumption that the capital cost of forming loans is bank-specific is motivated by the classical Monti-Klein oligopolistic banking model, or recent work on banking market power in deposit markets, such as Drechsler et al. (2017), and deliberate spatial sorting into markets with cheap deposits,

[3] I will make the simplifying assumption that banks are sufficiently well funded (that is, have a sufficiently large enough $N_{i,t}$) to cover loan losses so that I may abstract away from the possibility of endogenous bank failures and inability to pay depositors or resulting bank runs

such as Oberfield et al. (2024).

Regulatory costs. Banks also pay a regulatory cost based on their expected risk-adjusted leverage ratio, $\mathbb{E}(\sigma^R \frac{L_{i,t}}{N_{i,t}})$.[4] The regulatory cost function, $\Gamma(\cdot)$, will be kept general for the remainder of the presentation of the analytical model, and is assumed to be increasing and weakly convex. More formally, I assume $\Gamma'(X) \geq 0$ and $\Gamma''(X) \geq 0$ for all $X \in \mathbb{R}$. The convex regulatory costs acknowledges the real presence of such costs borne by banks, as well as establishes a connection between a bank's net worth, $N_{i,t}$, and ability to make loans. Moreover, I model regulation as a convex cost rather than a hard constraint, as in Gabaix and Maggiori (2015), to preserve differentiability and to allow regulatory tightness to map continuously into loan pricing.

By explicitly incorporating the expected portfolio return risk, the regulatory cost term captures a reduced-form risk-averse asset pricing dynamics, mirroring those highlighted by the intermediary asset pricing literature (namely the seminal works by He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014), or more recent work such as Akinci et al. (2022)).

Trade costs. There is a fixed cost of operating a bank each period f, and a fixed cost of operating in markets outside a bank's headquarters market, $f(d_{i,s})$, that is increasing in distance from the headquarters and non-headquarters market, $d_{i,s}$. The distance cost of operating in a bank's headquarters region is zero. These spatial fixed costs are central to generating endogenous regional concentration and time-varying loan market structure. The distance-based cost is similar in spirit to an iceberg trade cost common in the trade models of endogenous regional market entry and exit, such as Melitz (2003), and recent spatial banking models, such as Oberfield et al. (2024) where it is a result of managerial span-of-control problems.

2.3.3 The Bank's screening technology

Banks form their loan portfolio each period by screening loan Applicants and offering credit to those it predicts will not default. However, with imperfect screening technology, the bank is not able to guarantee that its borrowers will not default. As a result, banks must charge a risk premia to compensate for expected losses. That is, the bank's expectations for, and ultimately realizations of, regional loan losses $\lambda_{i,s,t}$ are determined by its ability to solve a loan-level screening problem. I next turn to describing this problem in detail, which is motivated by the seminal Hauswald and Marquez (2003) model of banking competition with asymmetric screening ability.

[4] Various authors take up a similar object of interest when formulating regulatory costs and constraints. For example, Gabaix and Maggiori (2015) focus on a liquidity ratio while Coimbra and Rey (2024) employ a leverage ratio.

Loan Applicants. Loan applicants are of two types, $\tau \in \{l, h\}$, such that l type borrowers default and h type do not. Each individual applicant in market s will receive an exit shock and default with probability $1 - q_{s,t}$. By the law of large numbers, $1 - q_{s,t}$ percent of loans in the bank's regional loan portfolio will default if all applicants are given loans.

Screening technology and productivity. Banks have access to an imperfect screening device that creates a private noisy signal, ι, that labels a loan applicant as an h type (indicating the firm will not be hit with the exit shock) or l type (indicating the firm will be hit with the exit shock). Banks only extend loans to applicants with an h type signal, $\iota = h$.

The probability that the borrower is actually h type given its signal is h type is determined by the bank's *screening ability*, which is the accuracy of the screening device and denoted ψ, such that:

$$Pr(\tau = h|\iota = h) = \frac{\psi q}{\psi q + (1 - \psi)(1 - q)}, \quad \psi \in [0.5, 1], \qquad q \in [0, 1] \tag{5}$$

where q is the *borrower quality*, the true proportion of the population that will not default. With $\psi = 1/2$, the signal is useless and the Bank relies exclusively on its log odds prior of successful screening ($\frac{q}{1-q}$). When $\psi = 1$, the signal is perfectly accurate and $Pr(\tau = h|\iota = h) = 1$.

I interpret a bank's screening ability as its lending productivity. Higher screening ability reduces expected loan losses by reallocating funds away from borrowers that are likely to default and toward solvent firms. In doing so, effective screening preserves financial capital that would otherwise be dissipated through defaults, thereby lowering aggregate financial risk and expanding the volume of funds that can be intermediated. This role is directly analogous to technological productivity in non-financial firms, which measures the extent to which a given set of inputs can generate a larger effective resource set for the economy. Improvements in screening technology similarly enlarge the economy's usable financial resources, making bank screening ability the financial-sector counterpart to standard notions of non-financial productivity.[5]

Loan application and offer timing. Banks do not know their own screening ability before making lending decisions, which creates a source of risk that the banks must bear to operate. The exact timing of the "morning" lending sub-period follows:

- Potential borrowers come to the bank and submit loan applications, becoming Applicants.

[5] Philippon (2015) makes a closely related argument for banks' monitoring costs, which are nearly isomorphic to my notion of screening ability. The key distinction is that he assumes perfect screening with a resource cost of monitoring that banks incur to assess risk and pass on to borrowers, whereas in my framework screening is imperfect and borrower costs arise endogenously as compensation for expected loan losses.

- Banks do not know the current efficacy of their screening ability, so form $E\psi_{i,t}$.
- Banks evaluate all Applicants and formulate their loan offer rate based on the expected regional portfolio default rate $\lambda_{i,s,t}$.[6]
- Applicants with h type signals are offered loans while those with l type signals are rejected.
- Accepted applicants accept or reject the bank's take-it-or-leave-it loan rate offer.

Banks and borrowers then learn the realized state of the world in the "afternoon" sub-period, which also implies the Bank's technological efficiency for the given period. A full timeline of events within a period are outlined at the end of this section.

2.3.4 Banking sector evolution

The banking sector is the time-varying set of banks $\mathcal{B}_t$. Along the intensive margin of the banking sector, individual banks are characterized by their endogenous net worth $N_{i,t}$, exogenous region-specific appeal $\bar{\eta} = \{\eta_{i,s,t}\}_{s\in S}$, screening ability $\psi_{i,t}$, and capital cost premium $G_{i,t}$. The banking sector can be characterized by the joint-distribution of these four bank-level and bank-region-level characteristics, $\mathcal{M}(N, G, \psi, \bar{\eta})$, where the density function $m(N, G, \psi, \bar{\eta})$ is the population share of banks characterized by the specific tuple of values $\{N, G, \psi, \bar{\eta}\}$.

Bank birth and death. Along the extensive margin, the cardinality of the banking sector $\mathcal{B}_t$ is held fixed for tractability. Banks are subject to an exogenous exit shock and are replaced one-for-one by new entrants upon exit.[7] This replacement mechanism prevents unbounded accumulation of bank net worth and ensures a stationary cross-sectional distribution of banks. Banks are ex ante identical at birth, draw a headquarters location uniformly across regions, and begin operating as incumbents upon entry. Entrants are endowed with initial net worth N_0.

Evolution. With banks entering, exiting, and growing, the banking sector distribution evolves along the extensive and intensive margin each period. The banking sector evolves according to the time-invariant law of motion $\mathcal{M}_{t+1} = \Phi(\mathcal{M}_t)$.

2.4 Timeline and Recursive Equilibrium

I conclude the description of the economic environment by summarizing the within-period timing and formally defining the economy's recursive equilibrium.

[6] Recall that all Applicants that will receive a loan offer are the same h signal type, thus pose the same perceived credit risk to the bank and will receive identical offer rates, so I will abstract pricing from the loan to portfolio level.

[7] This is similar to the exogenous banker exit and replacement strategy in Gertler and Kiyotaki (2010, 2015).

Timeline within a Period. Each period t is divided into a financing morning and a real-activity afternoon. The sequence of events is as follows:

- *Partial states are realized.* Regional productivity, bank-region appeal, and banks' capital cost premiums are observed.
- *Morning: Bank decisions and lending.* Given the policy rate and current states, banks choose which regions to enter, raise funding, screen loan applicants, and post take-it-or-leave-it loan rates. Firms submit loan applications and accept offered credit.
- *Remaining states are realized.* Banks' screening abilities and firms' exit rates are realized.
- *Afternoon: Production and repayment.* Financed firms hire labor and produce output. Exogenous exit shocks are realized, defaults occur, and surviving firms repay loans. Banks realize profits, net worth is updated, and exogenous bank exit occurs. Finally, the Central Bank sets the policy interest rate.

This timing ensures that banks' loan pricing and market entry decisions are forward-looking, embedding expectations over borrower risk, screening outcomes, and future net worth.

Recursive Equilibrium. Let the aggregate state at time t be $(\mathcal{M}_t, \Omega_t)$, where $\mathcal{M}_t$ denotes the cross-sectional distribution of banks over net worth, funding costs, screening ability, and regional appeal, and Ω_t collects the distribution of regional fundamentals, including productivity, firm exit rates, and loan market frictions.

A recursive equilibrium is a set of policies for banks, firms, households, and retailers such that, given $(\mathcal{M}_t, \Omega_t)$:

- Banks choose loan supply and regional entry to maximize expected discounted net worth, subject to balance sheet, regulatory, and screening constraints.
- Firms choose loan demand and labor inputs to maximize profits, taking loan rates and wages as given.
- Households and retailers solve their respective optimization problems.
- The Central Bank sets the policy rate according to its monetary policy rule.
- All regional loan, labor, and goods markets clear.

- The banking sector distribution evolves according to

$$\mathcal{M}_{t+1} = \Phi(\mathcal{M}_t),$$

as induced by banks' optimal decisions, realized loan outcomes, and bank exit and entry.

There are two facts about the recursive equilibrium that will be important to consider in the following empirical estimations and quantitative simulations. First, this recursive equilibrium assumes that regional loan markets are "large enough" that the entry of any single bank does not materially alter average bank fundamentals in the region. In smaller regions, bank entry would constitute a dynamic game, requiring a modified equilibrium concept (see Weintraub et al. (2008)). Further discussion is provided in Section 3. Second, although the economic environment admits a unified stochastic law of motion for the full state vector, in the empirical implementation I estimate the model in two layers. First, I estimate the laws of motion for aggregate states using economy-wide moments. Second, taking these aggregate dynamics as given, I estimate bank- and region-specific deviations from the aggregate that govern idiosyncratic fundamentals and outcomes.

3 Analytically Characterizing Bank Dynamics

Having outlined the economic environment, I next turn to analytically exploring how decisions are made and what determines their outcomes in this economy. I begin by characterizing bank-level regional entry and exit, competition, and lending outcomes, before turning to considering how these firm dynamics shape the distribution of activity across the economy.

3.1 Characterizing bank-level outcomes

I will first analytically characterize bank dynamics and the fundamentals and frictions that shape these firm-level outcomes. Additional bank-level outcomes, including bank's region-specific loan demand, portfolio return risk, and profit margins, along with proofs for each result discussed here are presented in Appendix A.

3.1.1 Loan losses

Banks know neither $\psi_{i,t}$ nor $q_{s,t}$ while screening applicants and must use $\mathbb{E}\psi_{i,t}$ and $\mathbb{E}q_{s,t}$, which leads to the probability that the bank's signal (of type h) is correct:

$$\mathbb{E}[Pr(\tau = h|\iota = h)] = \frac{\mathbb{E}\psi_{i,t}\mathbb{E}q_{s,t}}{\mathbb{E}\psi_{i,t}\mathbb{E}q_{s,t} + (1 - \mathbb{E}\psi_{i,t})(1 - \mathbb{E}q_{s,t})} \tag{6}$$

implying the expected default rate $\mathbb{E}\lambda_{i,s,t} = 1 - \mathbb{E}[Pr(\tau = h|\iota = h))]$.

The realized default rate will then reflect the realized screening accuracy across the bank's portfolio of loans, and assuming a "large enough" portfolio that the law of large numbers applies, will be:

$$\lambda_{i,s,t} = 1 - \mathbb{E}[Pr(\tau = h|\iota = h)|\psi] = 1 - \frac{\psi_{i,t} q_{s,t}}{\psi_{i,t} q_{s,t} + (1 - \psi_{i,t})(1 - q_{s,t})} \tag{7}$$

so that the resulting difference between the expected and realized loan default rate is:

$$\lambda_{i,s,t} - \mathbb{E}\lambda_{i,s,t} = \mathbb{E}[P(\tau = h|\iota = h)|\psi_{i,t}] - \mathbb{E}[Pr(\tau = h|\iota = h)] \tag{8}$$

a function of the difference in expected and realized screening ability. Moreover, Equation 7 shows that as $\psi_{i,t}$ approaches one, the screening device is perfect and the loan default rate falls to zero as banks are able to filter out all would-be defaulting borrowers. Similarly, as the borrower quality $q_{s,t}$ approaches one, loan defaults fall to zero because no firms are exiting.

3.1.2 Loan rates

The Bank solves Equation 4 by charging the region-specific loan rate:

$$R_{i,s,t} = \underbrace{\frac{1}{1 - E\lambda_{i,s,t}}}_{\text{risk premium}} \cdot \underbrace{\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}}}_{\text{markup}} \cdot \underbrace{\left(C_{i,t} + \Gamma'(\sigma^R \frac{L_{i,t}}{N_{i,t}})\right)}_{\text{marginal cost}} \tag{9}$$

so that as the expected default rate, $\lambda_{i,s,t}$, bank-region market share $s_{i,s,t}$, cost of capital $C_{i,t}$, or regional market frictions $\sigma_{s,t}$ increases, so does the interest rate charged to the market. Moreover, expanding the expected default rate, regulatory cost, and capital cost yields reveals:

$$R_{i,s,t} = \underbrace{\left[1 + \mathbb{E}_t\left(\frac{1 - q_{s,t}}{q_{s,t}}\right) \cdot \mathbb{E}_t\left(\frac{1 - \psi_{i,t}}{\psi_{i,t}}\right)\right]}_{\text{risk premium}} \cdot \underbrace{\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}}}_{\text{markup}} \cdot \underbrace{\left(I_t G_{i,t} + \zeta_t \frac{\sigma^R}{N_{i,t}}\right)}_{\text{marginal cost}} \tag{10}$$

where the regulatory cost function is linear is assumed to linear in a time-varying cost coefficient ζ_t, so that $\Gamma(\cdot) = \zeta_t(\sigma^R \frac{L_{i,t}}{N_{i,t}})$. Loan rates decrease as expected borrower quality $q_{s,t}$ or bank's expected ability $\phi_{i,t}$ increases, while increasing with the bank's funding cost premium $G_{i,t}$. Conversely, as the size (measured by net worth) of the bank increases the loan rate decreases and the quantity supplied increases, in line with standard macro-banking models (for example the seminal Gertler and Kiyotaki 2010 model) and empirical evidence (Ottonello and Song 2025).

I rewrite the equation one last time into a "finance view" of the loan rate to emphasize that even though the rate is born of a firm dynamics model, it still captures the common features of loan rates explored in the finance and IO literature. The loan rate can be rewritten as follows:

$$R_{i,s,t} = \left[\underbrace{\frac{\sigma_{s,t}}{(\sigma_{s,t}-1)}}_{\substack{\text{exog.}\\ \text{markup}}} \cdot \underbrace{\frac{1}{(1-s_{i,s,t})}}_{\substack{\text{endog.}\\ \text{markup}}} \right] \cdot \left[\underbrace{\mathbb{E}_t\left(\frac{1}{1-\lambda_{i,s,t}}\right)}_{\substack{\text{risk neutral comp.}\\ \text{for expected defaults}}} + \underbrace{\frac{\sigma^R}{1-\mathbb{E}\lambda_{i,s,t}}}_{\text{quantity of risk}} \cdot \underbrace{\frac{\zeta_t}{N_{i,t}}}_{\substack{\text{price}\\ \text{of risk}}} \right] \cdot \underbrace{C_{i,t}}_{\substack{\text{capital}\\ \text{cost}}} \tag{11}$$

which emphasizes two key features of banks' pricing behavior.

Markups. First, profitable markups exist, are both exogenous and endogenous to the bank's behavior. The exogenous markup arises from loan market frictions that stand in for search and matching frictions which might lead to sticky borrower-lender relationships. That is, a portion of the markup is likely attributable to the premium extracted from sticky lending relationships. The endogenous markup arises from the bank's market share, thus is an endogenous oligopolistic rent that accrues due to the bank's decision to enter and compete in regional loan markets. The oligopolistic markups are a key feature of banking sector dynamics models such as Corbae and D'Erasmo (2021, 2025).

Compensation for risk. Second, the compensation for risk is a function of the level of expected defaults as well as terms that capture dynamics similar to standard measures of the quantity of risk (the loan portfolio return variance divided by the expected return, here the solvency return rate) and the price of risk (a time-varying coefficient divided by the wealth of the investor, in this case, the net worth of the bank). In this view, the time-varying regulatory cost coefficient would in fact be interpreted as the bank's time-varying risk aversion, which later empirical estimates will corroborate. The presence of the price and quantity of risk means that a bank prices its loan rates as if it was a risk-averse investor pricing risky assets. This captures the key dynamics of the risk-averse intermediary asset pricing literature. The compensation for risk term also highlights that the bank requires compensation for risk over and above the risk-neutral compensation for expected defaults. This means that the price and quantity of risk term may also be interpreted as the Excess Loan Premium (ELP), analogous Gilchrist and Zakrajšek (2012) Excess Bond Premium (EBP).

3.1.3 Market share

I next turn from prices to quantities and analytically characterize banks' competitive outcomes. A bank's log regional loan market share (i.e. relative regional lending volume) is:

$$\log(s_{i,s,t}) = \frac{\sigma_{s,t}\log\tilde{s}_{s,t}}{\sigma_{s,t}\log\tilde{s}_{s,t}+1}[(1/\sigma_{s,t})(\log(\eta_{i,t})-\tilde{\eta}_{s,t})+(\tilde{C}_{s,t}-\log(C_{i,t}))+\ldots$$
$$\ldots+(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t}))+(\mathbb{E}\tilde{\chi}^{\psi}_{s,t}-\mathbb{E}\log(\chi^{\psi}_{i,t}))] \quad (12)$$

where $\chi^{\psi}_{i,t} = \frac{1-\psi_{i,t}}{\psi_{i,t}}$. A bank's market share is a function of its fundamentals relative to the (geometric) average fundamentals of other banks competing within the regional loan market. The bank's market share is an increasing function in its appeal relative to the average appeal, borrower quality relative to the average borrower quality, ability relative to the average ability, but decrease in capital costs relative to average capital costs. The fact that banks with the strongest fundamentals (i.e. the most productive banks) will garner the largest market share is in line with the firm dynamics in Melitz (2003) models of imperfect export competition.

However, a bank's fundamentals are not the only factors that matter for determining its market share. The bank's relative regulatory costs appear in Equation 12, meaning that a bank's pre-existing size (i.e. balance sheet conditions) also determines its ability to compete for market share. The greater the bank's size, the smaller its regulatory costs, and the more market share it will claim.

3.1.4 Regional market entry

I lastly describe the bank-level decision to enter or exit regional lending market. A bank will enter a regional market if the expected profits of operating in the region are positive. Given oligopolistic structure of the regional loan markets, the condition in turn depends on whether the bank can make a large enough profit per loan to pay its fixed cost of operating in the given region. For a bank's headquarters region, then the fixed cost is zero, so I can attribute the constant fixed cost of operating in any period to the headquarters region, $f_{i,s,t} = f$, following Melitz (2003). If the region is not the headquarters region then the fixed cost is $f_{i,s,t}f(d_{i,s,t})$. The point at which a bank will be indifferent between entering a regional market or not then follows from the expected profit condition:

$$\mathbb{E}log\bar{\Pi}_{i,s,t} = \mathbb{E}\left\{\log\frac{\sigma_{s,t}}{\sigma_{s,t}-1}+s_{i,s,t}+\log s_{i,s,t}+w_{s,t}+\frac{1}{\alpha}y_{s,t}-\frac{1}{\alpha}a_{s,t}\right\} = \log f_{i,s,t} \quad (13)$$

where $log\bar{\Pi}_{i,s,t}$ is the cutoff expected profit at which a bank will enter a region. This condition is based on four primary forces, the banks expected competitiveness, local market frictions, depth of the regional market, and the distance from headquarters.

Spatial sorting factors. First, market entry is a function of the bank's expected market share (and oligopolistic profits), thus the more competitive a bank is expected to be, the more likely it will enter the market. This is the productive exporter motivation that is seen in trade models such as Melitz (2003). Second, the larger the regional economy, as measured by its log output, $y_{s,t}$, the more likely a bank will be to enter the market. Third, the farther the region is from the bank's headquarters, the more expensive it will be to start operations there, and the less likely the bank will be to move into the new region. The second force represents an agglomeration effect while the third force may represent a reduced-form managerial span-of-control cost, both of which are commonly present in spatial sorting models such as Oberfield et al. (2024) or Oberfield et al. (2024). Lastly, the greater the local market frictions, the greater the exogenous markups and profit the potential entrant can expect to make in the region. Therefore, as local market frictions rise, the more likely banks are to enter the region, all else equal.

Entry game. The dependence on the bank's expected relative fundamentals (i.e. its competitiveness in the local banking sector) introduces a potentially high dimensional game among banks. On the one hand, if a regional market is "large" then the entry of a single bank will not meaningfully change the industry average, so the potential entrant can use the current distribution of fundamentals to make its entry calculations (this follows from the Oblivious Equilibrium concept presented by Weintraub et al. (2008) and is common in the IO literature). On the other hand, if the region is "small" then the entry of an individual bank may substantially change its industry fundamentals and entering banks then need to consider which other banks in the economy could enter in a way that would make their own entry unprofitable. I will assume that regions are equivalent to U.S. states and satisfy the properties of "large" markets for the duration of this paper.[8]

3.2 Banks and the real economy

The fundamentals and frictions that drive bank dynamics work through the price and quantity of loans to also shape aggregate business cycles and the regional distribution of activity. Bank dynamics effect the real economy through one direct and two indirect channels.

3.2.1 Fundamental shocks

First and most directly, banks acts a direct source of shocks that effect regional employment and production through the price of working capital. Total regional production is a function of the

[8] I will present empirical evidence in Section 4.3 that supports this assumption.

firm's technology, labor inputs, and firm exits:

$$Y_{s,t} = (1 - \lambda_{s,t}) \cdot A_{s,t} \cdot \left(\frac{\alpha A_{s,t}}{R_{s,t} W_{s,t}} \right)^{\frac{\alpha}{1-\alpha}} \tag{14}$$

where $\lambda_{s,t}$ is the producing firm exit rate in region s at time t. Note that regional production is increasing in regional productivity $A_{s,t}$, but decreasing in regional loan rates $R_{s,t}$, wages $W_{s,t}$, and producing firm exits $\lambda_{i,t}$ which are equivalently the region's loan default rate.

Therefore, as shocks to banks' fundamentals or loan market frictions affect loans rates they in turn impact regional production and employment. An increase in banks' screening ability, funding technology, or decline in market frictions will decrease loan rates and increase regional activity. This can lead to either aggregate fluctuations in employment and GDP if financial shocks are common across banks. However, this direct channel can also lead to variation in output across regions if banks' productivity shocks are regionally concentrated (similar to the regional non-financial productivity shocks studied in Caliendo et al. 2018) or if individual banks receive idiosyncratic shocks and have a granular effect in the regions where they operate (as in Gabaix 2011 and Gabaix and Koijen 2020). The financial sector can further act as a source of heterogeneous regional outcomes through two additional channels.

3.2.2 The credit exposure channel

The financial sector acts as an asymmetric propagation mechanism for regional shocks through the *credit exposure channel*, leading to heterogeneous regional economic outcomes. The intuition is straightforward. Multi-region banks may be impacted by firm-specific or region-specific shocks, which counterfactually impair (or improve) its profits in a given year, leading to a change in marginal regulatory costs that gets passed along across the markets it serves in the next period. Given that banks may have different geographic footprints, banks will propagate shocks through their loan rates to a patchwork of regions, potentially leading to heterogeneity across regional outcomes. The following outlines the channel more formally.

First, a bank's region-specific loan rate is:

$$R_{i,s,t} = \left[1 + \mathbb{E}_t \left(\frac{1 - q_{s,t}}{q_{s,t}} \right) \cdot \mathbb{E}_t \left(\frac{1 - \phi_{i,t}}{\phi_{i,t}} \right) \right] \cdot \frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}} \cdot \left[C_{i,t} + \Gamma' \left(\frac{L_{i,t}}{N_{i,t}} \right) \right]$$

which depends upon, among other factors, the bank's net worth $N_{i,t}$. A bank's net worth evolves according to the simple law of motion: $N_{i,t} = \Pi_{i,t} + N_{i,t-1}$, for periods in which the bank does not

exit. That is, the change in net worth, thus a loan rates, is determined by the bank's profits.

A bank's profit in a given region is given by the expression:

$$\Pi_{i,s,t} = (r^*_{i,s,t} - c_{i,t} - \Gamma'_{i,t})L_{i,s,t} - f_{i,s,t}$$

which is a function of both the bank's fundamentals through net returns $r^*_{i,s,t}$ and capital costs $c_{i,t}$, as well as the region's fundamentals through local demand for credit, $L_{i,s,t} = W_{s,t}M_{s,t}$.

The bank's profits and next periods net worth are thus a function of the bank's fundamentals which determine profitability and regional fundamentals which determine credit demand. As a result, a shock that impact either the bank's or region's fundamentals will in turn impact regional profits and total profits. If these shocks increase profits then the net worth increases, decreasing loan rates across all regions the bank operates in next periods, while if the shocks decrease profits then the bank will increase loans rates across all the regions it operates in next period. This loan pricing behavior will result in the asymmetric propagation of shocks across regions based on the operating footprint of banks passing along the shocks. This behavior is analogous to banks increasing loan rates for all firms in its portfolio to make up for losses from one defaulter, which has been empirically documented in works such as Amiti and Weinstein (2018).

3.2.3 The credit elasticity channel

The financial sector mediates the regional elasticity to exogenous shocks through the *credit elasticity channel*. The worse (better) fundamentals or frictions are, the greater (smaller) the region's response to shocks, all else equal. This is because the bank-level loan rates that make up the regional cost of borrowing are more sensitive to shocks when bank fundamentals are weak or loan market frictions are high. The following formally characterizes this behavior, using monetary policy shocks for concreteness of exposition.

Taking the derivative of bank-level loan rates with respect to an increase in the federal funds rate:

$$\frac{\partial R_{i,s,t}}{\partial I_t} = \mathbb{E}\left(\frac{1}{1-\lambda_{i,s,t}}\right)\frac{\sigma_{s,t}}{1-\sigma_{s,t}}\frac{1}{1-s_{i,s,t}}G_{i,t}$$

It then follows that the response of the loan rate to the monetary policy shock is a function of the bank's fundamentals, where stronger fundamentals, those that would decrease $R_{i,s,t}$ will dampen the response to a monetary policy shock. In contrast, weak fundamentals, those that would elevate $R_{i,s,t}$ will amplify the response to the monetary policy shock. Moreover, market share, a function of

fundamentals, then further reinforces the amplification or mitigation through as an indirect channel.

The regional loan rate $R_{s,t}$ is the CES aggregation of appeal-weighed bank-level loan rates $R_{i,s,t}$. Therefore, the common monetary policy shock will increase all bank-level loan rates, and in turn the regional loan rate. It also follows that if the fundamentals of banks operating in a given region are high or low relative to some baseline, the the response of the regional loan rate will be dampened or amplified accordingly.

Finally, take the derivative of region s's output with respect to the regional loan rates $R_{s,t}$:

$$\frac{\partial Y_{s,t}}{\partial R_{s,t}} = \frac{\lambda_{s,t} - 1}{W_{s,t}} A_{s,t} \frac{\alpha}{1-\alpha} \left(\frac{\alpha A_{s,t}}{W_{s,t}} \right)^{\frac{2\alpha-1}{1-\alpha}} R_{s,t}^{2(\frac{1-2\alpha}{1-\alpha})} \tag{15}$$

which shows that regional output declines as loan rates increase but the size of the change depends on $R_{s,t}$ and α. Equation 15 is decreasing in $R_{s,t}$ for $\alpha > 0.5$ and increasing in $R_{s,t}$ when $\alpha \leq 0.5$. Taking the standard labor share of output, $\alpha = 1/3$, implies that the change in output with respect to changes in the effective regional loan rate are increasing in the loan rate. Therefore, as fundamentals amplify or dampen the bank-level loan rate response to monetary policy shocks, they also amplify or dampen the response of regional output to the monetary policy shock.

Moreover, changes in the bank's net worth will also affect the transmission of monetary policy. As a bank's net worth increases its regulatory cost falls, which increases market share and dampens loan rates all else equal. That is, banks growing larger will dampen the response to monetary policy, all else equal, while banks shrinking will amplify their response. This falls in line with empirical findings by Jiménez et al. (2014) that small banks react more strongly to monetary policy tightening, as well as underscore the importance of balance sheet conditions.

Policy consequences. As regions have heterogeneous average banking sector average fundamentals, they will in turn have heterogeneous responses to common policy shocks through the credit elasticity channel. Moreover, as regions respond asymmetrically to the common shock, the fortunes of banks operating in different regions will vary and in turn be differentially propagated though the banking sector leading to persistent differences in regional outcomes due to a common, transitory, monetary policy shock and the credit exposure channel.

Table 1: Notation for Bank Fundamentals and Regional Frictions

Unobservable Fundamentals, Frictions, and Distortions		*Observable Bank and Regional Outcomes*	
$\psi_{i,t}$	Bank i's screening ability	$E_{s,t}$	Regional firm survival rate
$\chi^{\psi}_{i,t}$	Log-odds transformation of $\psi_{i,t}$	$\lambda_{i,t}$	Bank i's loan default rate
$G_{i,t}$	Bank i's capital cost premium	$s_{i,s,t}$	Bank i's market share in region s
$q_{s,t}$	Regional borrower quality	i_t	Net nominal federal funds rate
$\chi^{q}_{s,t}$	Log-odds transformation of $q_{s,t}$	$\tilde{G}_{i,t}$	Bank i's capital cost premium
$\eta_{i,s,t}$	Bank-region loan appeal		
$\sigma_{s,t}$	Regional elasticity of loan substitution		
ζ_t	Time-varying regulatory cost coefficient		
$A_{s,t}$	Regional non-financial productivity shifter		

4 Measuring Bank Fundamentals and Market Frictions

I next turn to estimating the unobservable fundamentals, regulatory distortions, and regional market frictions that drive bank-level dynamics and lending outcomes. To do so, I develop a theoretically disciplined state-space model that uses equilibria conditions from the structural model to map observable bank and regional outcomes to unobservable frictions, distortions, and fundamentals. Meanwhile frictions and distortions will be estimated with the cross-section of interest income adjusted for bank and regional fundamentals.

However, this raises the problem that the identification and measurement of both fundamentals and market frictions simultaneously rely on one another. Therefore, the measurement procedure will jointly estimate fundamentals and frictions with an Expectation-Maximization (EM) algorithm, which effectively estimates the unobservable fundamentals in the E-step, and the regional frictions and distortions in the M-step. The algorithm iterates between the two steps until estimated fundamentals and frictions converge. The following will outline these two steps in turn. I leave a formal description of the algorithm and estimating the non-financial regional productivity wedge, $A_{s,t}$ to Appendix C. Table 1 lists the variables and parameters to be estimated for convenience.

4.1 E-step: estimating fundamentals

I use a state-space model to jointly estimate bank fundamentals, $\phi_{i,t}$, $g_{i,t}$, region fundamentals $q_{s,t}$, and bank-region fundamental $\eta_{i,s,t}$. The state transition equation is a standard AR(1) process with *i.i.d.* Gaussian errors:

$$X_t = \Gamma X_{t-1} + \epsilon_t, \qquad \epsilon_t \sim N(0, Q), \tag{16}$$

where the state vector is the stacked partitioned vector $X_t = \{\log \chi_t^{\phi} \ \log G_t \ \log \chi_t^{q} \ \log \eta_t\}$ in which $\chi_t^{\phi} = \{\chi_{i,t}^{\phi}\}_{i\in\mathcal{B}}$, $G_t = \{G_{i,t}\}_{i\in\mathcal{B}}$, $\chi_t^{q} = \{\chi_{i,t}^{q}\}_{s\in S}$, $\eta_t = \{\eta_{i,s,t}\}_{i\in\mathcal{B},s\in S}$; Γ is the conformable matrix of auto-regressive coefficients, and ϵ_t is the analogously stacked vector of fundamental shocks. The shocks are *i.i.d.* across and within banks and regions, thus Q is a diagonal matrix. Note that I use the log-odds transformation of both screening ability $\chi^{\phi} = \frac{1-\phi}{\phi}$ and borrower quality $\chi^{q} = \frac{1-q}{q}$ rather than the direct objects of interest $\phi_{i,t}$ and $q_{s,t}$. This transformation of the latent states is necessary to maintain the linear structure of the measurement equations.

The unobservable fundamentals are then related to a set of observable bank and region characteristics: the regional firm survival odds, $E_{s,t} = \frac{1-q_{s,t}}{q_{s,t}}$, bank loan losses $\lambda_{i,t}$, bank capital costs premium $\log \tilde{G}_{i,t}$, and bank-region market share $s_{i,s,t}$.[9] The measurement equation relating the latent states (i.e. unobservable fundamentals) to the observable data then follows:

$$Y_t = \gamma + \underbrace{H_t X_t}_{\substack{\text{bank/region} \\ \text{fundamentals}}} + \underbrace{\tilde{H}_t X_t}_{\substack{\text{regional} \\ \text{banking sector} \\ \text{characteristics}}} + \underbrace{B_t Z_t}_{\substack{\text{exogenous} \\ \text{terms}}} + \underbrace{\nu_t}_{\substack{\text{meas.} \\ \text{error}}} \qquad \nu_t \sim N(0, R) \tag{17}$$

where the stacked vector of observables is $Y = \{\log E_t \ \log \lambda_t \ \log \tilde{G}_t \ \log s_t\}$ in which $\lambda_t = \{\lambda_{i,t}\}_{i\in\mathcal{B}}$, $\tilde{G}_t = \{\tilde{G}_{i,t}\}_{i\in\mathcal{B}}$, $E_t = \{E_{s,t}\}_{s\in S}$, $s_t = \{s_{i,s,t}\}_{i\in\mathcal{B},s\in S}$; the matrix H_t relates the latent fundamentals to the observable outcomes; $\tilde{H}_t$ is the conformable matrix such that $\tilde{H}_t X_t$ creates a stacked vector of region-specific banking industry (geometric) average fundamentals for calculating the bank-region market share following Equation 12; the vector Z_t is the stacked vector of exogenous variables (the bank-level marginal regulatory costs and nominal federal funds rate), while ν_t captures the possible measurement error and γ the vector of constants.[10] The measurement matrices are time-varying because the parameters in the market share equation (Equation 12) are time-varying functions of regional frictions, $\sigma_{s,t}$.

Consistent with the model structure, estimation proceeds in two steps: an aggregate component that identifies the laws of motion for economy-wide states, and an idiosyncratic component that disciplines bank- and region-level deviations from these aggregates. The bifurcated estimation is discussed in detail in Appendix C.

[9] Note that the bank's capital cost markup $G_{i,t}$ may be approximate observed using the bank's income statement. Therefore I will use $\log \tilde{G}_t$ to denote the observable approximation of the latent state $\log G_{i,t}$ to not confuse the observable proxy and true fundamental in exposition.

[10] The regional banking sector matrix $\tilde{H}_t$ additionally introduces the possibility of regional spillovers, and resembles the spatial autocorrelation term present in spatial vector-autoregression models, Beenstock and Felsenstein (2019).

Identification. The identification of bank and region fundamentals come from the theoretically disciplined measurement equation. For exposition, consider the model with only one bank and one region so that that the system would follow as:

$$\mathbf{X}_t = \begin{bmatrix} \log G_{i,t} \\ \log \eta_{i,s,t} \\ \log \chi^{\psi}_{i,t} \\ \log \chi^{q}_{s,t} \end{bmatrix}, \quad \mathbf{Y}_t = \begin{bmatrix} \log \tilde{G}_{i,t} \\ \log \lambda_{i,t} \\ \log E_{s,t} \\ \log s_{i,s,t} \end{bmatrix} = \gamma_t + \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ \bar{\sigma}_{s,t} & 0 & \frac{\bar{\sigma}_{s,t}}{\sigma_{s,t}} & \bar{\rho}^{\psi}_{s,t} & 0 \end{bmatrix}}_{\equiv \mathbf{H}_t} \mathbf{X}_t + \tilde{H}_t X_t + B_t Z_t + \nu_t \quad (18)$$

$$\bar{\sigma}_{s,t} \equiv \frac{\sigma_{s,t} \log \tilde{s}_{s,t}}{\sigma_{s,t} \log \tilde{s}_{s,t} + 1} \qquad \rho^{\psi}_{i,t} \equiv \rho^{\psi} \bar{\sigma}_{s,t}$$

where both s and i are equal to one. The observable variation used to identify each fundamental then becomes clear. First, the bank-level capital cost premium is approximately observed on the banks balance sheet, thus the fundamental is actually well proxied by observable data. Second, from public U.S. Census data one can construct an observable proxy for the state-level firm survival rate, $E_{s,t}$. From the model, the log-odds survival rate is solely a function of the exogenous regional fundamental $q_{s,t}$, thus the observable proxy for the odds of survival uniquely identifies $\chi^{q}_{s,t}$ (after explicitly accounting for measurement error in the state-space model). Third, the structural equation for a bank's loan default rate, Equation 7, shows that log loan default rate is a function of both the regional exit rate, $1 - q_{s,t}$, and the bank's screening ability $\psi_{i,t}$. However, with the regional exit rate pinned down by the observable firm survival odds, the residual loan default rate is accounted for by the bank's screening rate. Fourth, with all three other fundamentals already uniquely identified, the residual regional market share is determined by, and thus measures, the bank-region loan appeal.

Threat to identification. The primary threat to identification is within-region sorting of borrower and lenders. This is because the variation used to identify and measure banks' screening ability follows from the equation for log loan losses:

$$\log \lambda_{i,s,t} = \log \frac{1 - q_{s,t}}{q_{s,t}} + \log \frac{1 - \psi_{i,t}}{\psi_{i,t}}$$

In other words, if the bank's loan portfolio is not a function of $q_{s,t}$ then the resulting estimate of $\phi_{i,t}$, conditional on $q_{s,t}$, may be biased.

The structural model assumes that banks face common borrower quality within a state-level loan

market. However, this assumption can fail for two primary reasons. First, banks operate in a sub-region of the state that is subject to its own idiosyncratic exit rate shocks. For example, banks operating in rural upstate New York may operate in communities with very different local economic conditions, and exit rate fluctuations, than in New York city and the aggregate state economy. Second, banks may serve a specific industry or type of borrower that has a systematically elevated or dampened risk profile than the aggregate state-level economy. For example, banks serving the oil and gas industry in Pennsylvania may serve borrowers with a systematically amplified boom-bust cycle than the well-diversified aggregate state economy. If either of these cases are true, then the use of $q_{s,t}$ in the measurement equation for log loan losses will not reflect the true borrower quality the bank faces and the resulting estimate of the banks screening ability will be biased.

To remedy this problem, I will estimate a bank-specific borrower quality, $q_{i,t}$, and use it in the subsequent estimation of the state-space model. To do so, I will proceed in two steps.

First, I create a proxy for the firm survival rate in a tight geographic region around a bank's branch footprints. That is, I assume banks lend to borrowers residing or operating close to the bank's physical branch locations and in turn create a proxy for the borrower quality of firms operating in that narrowly defined banking market. I calculate the narrowly defined $\bar{q}_{i,t}$ as the (deposit-weighted) average county-level firm survival rate for counties with active bank branches. Formally, the proxy is calculated as:

$$\bar{q}_{i,t} = \sum_{c \in C_{i,t}} \left(\frac{d_{i,c,t}}{\sum_{j \in C_{i,t}} d_{i,j,t}} \right) q_{c,t}$$

where $C_{i,t}$ is the set of counties a bank is operating in at time t, $d_{i,c,t}$ is the number of deposits a bank holds in branches located in county c, and $q_{c,t}$ is the firm survival rate of county c. This first steps ameliorates concerns that banks are exposed to credit markets that differ in their regional exit rate shocks compared to the broader state-wide economy.

Second, I allow banks to have an amplified or attenuated response to the regional exit rate fluctuations to ameliorate concerns about exposure to scaled risk profiles. To do so, I estimate a bank-specific elasticity of log loan default rate to local borrower quality. However, I cannot simply project the proxy for local borrower quality onto the bank's log loan default rate because of the potential presence of shocks that impact both the exit rate and the bank's screening ability simultaneously —that is, the threat of omitted variable bias. For example, drones have been used by both farmers and banks to monitor crops, making their development a type of positive technology shock boosting both farmers productivity and banks ability to assess risk in real time (Blickle et al. 2024).

I therefore turn to an instrumental variable approach and estimate the the following model:

$$\log \lambda_{i,t} = \alpha_i \left[\log \widehat{\frac{1-\bar{q}_{i,t}}{\bar{q}_{i,t}}} \right] + \epsilon_{i,t}, \qquad \epsilon_{i,t} \equiv \log \frac{1-\psi_{i,t}}{\psi_{i,t}} + e_{i,t}$$

$$\log \frac{1-\hat{q}_{i,t}}{\hat{q}_{i,t}} = a + b x_t + \nu_{i,t} \qquad \nu_{i,t} \perp e_{i,t} \text{ and } \nu_{i,t} \perp \psi_{i,t}$$

where the elasticity of the default rate to local borrower quality, α_i, is the statistic of interest, $\hat{q}_{i,t}$ is the observable proxy for borrower quality $q_{i,t}$ estimated in the previous step, $\widehat{\frac{1-\bar{q}_{i,t}}{\bar{q}_{i,t}}}$ is the predicted borrower quality term, x_t is the instrument for borrower quality shocks, and $\epsilon_{i,t}$ is the residual that contains the unobservable log odds screening ability and measurement error $e_{i,t}$. The instrument must satisfy the standard relevance condition, as well as a two-faced exclusion restriction:

$$\mathbb{E}[x_t e_{i,t}] = 0, \quad \mathbb{E}[x_t \psi_{i,t}] = 0.$$

That is, the instrument must be exogenous (statistically independent) to both the measurement error $e_{i,t}$ and the bank-specific productivity $\psi_{i,t}$. Estimating the model will result in a consistent estimate of α_i and isolate the unobservable bank productivity in the residual loan default rate.

I use Känzig (2021) oil news shocks as the instrument x_t.[11] These shocks easily satisfy the relevance condition both theoretically (oil price shocks are effectively a marginal cost shock for non-financial firms that induce exits in standard Hopenhayn (1992) style models of firm dynamics), as well empirically (the first stage F-stat of the pooled model is 11042.89). The oil news shocks additionally satisfy the exclusion restriction —bank's screening technology is reasonably not impacted by oil price shocks.[12] See Appendix C for a detailed discussion of IV diagnostics and results.

As a result, I create an observable proxy for the bank's log borrower quality term and residualized loan losses. The former is the fitted value while the latter is the residual of the second stage regression. With an observable proxy for loan losses that is only a function of the bank's log productivity term and measurement error, I can update the state-space model of the bank-level fundamentals such that the residualized loan losses is used as the observable in place of the total loan losses. The bank's log borrower quality term then no longer loads onto the observable,

[11] Känzig (2021) identifies oil supply news shocks by exploiting high-frequency changes in oil futures prices around OPEC announcements. By focusing on asset price fluctuations during a tight window around the macroeconomic news announcement, he isolates variation in oil prices to supply shocks.

[12] This condition holds in the context of the theoretical framework guiding the empirical estimation. However, it may fail under deviations from the full-information rational expectations paradigm. For example, if banks exhibit bounded attention, large oil price shocks may divert resources away from screening and toward processing aggregate information, violating the exclusion condition.

removing the potential identification problem and allowing for the consistent estimation of the latent bank screening ability.

4.2 M-step: estimating regional frictions and industry distortions

While distortions and frictions are not directly estimated in the bank-level state-space model, they can be inferred through the lens of the bank's interest income. The following will describe how, given an estimate of bank-level fundamentals, the regulatory cost function and elasticity of intermediate loan substitution can be consistently estimated.

Bank-level net interest income can be partitioned into observable and unobservable components:

$$\underbrace{r^*_{i,t} - \sum_{s \in S_{i,t}} \frac{L_{i,s,t}}{L_{i,t}} \left[s_{i,s,t} + c_{i,t} - (\lambda_{i,s,t} - \mathbb{E}\lambda_{i,s,t}) \right]}_{\equiv \xi_{i,t}} = \zeta_t \left(\frac{\sigma^R}{N_{i,t}} \right) + \log \left(\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \right)$$

where $\xi_{i,t}$ is the portfolio loan-weighted average interest income adjusted for capital costs, endogenous markups, and the net compensation for loan defaults, ζ_t is the time-varying regulatory cost coefficient from assuming a linear functional form for $\Gamma'(\mathbb{E}\sigma^{R^*} \frac{L_{i,t}}{N_{i,t}})$, and $\sigma_{s,t}$ is the regional elasticity of loan substitution. The left hand side, $\xi_{i,t}$, is observable or can be estimated with bank fundamentals measured in the E-step. Therefore, I can jointly estimate the regulatory cost function and elasticity of substitution in two steps.

I first estimate the following bank-level interest income process:

$$\xi_{i,t} = \zeta_t \left(\frac{\sigma^R}{N_{i,t}} \right) + \log \left(\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \right) + \epsilon^{\xi}_{i,t}$$

where $\epsilon^{\xi}_{i,t}$ is the idiosyncratic measurement error. the regulatory cost coefficient ζ_t can be consistently estimated by projecting observable $\xi_{i,t}$ onto the bank's inverse log net worth, maintaining the identifying assumption that the regional elasticity of substitution and idiosyncratic measurement error is not correlated with the firms time t net worth. The resulting term $\zeta_t \left(\frac{\sigma^R}{N_{i,t}} \right)$ then gives the estimated regulatory cost of bank i at time t.

Second, I recover the elasticity of substitution $\sigma_{s,t}$ by taking the mean residual of banks only operating in region s at time t. Assuming mean zero measurement error, the average residual then represents the exogenous markup from regional frictions. This approach is equivalent to estimating

$\sigma_{s,t}$ by matching the first moment of the observed realized return distribution with the return equation implied by the banks' FOC conditions, and follows the "demand function approach" widely employed by the industrial organization literature.[13]

Identification. The regulatory cost function is identified by the covariance between the bank's net worth and its net interest income residualized against fundamentals. The intuition is that distortions in the banks interest income, after accounting for its own and borrower fundamentals, must be coming from either regulatory distortions, which realistically vary with bank size, or from market frictions. The former is therefore identified by the cross-sectional correlation between bank-level residualized income and size, while the latter is identified by the common bias in residualized income of banks operating within the same region.

Threats to identification. The threats to identification come from a failure of the conditional independence assumption in the ordinary least squares estimating the regulatory cost function. The residual term from the estimation is made up of the region-specific friction term and possible bank-level measurement error. The region friction is common across all banks operating in the region, thus its covariance with the regressor is zero. The estimation of the regulatory cost function is then unbiased as long as measurement error in the residualized interest income is uncorrelated with bank size. In an ex post analysis we can observe the measurement error in fundamentals estimated by the state-space model and can confirm this assumption.

4.3 Data

I will briefly describe the data used to estimate the fundamentals and frictions. In particular, I need to construct observable proxies for banks' loan rates, net worth, default rates, realized return rates, regional market shares, regional firm survival rates, and regional real wages.

Bank balance sheet and income statement data are obtained from U.S. regulatory Call Reports. Banks' net worth and the size of their loan portfolios are directly observable from the balance sheet. Other key objects of interest, including capital costs, loan default rates, and realized returns, are not directly observed and must be constructed from available accounting variables. Capital costs are proxied by interest expenses divided by total loans, capturing the average interest expense required to finance loans on the bank's balance sheet. A bank's loan default rate is defined as net

[13] See De Loecker et al. (2020) for a practical overview of the three primary strategies for estimating markups. I take the "demand function approach" in contrast to Corbae and D'Erasmo (2021) or Jamilov and Monacelli (2025) who use the "production function approach" because the former subsumes all unexplained variation in loan rates, over and above the marginal cost, and defines it as a bank's markup, leaving no room for risk compensation, market power, or regional frictions to play a role.

Table 2: Summary statistics of observable bank variables

Variable	Units	Mean	P(5)	P(25)	P(50)	P(75)	P(95)
Avg. capital cost	%	8.41	0.43	1.29	3.06	4.92	7.38
Loan rate	%	7.59	4.42	5.73	7.10	8.58	10.10
Loan default rate	%	0.51	0.00	0.02	0.12	0.36	1.54
Net worth	$ Bn.	17.32	0.20	0.55	1.16	2.66	14.17
Loans	$ Bn.	9409.48	91.34	291.38	692.85	1717.23	9528.16

Notes: This table presents the summary statistics for the panel of banks used to estimate bank-level fundamentals, regulatory costs, and regional loan market frictions. There are 11,682 unique banks in the unbalanced panel. Data is from 1995 to 2020.

loan charge-offs divided by total loan assets.[14] Realized loan returns are interest income from loans divided by total loans, capturing the loan-weighted portfolio return.

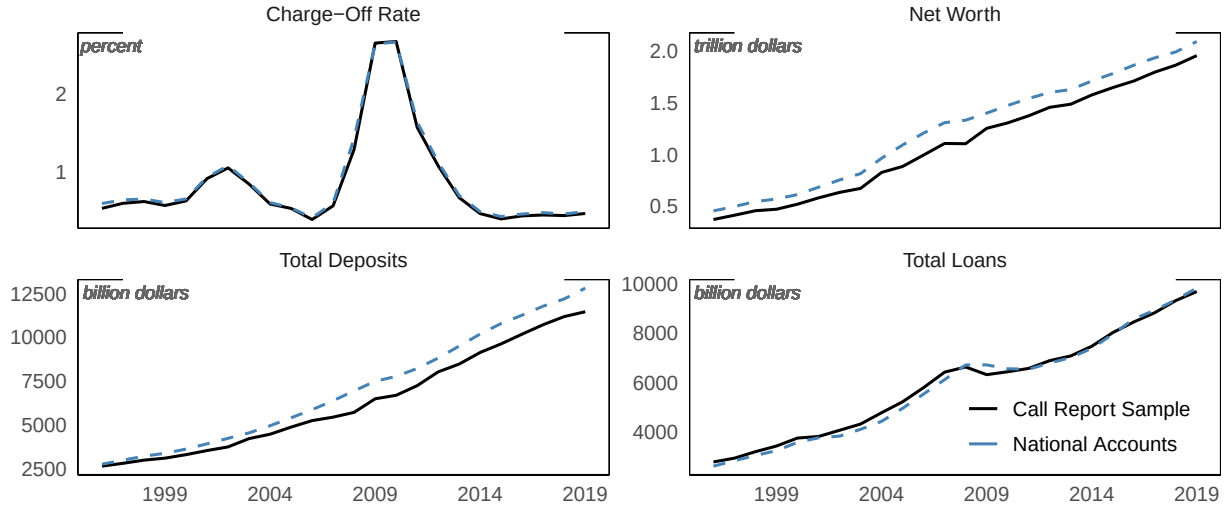
The panel of banks display substantial heterogeneity in both balance sheet size, income, and cost behavior. Table 2 summarizes the pooled distribution of key observable characteristics in the bank sample. The median bank holds 693 billion dollars in loans and 1.16 billion dollars in net worth, while the right tail is pronounced: banks at the 95th percentile hold more than 9.5 trillion dollars in loans and over 14 billion dollars in net worth. On the pricing side, the median loan rate is 7.1 percent, with an interquartile range from 5.7 to 8.6 percent, whereas average capital costs are lower and more dispersed, reflecting meaningful cross-sectional variation in funding conditions. Credit risk is modest for the typical bank, the median default rate is 0.12 percent, but rises sharply in the upper tail, reaching 1.54 percent at the 95th percentile.

Call Reports also lack direct observations of loan rates, $R_{i,t}$. Instead, the theoretical framework must provide guidance. Following from definitions in our model of the economy, the loan rate can be written as $R^* = (1-\lambda)R + \lambda \cdot 0$, so then, $R = \frac{R^*}{1-\lambda}$. Therefore, one might infer loan rates from the realized returns and charge-offs, which are observable in Call Reports.[15]

[14] Charge-offs are measured net of recoverable assets, thus reflect the net losses to the bank due to the default of a given loan. Therefore, where the analytical model may be unrealistic in ignoring the possibility of recoverable collateral or liens, the empirical exercises allow for this realistic possibility.

[15] Recall that λ is the portfolio average loan-level haircut, net recoveries, and so incorporates the average level of recoverable collateral. This implies that in an empirical setting one can assume a loan yields $R = 0$ if $\lambda = 1$, because the actual collateral recovery rate is incorporated into the average non-default yield via λ.

Figure 2: Banking industry size and loan performance, sample vs national accounts

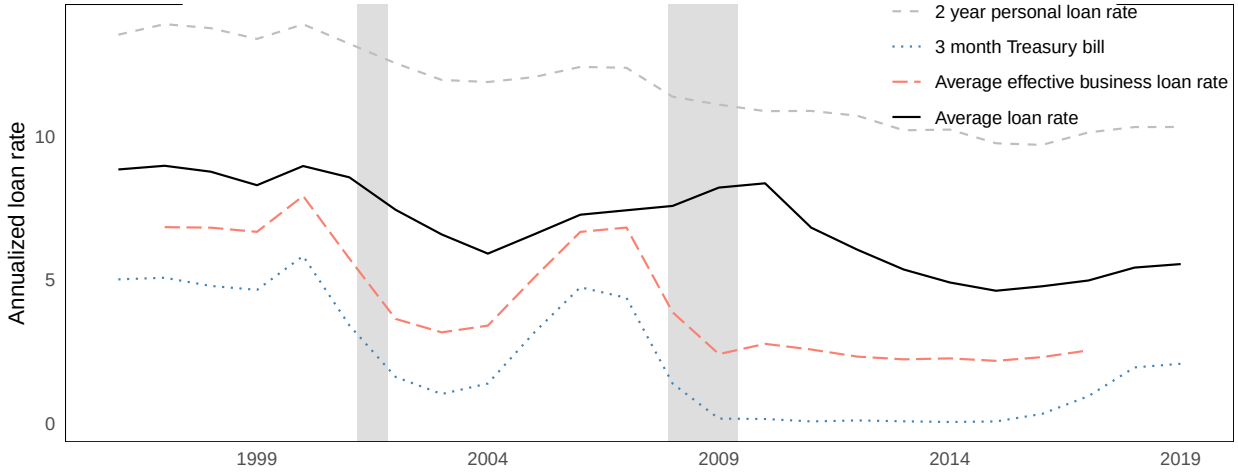


Notes: These plots compare the charge-off ratio, total loans, total deposits, and banking sector net worth in the sample to those reported by National Accounts. The sample covers the set of U.S. commercial banks that report FFIEC Call Reports and operate physical branch locations. The National Account values are from the H.8 Assets and Liabilities of Commercial Banks in the United States, reported by the Federal Reserve Board. Charge-offs are net of recoveries. Data are annual from 1995 through 2019.

Figure 3 shows the industry-wide (loan-weighted) average loan rate, and compares it to the risk-free rate proxied by three month Treasury Bill yield, the Federal Reserve reported two year personal loan rate, and weighted-average effective corporate loan rate. Notably, loan rates have been in secular decline since the beginning of the sample. All four loan rates have been in secular decline and largely co-move through the business cycle.

A bank's regional market share is calculated by the bank's loan share of all loans in a U.S. state. My algorithm for calculating a bank's regional market share is as follows: For banks that only have branch locations in one state, then the bank's entire loan portfolio is assumed to be made to borrowers in that state. For banks with branch locations across multiple states, the bank's loan portfolio is allocated across states based on the bank's deposit holdings by state, following Nguyen (2019). That is, if a bank has raised half of its deposits in Virginia and half in Maryland, then half of its loan portfolio is counted as its Virginia loan portfolio and the other half as its Maryland loan portfolio. These regional loan portfolios are then summed across all banks operating in a given region to determine the total region loan market size. Dividing the bank's regional loan portfolio by the total regional loan market size gives the bank's regional loan market. Bank branch location and deposit data are collected from the FDIC's Summary of Deposits (SOD).

Figure 3: Bank loan rates

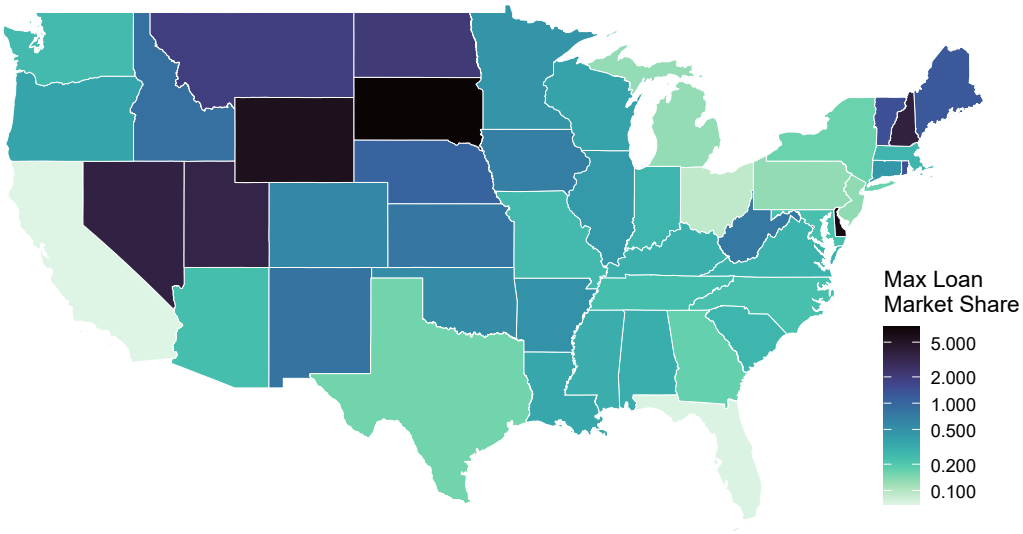


Notes: This plot compares the sample implied loan-weighted loan rate to published risk free and consumer loan rates. The bank-level sample loan rate is calculated as the reported quarterly loan income divided by total loans, scaled by the inverse one minus net charge-off ratio. The average loan rate is then the loan-weighted average of bank-level loan rates over the cross-section of banks in the given quarter. The risk-free rate is proxied by the US 3-month T-bill. The consumer loan rate is the Finance Rate on Personal Loans at Commercial Banks, 24 Month Loan, reported by the Federal Reserve Board's G.19 Consumer Credit. The average business loan rate is weighted-average effective loan rate for all commercial and industry Loans reported in the Federal Reserve Board's E.2 Survey of Terms of Business Lending. Gray shaded regions mark NBER dated recessions. Data are annual from 1995 through 2019.

The key assumption to calculating banks' regional market shares is that banks lend where they raise deposits. Banks are statutorily required by the Community Reinvestment Act to lend where they raise deposits. However, Oberfield et al. (2024) also shows that banks sort into localities with excess deposits to lending ratios so they can use the additional deposits to fund loans in higher margin markets. I therefore test the validity of the assumption in Appendix B and find a high spatial correlation between the average bank's small business, mortgage lending and deposits, empirically justifying the assumption.

The estimated banking sector concentration is muted at the state-level. Figure 19 shows that the greatest state-level market shares reach approximately five percent of local loans. This fact is empirical justification for assuming that regional loan markets are "large" and in turn justifies the recursive equilibrium conditions and the use of an Oblivious Equilibrium, à la Weintraub et al. (2008), when solving the quantitative model.

Figure 4: Maximum bank market share by state

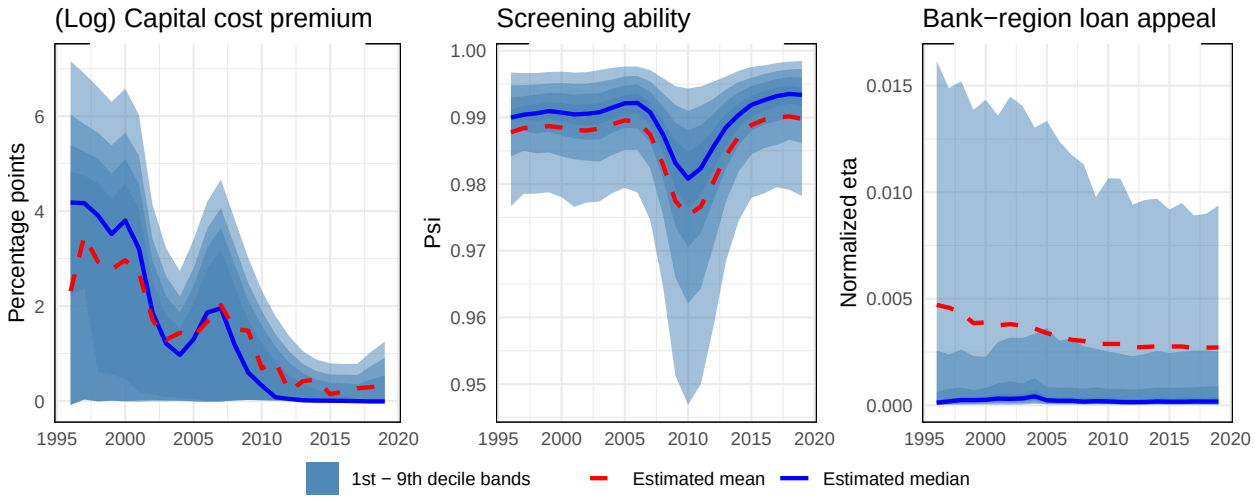


Notes: This plot shows the maximum loan market share of any bank in a given U.S. state. Market share is expressed in percent of state-level loan market. Data are annual from 1995 through 2019.

The remaining objects of interest, regional firm log-odds of survival and state-level wages are both readily observable. State- and county-level establishment exit rates are reported by the U.S. Census Bureau's Business Dynamics Statistics, while region-level wages are proxied for by the real median household income reported by the same agency.

The ultimate sample covers the nearly the universe of U.S. commercial banks with FDIC-insured deposits and at least one physical branch location, and all 50 U.S. states. Data are annual from 1994 through 2020 and cover 11,682 unique banks in total over the unbalanced panel. Despite requiring banks have at least one FDIC insured physical branch, the sample is highly representative of banking activity in the U.S., and covers 89 percent of deposits, 87 percent of net worth, and 102.2 percent of loans reported by national accounts (as shown by Figure 2). Banks are defined at the bank holding company (BHC) level when applicable, or to the standalone bank level for institutions without a holding company, following standard practices in the micro-banking literature (see for example Corbae and D'Erasmo 2021). I start the sample in 1995 for two reasons. First, that is when SOD data becomes reliably available. Second, that is when the Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994 started to go into effect, ending restrictions on interstate banking and beginning the regulatory regime most aligned with my model of the banking sector. I end the data in 2019 to remove the effects of the COVID-19 pandemic and the large policy interventions at

Figure 5: Distribution of estimated bank fundamentals



Notes: This figure shows the evolution of estimated bank-level fundamentals. The blue bands mark the first through ninth deciles of each distribution. Data are annual from 1996 to 2020.

that time. See Appendix B for additional descriptions of the data.

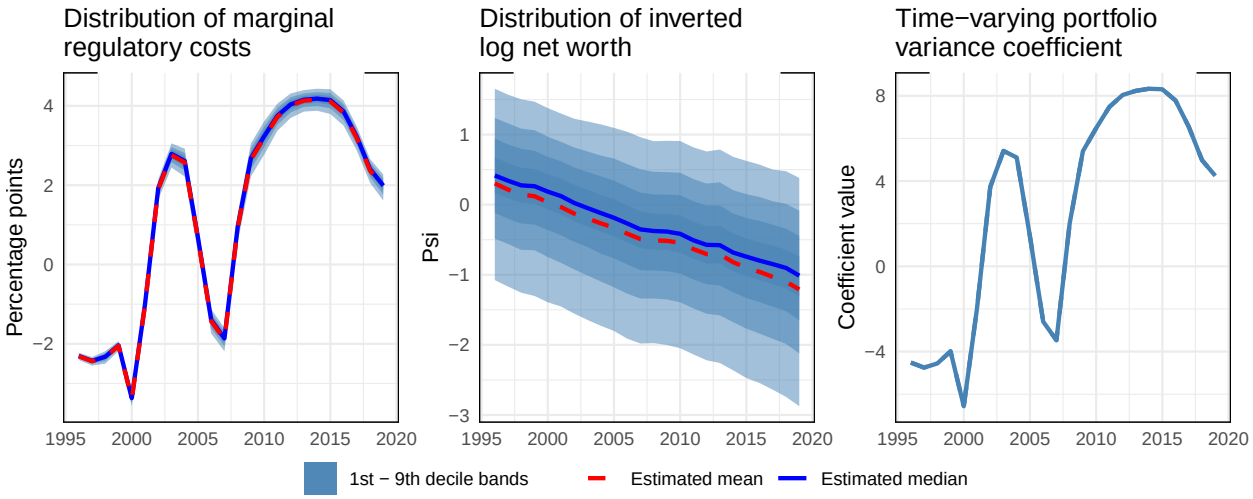
4.4 Describing fundamentals

I will lastly describe the estimated bank fundamentals, regulatory costs, and loan market frictions. The estimated regional fundamentals are described in Appendix D.

Fundamentals. The bank fundamentals, capital cost premium (marginal cost), screening ability (lending productivity), and loan appeal (firm-specific demand), all vary across bank and through time. Figure 5 presents the three banks fundamentals. Capital costs have been falling on average over time. The left panel shows that the average level of capital cost premiums have been in secular decline since 1995. Falling capital costs can be attributed to several potential sources, including growing market power in deposit markets (Drechsler et al. 2017) or spatial sorting of branches within regions to harvest cheap deposits (Oberfield et al. 2024). Moreover, it is a well documented fact that as banks grow their cost basis falls (see Berger and Mester 1997), so the falling capital cost may also reflect the secular growth in the modal bank's net worth.

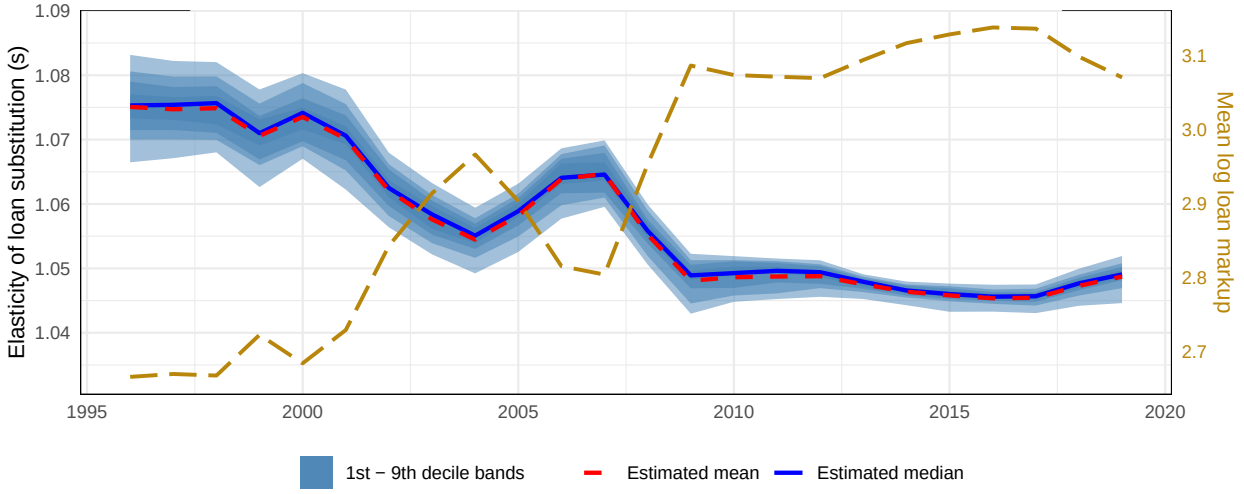
The average screening ability, or lending productivity, is stable over time. This is in line with Philippon (2015, 2022) finding that the average monitoring cost of the US financial sector has been relatively stable since the early 1900s. However, there is a noticeable, coordinated decline in screening ability during the global financial crisis. That is, the financial crisis coincided with a

Figure 6: Banks regulatory costs



Notes: This figure shows the evolution of the estimated regulatory cost and its underlying factors. The blue bands mark the first through ninth deciles of each distribution. The dashed red line marks the equally weighted mean while blue marks the median. Data are annual from 1996 through 2019.

Figure 7: Regional loan market frictions



Notes: This figure shows the evolution of the estimated regional loan market frictions and the implied loan rate markup. The left axis marks the elasticity of loan substitution, the direct proxy for loan market frictions. The blue bands mark the first through ninth deciles of each distribution. The dashed red line marks the equally weighted mean and the blue line marks the median. The right axis marks the mean log loan rate markup from frictions, $\log(\frac{\sigma_{s,t}}{\sigma_{s,t}-1})$, which is represented by the dark gold dashed line. Data are annual from 1996 through 2019.

large and wide-spread decline in banks ability to assess risk in their loan portfolios, an observation echoed by regulatory narrative accounts of the crisis.[16]

Meanwhile, the dispersion of banks screening ability is not stable through time and the evolution is not symmetrical. There is a right tail of high performing banks that throughout the sample history have close to a perfect screening ability. However, the left tail, and in particular the bottom decile of banks have a much lower, and variable screening ability through time. For example, while the top decile of banks saw screening ability fall by less than 0.1 percent during the GFC, the bottom decile fell from 0.975 to 0.945 or approximately 3 percent. The dynamics of the right and left tail during the crisis highlights the asymmetry in banks fundamentals that would otherwise be lost in aggregate, along with its potential implications for asymmetric credit allocations in downturns.

Both the mean and dispersion of bank loan appeal have declined over time, while the median has remained largely stable. This pattern indicates convergence driven primarily by a relative erosion in the advantage of the most attractive banks, rather than by broad-based changes across the distribution. One plausible explanation is the rise of internet and mobile banking, which has reduced the value of location-based amenities such as dense branch networks or ATM access. Consistent with this interpretation, the decline in appeal dispersion coincides with a secular contraction in the number of bank branches in the United States.

Regulatory costs. The regulatory cost is a significant, counter-cyclical, driver of loan premia. Figure 5 shows the net regulatory cost is negative, that is, acting as a subsidy, as banks seek to acquire risk during the late 1990s and the mid-2000s. These periods also coincide with commonly labeled Dot-com and mortgage bubbles. In comparison, the regulatory cost increases significantly during the 2001 recession (precipitated by the Dot-com bust) and the GFC (precipitated by the mortgage bubble bursting). The regulatory cost remains elevated through the 2010s and does not show signs of recovery until 2015. Dispersion in regulatory costs is narrow throughout the sample, though slightly increasing at peaks and troughs.

The counter-cyclical characterization of the regulatory cost points to the interpretation that the cost is capturing, on net, the banks demanded compensation for holding risky assets rather than a stable regulatory cost of doing business. Figure 6 shows the decomposition of the banks regulatory costs into the key components, log (inverse) net worth and the regulatory cost coefficient, which can be

[16]Faced with uncertainty about the value of specific instruments and mindful of the higher volatility of assets more generally, lenders demanded substantial cushions... on the assets they were willing to finance. (Senior Supervisors Group, 2009; Risk Management Lessons from the Global Banking Crisis of 2008)

interpreted as a time-varying risk aversion parameter. Because the distribution of inverse net worth has been in secular decline since 1995, it is clear that the counter-cyclical behavior of the regulatory cost is being driven by the estimated proxy for time-varying risk aversion.

Regional frictions. The regional loan market frictions have been increasing over time as the elasticity of loan substitution has been falling. Figure 7 shows that the elasticity of loan substitution has fallen from approximately 1.075 to 1.04, or three percent, on average across regions since 1995. As a result, the exogenous markup on loans has been increasing over time, rising from approximately 2.5 percentage points to 3.07 percentage points in 2019.

The increase in loan market frictions coincides with the decline in the number of both banks and branch locations in the United States (see figure 13 and Figure 20 respectively). This is consistent with the interpretation that the market frictions, which are theoretically frictions that make loans less substitutable, are reduced-form search and matching frictions that generate sticky borrower-lender relationships. It is documented that borrowers still need to go to branches (see for example Sakong and Zentefis 2023), so the decline in branch locations is making it more expensive to substitute loans from one bank to another since it is likely that there is a limited number of other banks with local branches close to borrowers (if any).

5 Determinants of Bank Firm Dynamics

With empirical measures for the fundamentals and market frictions that theoretically drive bank behavior in hand, I can answer the first question: what forces shape the firm dynamics of banks?

I rely on two exercises, an empirical accounting decomposition based on observable data and a structural variance decomposition based on quantitative simulations. By appealing directly to the data, the empirical decomposition will characterize the full realized distribution of bank outcomes. For example, the empirical decomposition will accurately represent the full size distribution of banks in the U.S., which has been heavily influenced by regulations determining the maximum size and geographic expansion of banks as well as ad hoc government interventions during crises. These historical regulatory and institutional features of the data are absent in the structural model. Therefore, relying solely on simulated data would not be able to accurately characterize the realized historical effects of banks' pre-existing size (measured by net worth) on market share through regulatory costs or loan rates through oligopolistic profits. However, the accounting decomposition is an analysis of variance and does not necessarily measure the causal relationships. The structural variance decomposition, on the other hand, does measure causal relationships free of influences

from historical institutional settings and regulations.

5.1 Empirical accounting decomposition

I first decompose the level, within- and across-bank variation of the observable bank-level loan spreads and market shares. For the level decomposition, I compute the gross contribution of each pricing factor i to the level of the outcome y for each bank, or formally:

$$\mu_{y|i} = \frac{|b_i x_i|}{|y|}, \qquad |y| = \sum_j |b_j x_j|,$$

where y denotes the outcome of interest (the loan spread pr regional market share) and

$$x = \left\{ g_{i,t}, \chi^{\psi}_{i,t}, \chi^{q}_{s,t}, \eta_{i,s,t}, \log\left(\frac{\sigma_{s,t}}{\sigma_{s,t}-1}\right), N_{i,t} \right\}$$

collects the model-implied pricing components which are scaled by the appropriate coefficient, b_i, determined by the structural equations characterizing the objects of interest. The object $\mu_{y|i}$ measures the gross share of factor i in the level of y, abstracting from offsetting effects across components.[17]

To decompose dispersion, I apply the total variance formula following Hottman et al. (2016) and Eslava et al. (2024). The share of variance in y attributable to factor i is given by

$$\sigma_{y|i} = \frac{\mathrm{Var}(x_i) + \sum_{j \neq i} \mathrm{Cov}(x_i, x_j)}{\mathrm{Var}(y)},$$

where x is the same vector of fundamentals and frictions used in the level decomposition. Across-bank variance decompositions are computed within each period across banks, while within-bank variance decompositions are computed for each bank over time.

Loan spreads. Table 3 shows the level and variance decompositions of loan spreads.[18] The level

[17] Two notes on the computational details of the decomposition. First, regulatory costs are not included as a separate component because they are endogenous functions of bank net worth, borrower quality, and screening ability through the portfolio risk term (see Appendix A.4 for details. I instead first decompose regulatory costs into these primitives and incorporate their contributions indirectly through the corresponding elements of x.) Second, the $\chi^{\psi}_{i,t}$ and $\chi^{q}_{s,t}$ terms enter the loan spread in a single term multiplying one another, so I calculate the contribution of the term and then allocate the contribution to either individual factor based on the estimated ration of $\chi^{\psi}_{i,t}$ to $\chi^{q}_{s,t}$.

[18] The spread decomposition includes an oligopolistic profits term that captures the contribution of market power, as reflected in the market share component of the net interest margin. This term can be removed by reallocating its contribution using the market share decomposition reported in Table 4. I retain the oligopolistic profits term because it is informative about the quantitative importance of market power.

Table 3: Decomposition of loan spreads into fundamentals and market frictions

	Oligopolistic profits	Regional frictions	Compensation for risk		
			Net worth	Bank ability	Borrower quality
Loan-weighted avg. loan spread	6.79	51.4	6.24	34.0	1.53
Equally-weighted avg. avg. loan spread	1.55	59.8	4.02	33.5	1.16
Loan-weighted cross-bank variance	0.33	7.33	30.9	49.0	9.66
Equally-weighted cross-bank variance	0.078	0.97	32.0	52.5	10.3
Loan-weighted within-bank variance	0.006	0.72	1.30	94.0	3.99
Equally-weighted within-bank variance	0.005	0.72	1.30	94.0	3.99

Notes: Each row decomposes the variance (or level) of banks' loan spreads into percent contributions by fundamentals and frictions. The difference between the sum of each row and 100 is the residual variation or level.

of spreads is primarily determined by regional frictions that generate sticky lender relationships. Regional market frictions account for approximately half of observable loan-weighted net interest margins, which increases to approximately 60 percent when equally weighting firms. The difference between the loan- and equally weighted decomposition suggests that small banks are even more sensitive to regional frictions in their pricing behavior than large banks. This is perhaps not surprising, because larger banks also demand an oligopolistic rent, which accounts for 6.79 percent of the spread when loan weighting compared to only 1.55 percent when equally weighting. The other significant factor accounting for observable spreads is the banks screening ability which accounts for approximately one third of the weighted and unweighted spreads. This indicates that screening ability accounts for a significant portion of banks pricing behavior across all bank sizes. The borrower quality of the credit market that a bank operates in, along with the size (net worth), both account for less than 10 percent of spreads.

Table 4: Decomposition of regional market shares into fundamentals

	Loan Appeal	Bank Ability	Capital Cost	Net Worth
Loan-weighted avg. market share	0.47	81.63	1.53	16.37
Equally-weighted avg. market share	1.65	34.07	1.15	63.14
Loan-weighted cross-bank variance	0.03	92.48	0.18	7.30
Equally-weighted cross-bank variance	0.02	56.45	0.47	43.05
Loan-weighted within-bank variance	0.66	74.75	0.71	23.89
Equally-weighted within-bank variance	0.01	80.68	0.26	19.05

Notes: Each row decomposes the variance (or level) of bank-region market shares into percent contributions from demand, ability, cost, and net worth components. The difference between the sum of each row and 100 is the residual variation or level.

In comparison the across-bank variance decomposition characterizes the variation in loan rates offered by different banks in the same time period. Approximately half of the variance across-banks can be accounted for by variation in the banks screening ability. In comparison, borrower quality only accounts for approximately 10 percent of the variation. That is, when we observe variation in the loan rates being offered by banks across the country, the majority of those differences are accounted for by differences in the banks ability to screen risk rather than because those banks are serving borrowers with different risk profiles.

Lastly, the within-bank variance decomposition characterizes the variation in loan rates given by the same bank over its lifetime. The within variation in spreads is largely dominated by the banks screening ability, accounting for 94 percent of the fluctuations in spreads over a banks life. That is, the change in loan rate spreads a bank charges throughout its life is primarily changing due to its ability to screen risk, not changes in other fundamentals, frictions, or the risk-profile of its borrowers.

Market share. Table 4 shows the corresponding level and variance decompositions for a bank's

regional market share. The average level of a banks market share is sensitive to whether the sample is loan- or equally-weighted. For the former sample, the average market share is largely accounted for by a bank's screening ability, while for the latter it is primarily accounted for by the banks size. This means that for large banks (who have an outsized role in the loan-weighted decomposition) fluctuations in ability have a larger marginal effect on their loan rates, thus competitiveness within a region. On the other hand, for small banks, changes in their relatively much smaller net worth have a much larger effect on the demanded excess loan premium, making their competitiveness more sensitive to fluctuations in net worth. Strikingly both cost and loan appeal (demand) do not account for significant portions of the average level of banks market share. Moreover, both across-bank and within-bank variation in regional market shares are additionally dominated by differences in screening ability. However, the variance decomposition is less sensitive to loan-weighting sample.

5.2 Quantitative structural decomposition

To isolate the causal relationship between bank-level fundamentals, market frictions and bank dynamics, I next turn to a simulation-based structural exercise where I can directly observe the changes in bank outcomes in response to structural shocks. To do so, I embed the empirically estimated processes from Section 4 in a quantitative calibration of the regional NK DSGE outlined in Section 2 and estimate the forecast error variance decomposition of bank-level firm dynamics.

5.2.1 Quantitative methodology

I start by calibrating the structural model outlined in Section 2. The bank- and region-level fundamentals' laws of motion are set using the empirical estimates from Section 4, while deep parameters are set to standard values from the literature. I then simulate the economy, starting with 50 ex ante identical regions and 500 banks headquartered uniformly across regions. See Appendix E for a full discussion of the model calibration, solution methods, simulation and setting details.

To then characterize the dynamic effects of structural disturbances in this nonlinear environment, I compute Generalized Impulse Response Functions (GIRFs) following the simulation-based approach of Koop et al. (1996). For each structural shock j, the GIRF at horizon h is defined as the difference in conditional expectations of an endogenous variable y between two economies that share the same information set Ω_t and the same future sequence of stochastic innovations, except for a one standard deviation shock to fundamental or friction j at time t:

$$\text{GIRF}_h(\varepsilon_j) = \mathbb{E}\left[y_{t+h} \mid \varepsilon_j, \Omega_t\right] - \mathbb{E}\left[y_{t+h} \mid \Omega_t\right]. \tag{19}$$

This procedure isolates the causal effect of the exogenous fluctuation in fundamental or friction while preserving the full nonlinear state dependence of the system.

To then assess the relative importance of each structural shock for the volatility of endogenous variables over time, I construct a Forecast Error Variance Decomposition (FEVD) based on the simulated GIRFs, in the spirit of Pesaran and Shin (1998). For variable i and horizon t, the contribution of shock j is given by

$$\text{FEVD}_{i,t}^{(j)} = \frac{\left(\text{GIRF}_{i,t}^{(j)}\right)^2}{\sum_k \left(\text{GIRF}_{i,t}^{(k)}\right)^2}, \tag{20}$$

where the denominator aggregates the squared responses to all structural shocks. The FEVD is the decomposition of innovations in endogenous variables (e.g. loan rates and regional market shares) into the relative contributions of exogenous fluctuations in fundamentals and frictions, allowing me to quantify how shocks account for business cycle fluctuations at different horizons.

After simulating the economy I estimate the effect of exogenous idiosyncratic fluctuations in bank-level fundamentals and regional loan market frictions. This captures the forces that drive bank-level outcomes *relative* to other banks. That is, I ask: what fundamentals and frictions differentiate banks and their competitive outcomes from one another?

5.2.2 Accounting for bank dynamics

A bank's relative net worth growth is primarily driven by shocks to bank-specific screening ability. Figure 8 shows the variance decomposition of relative banks' relative outcomes in contributions from fundamentals and frictions over the medium run (the average FEVD over five years after realizations of the structural shocks). Approximately 79.1 percent of the medium run variation in a bank's relative net worth growth is driven by bank-specific productivity shocks, leaving approximately 13 percent of the medium run variation to the bank's appeal, capital cost premium, and regional market frictions. Meanwhile the remaining 7 percent of the bank's lending and market share variation is due to non-bank fluctuations, including regional firm exit rates (borrower quality) and regional non-financial productivity.

A bank's relative lending (thus their relative loan market share) is also largely determined by fluctuations in the banks specific screening ability (approximately 42.1 percent), although with slightly larger influence from both other bank-specific characteristics and regional fluctuations. Since net worth grows over time based on the bank's accumulated profits, the difference between

factors driving the bank's relative lending volume and its net worth is the differential contribution of factors to the quantity and price of loans.

Similar to a bank's relative net worth growth, variation in a bank's relative loan spread is largely dominated by fluctuations in its ability to screen risk. Figure 8 shows that 75.6 percent of the medium run variation in relative loan spreads is attributable to variation in screening ability. This is larger than the approximately 50 percent of across-bank loan rate variation accounted for screening ability in the empirical accounting decomposition, where screening ability was too the dominant force behind across-bank variation in observable loan spreads.

In comparison, a bank's loan rates are not dominated by any one factor. Bank-specific characteristics account for approximately 50 percent of loan rate fluctuations, with capital costs accounting for slightly more than either the bank's appeal or screening ability. External factors, including the borrower quality, regional productivity, and loan market frictions then make up the other 50 percent of loan rate fluctuations. The increased importance of external factors on the loan rates reflects their ability to influence the macroeconomy and in turn the policy rate.

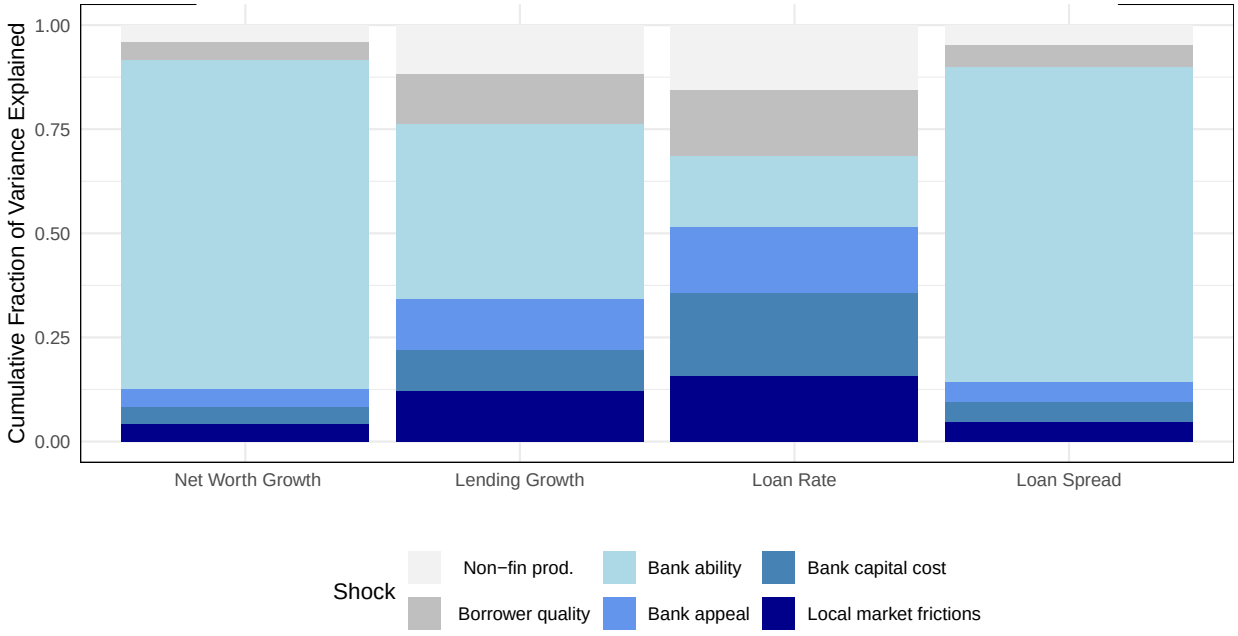
5.3 Discussion and policy implications

Screening ability is the primary determinant of bank growth and market share, while loan market frictions account for more than half of aggregate loan spreads. These findings have important implications for both the modeling of banks and the effects of banking competition policy.

Modeling banks. The empirical decomposition of net interest margins highlights the prominent role of regional credit market frictions and banks' ability to assess risk, while casting a shadow on the importance of oligopolistic rents, size, and borrower quality in determining these outcomes. Moreover, the comparative importance of these features in accounting for observable spreads can be used to discriminate among models of banks for those with the most empirically relevant features and mechanisms. For example, the role of credit market frictions that may generate sticky borrower-lender relationships is in line with empirical studies that both document this feature of U.S. credit markets, as well as measure the asymmetric effects of credit supply shocks on borrowers due to these frictions, such as Chodorow-Reich (2014), Siemer (2019), or Huber (2018). However, there are few dynamic macro-banking models with explicit sticky borrower-lender relationships.[19] Moreover, the structural decomposition exercises shows that only ten percent of

[19] Herreno (2023) presents a spatial model with sticky lender-borrower relationships, which is perhaps the closest exception, but this work largely foregoes analyzing the dynamic effects of this friction.

Figure 8: Structural variance decomposition of bank dynamics



Note: This figure presents the structural variance decomposition of bank dynamics into contributions from shocks to fundamentals and market frictions. The forecast error variance decompositions are based on GIRFs that are the average response of 100 treated economies less the average response of 100 untreated economies. The decomposition is the average contribution by shock to the endogenous outcome over five years after the realization of the structural shocks.

relative loan spreads are driven by non-financial shocks. This evidence supports neither macro-banking paradigms that treat banks as pass-through entities, such as Bernanke et al. (1999), Gertler and Kiyotaki (2010), or Christiano et al. (2014), nor demand driven narratives of the GFC, such as Mian and Sufi (2018). In comparison, the emphasis on banks' screening ability in their loan pricing decisions and the estimated large aggregate decline in screening ability during the GFC instead centers the focus on banks as the source of both business cycle fluctuations as well as the crisis.

Banks' screening ability is a key force explaining both across- and within-bank variation in spreads and market share. That is, even if bank's screening ability is on average very stable over time, as presented in works such as Basu et al. (2011) or Philippon (2015, 2022), it is key source of heterogeneity within the banking sector. This is in contrast to the limited role of loan appeal in empirically accounting either loan spreads or market share. Banks screening ability (i.e. loan productivity) driving firm heterogeneity rather than loan appeal (i.e. bank-specific demand) is a sharp break from the dynamics that characterize the non-financial firm dynamics, which empirically find a dominant role for product appeal in shaping firm-level growth (see for example, Foster et al.

(2008), Hottman et al. (2016) or Eslava et al. (2024))

Competition policy. These results have a stark implication for banking sector competition policy: limiting the growth or regional market entry of large banks is also stopping the growth or expansion of banks with the best screening ability. This may lead to a misallocation of credit and counterfactually greater levels of loan defaults, as market share naturally rises for banks with greater screening ability, those that can keep credit from those that will default on their loans, so when growth of these high ability banks is stymied, the credit demand and market share goes to banks who cannot screen as effectively, and counterfactually more creditors will default than if the high ability banks had been allowed to take market share from those with weak fundamentals. However, there are of course other considerations to micro-prudential policy, such as allowing too much concentration in turn decreases financial stability more broadly. I do not take up a normative analysis of size-based banking policies in this work, but think it is an important question for future research.

6 Bank Dynamics in National and Regional Business Cycles

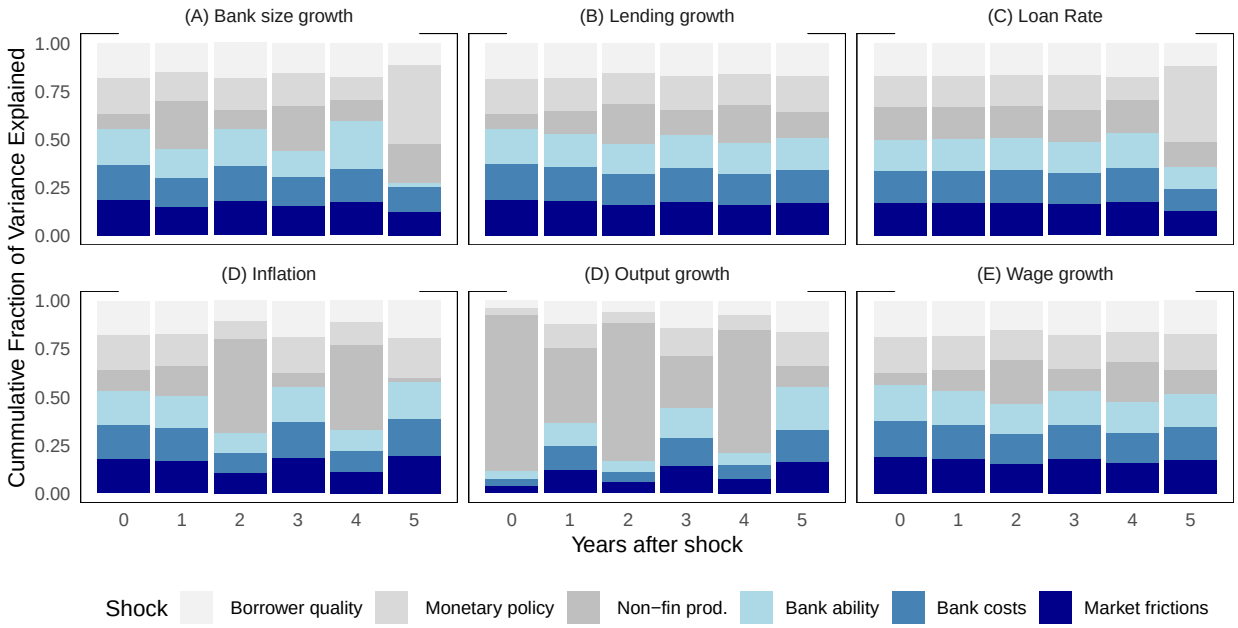
Having characterized the forces that determine bank dynamics, I can address my second question: how do these dynamics shape aggregate and regional business cycles? To answer this question, I continue forward with the calibrated NK DSGE model to isolate the causal relationship between bank dynamics and non-financial economic outcomes. The remainder of this section will consider bank dynamics as a direct source of aggregate and regional business cycle fluctuations, while the following section will consider bank dynamics as an indirect source of regional heterogeneity

Exogenous fluctuations in banking sector average fundamentals are an economically significant source of aggregate business cycle fluctuations in activity and prices. Figure 9 shows the variance decomposition of aggregate outcomes in the economy.[20] Aggregate shocks to the banking sector and to loan market frictions make up 30.7 percent of medium run fluctuations in output.

This is in line with other structural estimations of how much financial shocks drive business cycle fluctuations in output. For example Ajello (2016) estimates exogenous shocks to the intermediation spread (i.e. loan spread) account for approximately 25 percent of output fluctuations while Jermann and Quadrini (2012) estimate financial shocks account for up to 46 percent of output fluctuations. This estimate is further in line with empirical estimates of bank credit supply shocks driving output, for example Broadbent et al. (2025) estimates lending standard shocks account for approximately

[20]Because of the within-region normalization of bank appeal, an aggregate appeal shock has no aggregate effect.

Figure 9: Structural variance decomposition of aggregate business cycle fluctuations



Note: This figure presents the structural variance decomposition of aggregate business cycle fluctuations into contributions from shocks to fundamentals, market frictions, and policy. The forecast error variance decompositions are based on GIRFs that are the average response of 100 treated economies less the average response of 100 untreated economies.

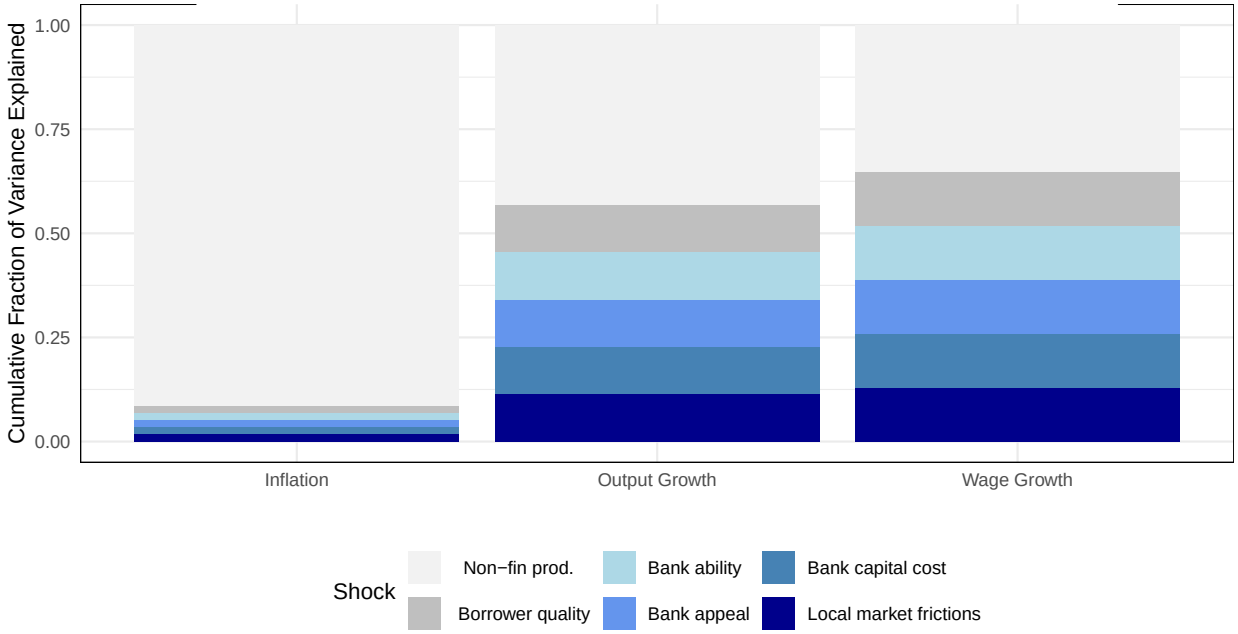
26 percent of real GDP fluctuations over the medium run.

However, the effect is not immediate, but rather builds over time. The effects of banking sector shocks are strongest five years after impact, and account for more than half of the fluctuations in output at this horizon. This persistent effect of financial shocks is in line with empirical findings that financial crises are longer lived than non-financial recessions, see for example Schularick and Taylor (2012), and Jordà et al. (2013).

In comparison, financial shocks' effect on prices, both inflation and wages, is immediate and persistent. The banking sector shocks to bank ability, costs, and loan market frictions account for approximately half of the fluctuations in both aggregate inflation and wage growth on impact and remains at this level through the medium run.

Turning to the banking sector and financial outcomes, there is a disconnect between the factors that drive banking sector net worth (size) and financial activity or prices. The top row of Figure 9 shows the variance decomposition of net worth growth, lending growth, and loan rates. Aggregate

Figure 10: Structural variance decomposition of region dynamics



Note: This figure presents the structural variance decomposition of regional business cycles fluctuations into contributions from shocks to fundamentals and market frictions. The forecast error variance decompositions are based on GIRFs that are the average response of 100 treated economies less the average response of 100 untreated economies. The decomposition is the average contribution by shock to the endogenous outcome over five years after the realization of the shocks.

financial shocks are immediately and strongly impact banking sector size, lending, and rates. However, there is a slight asymmetry across these effects. The effect of financial shocks on bank size growth decays over time, accounting for less than 20 percent of growth five years after the shocks are realized. That is, the growth of the banking sector is driven by bank fundamentals and financial frictions in the short run, but by broader economic fundamentals and policy in the medium run. In comparison, the effect of financial shocks on loan growth and loan rates is persistent. That is, lending volume and loan rates are driven by both financial and non-financial conditions.

I next turn to the regional effects of financial shocks. While the previous exercise captured the effect of financial shocks on *aggregate* economic outcomes, this exercise captures the effect of financial shocks on the *distribution* of economic outcomes.

Regionally concentrated financial shocks —shocks to a region's loan market frictions or shocks banks' ability, appeal, or costs that effect all banks operating in a given region— are important drivers of output and wage growth, but not inflation in consumption prices. Figure 10 shows that

45.2 percent of medium run variation in output growth is driven by shocks to bank ability, cost, appeal, and loan market frictions. Likewise, more than half of the medium run regional variation observed in wage growth is attributed to local financial shock. Together these results highlight that local financial conditions account for approximately half of the regional inequality observed in both GDP growth and wage growth.

These findings build on empirical studies that show bank credit supply shocks drive differences in regional economic activity by putting them in context relative to real shocks that may influence the same outcomes. For example, Davis and Haltiwanger (2024) find that differences in banks' willingness to supply credit impacts both the extensive margin, firm entry and entrepreneurial activity, as well as the intensive margin of local dynamism, namely young firm growth. This exercise adds structural evidence of the same qualitative result, regional differences in bank credit supply matter for regional activity, as well as contextualize that financial forces are approximately equal in driving differences in output growth as regional non-financial productivity shocks (i.e. real forces), and even more for medium run wage growth.

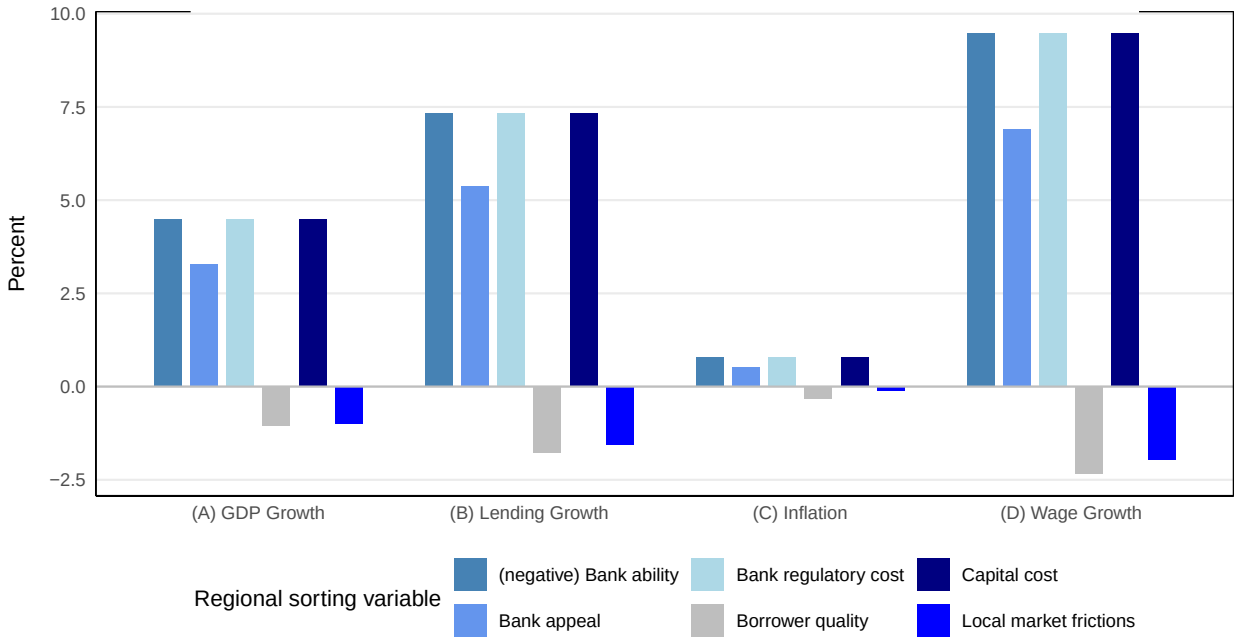
7 Bank Dynamics and Monetary Policy Transmission

While shocks originating in the banking sector may directly precipitate aggregate and regional business cycles, the state of banks fundamentals and market frictions are likewise an *indirect* force that shape economic outcomes through amplifying and dampening regional responses to business cycle and policy shocks. Banks indirectly effect short run fluctuations through two mechanism previously discussed, the credit exposure channel and the credit elasticity channel. I will next quantify the effect of either channel on the economy specifically through their effect on the regional transmission of monetary policy.

7.1 The credit elasticity effect

To quantify the credit elasticity channel I will continue with the quantitative calibration of the model used in Sections 5 and 6. In particular, Figure 11 shows the difference in regional responses to tightening monetary policy shock, based on high versus low local banking sector characteristics. I shock the economy with a 25 basis point increase in the federal funds rate and then take the difference in the immediate response of regions with the top decile of average local banking sector characteristics and those with the bottom decile of average local banking sector characteristics. For example, I find the average (GDP-weighted) response of regions with local banking sector average screening ability in the top decile of all local banking sector average screening ability and then

Figure 11: Heterogeneous effects of a monetary policy by regional banking sector characteristics



Notes: The figure plots the difference in responses to a 25 basis point monetary policy tightening between regions at the 90th and 10th percentiles of the relevant sorting variable. Sorting variables capture negative bank ability (higher values imply higher loan spreads), capital cost premia, bank appeal, regulatory costs, borrower quality, and local loan market frictions.

subtract the analogous response for regions with the lowest average screening ability.

Regions with weaker fundamentals have a stronger response to common monetary policy shocks. Figure 11 shows that regions with weaker bank ability, appeal, or costs have a larger response to monetary policy shocks across GDP growth, lending growth, inflation, and wage growth. However, the effect is neither symmetric across bank fundamental nor economic outcome. The difference in inflation is muted, measuring at less than one percent. Meanwhile, the difference in GDP growth is approximately 4.5 percent, lending growth is 7.5, and wage growth almost 10 percent when sorting on bank ability or capital costs. The difference in responses when sorting on bank appeal is much smaller than when sorting on either ability or costs. Although this is perhaps not surprising given the limited role that bank ability appears to play in driving loan rate or lending volumes.

In contrast with banking sector fundamentals, Figure 11 shows that greater regional loan market frictions lead to relatively attenuated responses to monetary policy shocks. Regions with greater frictions see wage growth respond almost 2.5 percent less than regions with lower frictions.

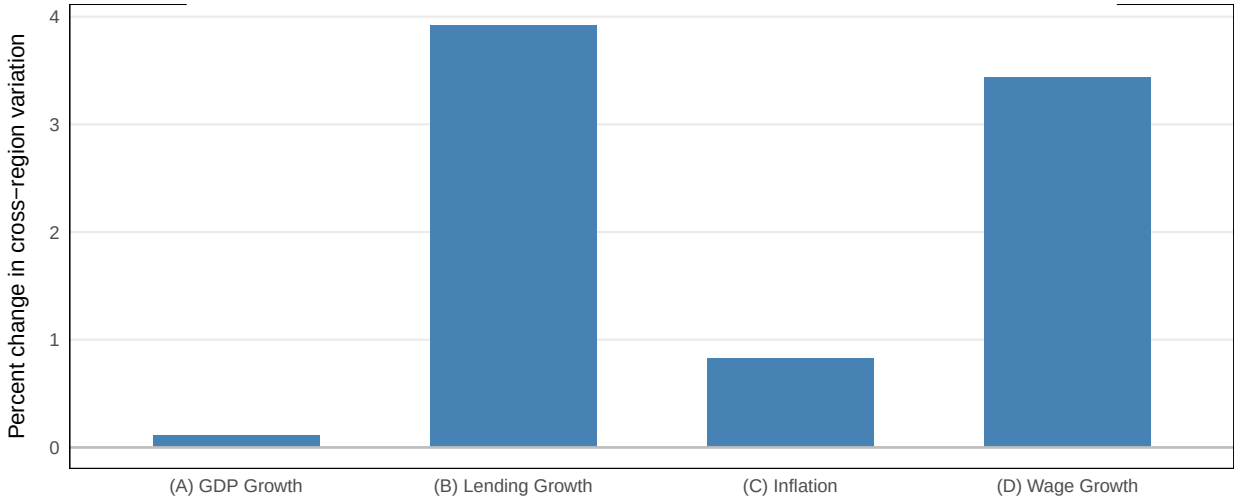
The reason that elevated frictions appear to dampen responses is because of the entry and exit dynamics and banking competition within regions with high frictions. That is, because of bank dynamics. The credit elasticity channel predicts that poor fundamentals or high frictions, either of which will increase the regional spread on loan rates holding all else equal, will amplify the regional response to shocks. In the case of weak fundamentals, recall from Section 3: Poor banking sector fundamentals increase loan rates, which puts the region lower on the DRS production curve, so the second derivative with respect to input prices is higher (ie greater supply elasticity to input prices), and a change in input prices from the monetary policy shock has a larger effect on regional output. However, in the case of regions with high loan market frictions: High frictions motivate competitive banks to enter region, so improving fundamentals and decreasing loan rates, increasing natural level of activity dampens elasticity of supply to input prices which leads to a smaller response to contractionary monetary policy. It is then a quantitative result that the entry of high performing banks and subsequent reduction in loan rates (thus dampening the credit elasticity channel) outweighs the increase in rates due to the elevated frictions.

7.2 The credit exposure effect

I next quantify the effect of the credit exposure channel on the regional transmission of monetary policy. In contrast to the credit elasticity channel, the credit exposure channel will effect the persistent difference in regional outcomes due to exogenous shocks, holding real conditions constant. Therefore I will quantify this effect by comparing the cross-sectional variation in regional outcomes after a monetary policy impact. Figure 12 shows the average percent increase in the cross-sectional variation of select regional outcomes from one to five years after a common monetary policy shock.

The common shock markedly increases dispersion in regional lending activity, with the cross-sectional variation increasing by 3.92 percent on average over the medium run. This may reflect that the credit exposure channel explicitly works through lending activity, and that is where the effect is strongest. However, the credit channel not only increases dispersion in financial activity, but also in composite input prices; dispersion in wages increases by 3.44 percent on average over the medium run. Here we see the effect dampened as it moves from financial activity to the labor markets, which is where loans are directly utilized in this model of the economy. The effect becomes attenuated as production moves further from lending activity though. The average increase in regional output dispersion is 11.6 basis points on average, while the increase in price dispersion (via inflation) increase by 82.6 basis points.

Figure 12: Persistent regional dispersion after a monetary policy shock



Notes: The figure plots the medium-run response of cross-regional dispersion to a 25 Bp. monetary policy shock. Each bar reports the average annual percent change in the cross-sectional variance of the indicated variable relative to its pre-shock level, averaged over horizons one through five years.

7.3 Policy implications

These quantitative exercises show that not only do the state of bank dynamics lead to an asymmetric transmission of a monetary policy shock, but the resulting dispersion is persistent through the medium run. The contemporaneous effect is directly due to the condition of bank-level fundamentals and how sensitive banks' pricing decisions are to external shocks because of those fundamentals, while the persistence effect is due to the asymmetric propagation of shocks through multi-regional banks' balance sheet conditions and resulting loan pricing behavior.

Therefore, as one observes heterogeneity in regional banking sector fundamentals and loan market frictions across the U.S., one will also observe the asymmetric regional transmission of national monetary policy. It also follows that if policymakers want to use monetary policy to combat regionally concentrated crises, such as the GFC or COVID-19 recession, then it is important for them to understand the distribution of banks' fundamentals and market frictions to understand the impact that policy will have on the region(s) in crises as well as the risk of overheating those not yet in recession. For example, if crisis regions are positively correlated with local banking sector fundamentals, then policy will have an attenuated effect on regions that need accommodating policy while buoyant regions will have an exaggerated response to the stimulative policy. Conversely, the opposite can be true and a smaller policy shock may be needed to achieve national employment and price stability. Moreover, either case can occur with the same aggregate banking sector net worth,

a common proxy for banking sector fundamentals in models of monetary policy (see for example in Gertler and Karadi (2011)). Therefore, banking sector aggregate conditions are not sufficient statistics for understanding the transmission and efficacy of monetary policy shocks in regionally concentrated crises.

8 Conclusion

This paper studies firm dynamics in the U.S. banking sector and how these dynamics shape credit allocation, business cycle fluctuations, and the regional transmission of monetary policy. More specifically, I ask what forces drive the firm dynamics of banks, and how do these dynamics shape aggregate and regional business cycles? To answer the first question: Screening ability emerges as the primary determinant of bank growth and market share, accounting for the majority of both cross-sectional and within-bank variation, while loan market frictions explain over half of aggregate loan spreads. To answer the second question: I show that shocks originating in the banking sector account for roughly one-third of medium-run output fluctuations and, even more so, generate substantial regional heterogeneity in wages and activity. Bank dynamics also amplify and (heterogeneously) propagate monetary policy through credit elasticity and banking network exposure channels, producing an economically meaningful asymmetric regional transmission of common policy shocks.

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A Theory

This appendix completes the description of the regional NK DSGE model, analytically characterizes non-financial outcomes in the model, and provides proofs for the financial outcomes discussed in Section 3.

A.1 Regional economies and the monetary authority

This section describes the regional economic agents (firms, retailers, and households) and the Central Bank. These agents are standard but presented here for completeness.

Firms. Firms produce a homogeneous consumption good to be sold in a perfectly competitive market by combining bank-specific credit products into a composite input (taken as a stand-in for working capital constraints on a variety of factor inputs, including labor, durable capital, and inventory) and expanding this input with a decreasing returns to scale technology. Firms are potentially subject to an exogenous exit shock each period. The following formalizes the individual and representative firm's problem.

There is a continuum of individual firms. Each firm applies to a local bank for a loan to fund its labor bill before starting production. However, each firm receives an exogenous exit shock with probability $1 - q_{s,t}$. If a firm is forced to exit, then it does not produce that period and defaults on its loan. If a firm is denied a loan by the bank then it cannot produce that period.[21]

The probability of receiving an exit shock is time-varying and region specific, following the law of motion:

$$q_{s,t} = \rho^q q_{s,t-1} + \epsilon^q_{s,t}$$

where ρ^q is the regional persistence of exit rates and $\epsilon^q_{s,t}$ is the stochastic exit shock, which may be correlated across islands.

The decisions and outcomes of the individual firms can be studied as a single regionally representative firm. The representative firm aggregates bank-specific loans into a composite credit product

[21] Because firms only match with and submit loan applications to a single bank each period, and there is no explicit persistence in borrower-lender relationships, there is no multiple-equilibria or "winners curse" present in this economy. These outcomes can arise in the presence of sticky borrower-lender relationships and asymmetric screening ability, as analyzed by works recently reviewed by Blickle et al. (2024).

$L_{s,t}$ using CES aggregator technology:

$$\bar{L}_{st} = \left(\int_{i \in \mathcal{B}_{s,t}} \eta_{ist}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}} d_s \right)^{\frac{\sigma_{s,t}}{\sigma_{s,t}-1}}, \qquad \prod_{\mathcal{B}_{s,t}} \eta_{i,s,t}^{\frac{1}{||\mathcal{B}_{s,t}||}} = 1,$$

where $\sigma_{s,t} > 1$ is the elasticity of substitution across bank loans, and $\eta_{ist} \geq 0$ is the firm's idiosyncratic preference weight for bank i. Note that the firms preference weights may change period-by-period as the set of banks operating in region s, $\mathcal{B}_{s,t}$, changes over time. The firm-specific preference weights over bank-specific loans represent the firm's borrowing requirements across different types of loans (for example commercial and industrial loans versus real estate), industrial specialization of the bank, or geographic locations of bank branches.

The firm uses the composite loan $\bar{L}_{s,t}$ to hire labor as a composite input to produce output with standard decreasing returns to scale technology and time-varying total factor productivity:

$$Y_{s,t} = A_{s,t} \cdot M_{s,t}^{\alpha}, \quad \text{with } \alpha \in (0,1),$$

$$a_{s,t} = \rho^a a_{s,t-1} + \epsilon_{s,t}^a \qquad a_{s,t} = \log A_{s,t}$$

where $\epsilon_{s,t}^a$ is the firm's productivity shock.

Combing the loan aggregation and output production technology, the firm's problem is to maximize profits by choosing labor inputs N_t, taking wages as given, while minimizing financing costs by choosing loans $\{L_{ist}\}_{i \in \mathcal{B}_{s,t}}$, taking interest rates $\{R_{i,s,t}\}_{i \in \mathcal{B}_{s,t}}$ as given:

$$\max_{M_{s,t},\ \{L_{is,t}\}_{i \in \mathcal{B}_{s,t}}} A_{s,t} M_{s,t}^{\alpha} - R_{s,t} L_{s,t} \quad \text{s.t} \tag{21}$$

$$L_{s,t} \geq W_{s,t} M_{s,t} \qquad L_{s,t} = \left(\sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}} \right)^{\frac{\alpha \sigma_{s,t}}{\sigma_{s,t}-1}}$$

where $L_{s,t} \geq W_{s,t} M_{s,t}$ is the condition that the Firm must pay its labor bill with working capital before production can begin.

Retailers. Each region is home to a continuum of retailers that purchase firms' homogenous output in a perfectly competitive market and use a linear technology, $\bar{Y}_{s,t}(j) = Y_{s,t}(j)$, to differentiate these goods. Retailers sell directly to households as a set of competitive monopolists subject to menu

cost nominal rigidities. The Retailers problem is:

$$\max_{P_{s,t}(j)} P_{s,t}(j)\bar{Y}_{s,t}(j) - Y_{s,t}(j) - \frac{\theta}{2}\left(\ln P_{s,t}(j) - \ln P_{s,t-1}(j)\right)^2 Y_{s,t}(j)$$

where θ is the adjustment cost of changing the retailer's nominal price, and captures the degree of nominal rigidity in the region.

Households. The region's continuum of households maximizes expected lifetime utility by supplying labor to firms for wages, purchasing a continuum of consumption goods, and saving in an intra-island bond market. Households in region s can be written as a representative agent, whose problem is:

$$V_H(B_{s,t-1}) = \max_{B_{s,t},M_{s,t},C_{s,t}} \{U(C_{s,t}, M_{s,t}) + \beta\mathbb{E}[V_H(B_{s,t})]\} \quad \text{s.t.} \tag{22}$$

$$C_{s,t} \equiv \left(\int_0^1 C_{s,t}(j)^{\frac{\epsilon}{\epsilon-1}}\right)^{\frac{\epsilon-1}{\epsilon}} \qquad U_{s,t} = \left(\frac{C_{s,t}^{1-\varrho}}{1-\varrho} - \frac{M_{s,t}^{1+\phi}}{1-\phi}\right)$$

$$P_{s,t}C_{s,t} + Q_{s,t}B_{s,t} \leq B_{s,t-1} + W_{s,t}M_{s,t} + D_{s,t} \qquad 0 \leq M_{s,t} \leq 1$$

where $C_{s,t}$ is the aggregate consumption in region s at time t, created by combining varieties $C_t(j)$ via CES aggregator, $P_{s,t}$ is the nominal aggregate price of the household's consumption bundle, $Q_{s,t}$ is the bond market interest rate, $B_{s,t}$ is the level of household savings in time t, $W_{s,t}$ are wages in time t, $M_{s,t}$ is the household's hours worked or its supply of labor, and $D_{s,t}$ are the dividends from firms, ϱ is the households coefficient of relative risk aversion ($1/\varrho$ is the elasticity of intertemporal substitution), ϵ is the elasticity of substitution between consumption goods, and ϕ is the curvature of the labor disutility (i.e. inverse Frisch elasticity). The firms intermediate good is the numeraire within each island.

The household's demand for total consumption, savings, and its labor supply follow from the standard conditions:

$$Q_{s,t} = \beta\mathbb{E}\left(\frac{U_{c,t+1}}{U_{c,t}}\right) \qquad M_{s,t}^{\phi}C_{s,t}^{\varrho} = W_{s,t}$$

and the intra-island bond market is in zero net supply, so $B_{s,t} = 0$ for all t. The household's demand for individual consumption varieties is standard and follows the inverse demand function:

$$c_{s,t}(j) = \left(\frac{P_{s,t}(j)}{P_{s,t}}\right)^{-\epsilon} C_{s,t} \qquad P_{s,t} = \left(\int_0^1 P_{s,t}^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}}$$

where is the regional price index $P_{s,t}$ is also standard.

By having firms pay wage bills with working capital loans, the household wage income is guaranteed despite some individual firms receiving an exit shock and not producing, thus not needing the labor. The structure of the economy effectively shifts the income risk from regional households to the banking sector.

A.2 Real activity and prices

Regional loan demand, employment, and GDP. Solving the regional representative firm's dual problem leads to the region's demand for bank i's loans and composite region-specific loan rate:

$$L_{i,s,t} = \eta_{ist} \cdot \left(\frac{R_{s,t}}{R_{i,s,t}} \right)^{\sigma_{s,t}} \cdot L_{s,t}, \qquad R_{s,t} = \left(\sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t} \cdot R_{i,s,t}^{1-\sigma_{s,t}} \right)^{\frac{1}{1-\sigma_{s,t}}}$$

while the primal profit maximization problem leads to the standard factor input demand:

$$M_{s,t} = \left(\frac{\alpha A_{s,t}}{R_{s,t} W_{s,t}} \right)^{\frac{1}{1-\alpha}}$$

where $M_{s,t}$ is in turn the regional labor demand and when markets clear, the region's employment level.

Total region production is then a function of the firm's technology, labor inputs, and firm exits.

$$Y_{s,t} = (1 - \lambda_{s,t}) \cdot A_{s,t} \cdot \left(\frac{\alpha A_{s,t}}{R_{s,t} W_{s,t}} \right)^{\frac{\alpha}{1-\alpha}}$$

where is the $\lambda_{s,t}$ is the firm exit rate in region s at time t. Note that regional production is increasing in regional productivity $A_{s,t}$, but decreasing in regional loan rates $R_{s,t}$, wages $W_{s,t}$, and firm exits $\lambda_{i,t}$ which are equivalently the region's loan default rate.

Given the structure of the economy and market clearing conditions, the regional equilibrium wages, labor, loan demand, and planned production can all be characterized analytically in terms of the regional loan rate $R_{s,t}$, economic fundamentals, and model parameters:

$$W_{s,t} = \left[A_{s,t}^{\varrho} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{\alpha\varrho+1}{1-\alpha}} \right]^{\frac{1}{\frac{\alpha\varrho+1}{1-\alpha}+1}} \tag{23}$$

$$M_{s,t} = \left(\frac{\alpha A_{s,t}}{R_{s,t}}\right)^{\frac{1}{1-\alpha}} \left[A_{s,t}^{\varrho}\left(\frac{\alpha A_{s,t}}{R_{s,t}}\right)^{\frac{\alpha\varrho+1}{1-\alpha}}\right]^{\frac{1}{\alpha\varrho+2-\alpha}} \tag{24}$$

$$L_{s,t} = A_{s,t}^{\frac{\varrho}{1+\varrho}}\left(\frac{\alpha A_{s,t}}{R_{s,t}}\right)^{\frac{\alpha\varrho+1}{(1-\alpha)(1+\varrho)}+\frac{1}{1-\alpha}} \tag{25}$$

$$Y_{s,t} = A_{s,t}N_{s,t}^{\alpha} = A_{s,t}\left(\frac{\alpha A_{s,t}}{R_{s,t}}\right)^{\frac{1}{1-\alpha}} \left[A_{s,t}^{\varrho}\left(\frac{\alpha A_{s,t}}{R_{s,t}}\right)^{\frac{\alpha\varrho+1}{1-\alpha}}\right]^{\frac{1}{\alpha\varrho+2-\alpha}} \tag{26}$$

which all reveal that as loan rates increase, economic activity, labor, and wages all decline in turn. This further highlights the transmission of monetary policy through loan rates. As the monetary authority tightens (eases) policy and increases (decreases) loan rates through the banks' cost of borrower capital funds, real economic activity and employment will slow.

Regional prices and inflation. Given two conditions (along with regional goods and financial market clearing conditions) one can analytically derive the the regional nominal price level and inflation: first $\epsilon = 1$, second, the intermediate goods firms produce with a decreasing returns to scale technology. Meeting these conditions, the the log nominal price level and inflation are:

$$\Pi_{s,t} = \ln(P_{s,t}/P_{s,t-1}) = \theta Y_{s,t} \implies \ln(P_{s,t}) = \theta Y_{s,t} + \ln P_{s,t-1} \tag{27}$$

where $Y_{s,t}$ is the level of regional intermediate output in terms of model primitives, as described in Equation 26. This implies that as the Central Bank tightens policy, inflation declines.

National activity and prices. The aggregate output, Y_t, and inflation rate Π_t are the equally-weighted average of region-level output and inflation. I abstract from differences in population, otherwise national activity and inflation would be population-weighted.

A.3 Additional financial outcomes

I next turn to analytically characterizing additional bank-level outcomes that complement those discussed more extensively in Section 3, including bank's loan demand, the sources of banks' portfolio return risk, profits, and comparative loan rates and quantities.

A.3.1 Bank-specific demand

The demand for a bank's loans within a region is necessarily the aggregation of demand for loans from each firm operating in the region. If we take the representative firm view of the region's

manufacturing sector, then the bank's loan demand follows from a CES demand for working capital.

Loan demand for bank i in region s at time t is therefore standard and given as:

$$L_{i,t} = \eta_{i,t}\left(\frac{R_t}{R_{i,t}}\right)^{\sigma_{s,t}} L_{s,t} \tag{28}$$

where $L_{i,t}$ is bank i's loans, $L_{s,t}$ are total regional loans, $R_{i,t}$ is bank i's portfolio average loan rate, $R_{s,t}$ is the effective regional loan rate, $\eta_{i,t}$ is the bank-specific appeal (i.e. the CES preference weight), and $\sigma_{s,t} \geq 1$ is the time-varying elasticity of intermediate loan substitution.[22]

A.3.2 Portfolio risk

The portfolio risk σ^R is the variance of portfolio returns rate, which are a function of the bank's borrower quality and screening ability given all other variables of Equation 10 are known at the time loan offers are made. As a result, the portfolio risk is simply the variance of sum of these two stochastic fundamentals:

$$\lambda_{i,t} = \underbrace{\frac{1-\psi_{i,t}}{\psi_{i,t}}}_{\equiv \chi^{\psi}_{i,t}} + \underbrace{\frac{1-q_{i,t}}{q_{i,t}}}_{\equiv \chi^{q}_{i,t}} \quad \Longrightarrow \quad \sigma^R = \sigma^{\psi} + \sigma^{q} + cov(\sigma^{\psi}, \sigma^{q})$$

where σ^{ψ}_t is the variance of $\chi^{\psi}_{i,t}$ and σ^{q}_t is the variance of $\chi^{q}_{i,t}$. A further transformation of variables may solve the exact mapping from the variance of ψ and q to the portfolio risk, but the log-odds terms will be used later in the quantitative estimation.

A.3.3 Loan returns and total profits

Realized loan return and net interest margin. The bank's realized return in region s at time t is:

$$R^*_{i,s,t} = \frac{\lambda_{i,s,t}}{1 - E\lambda_{i,s,t}} \cdot \frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}} \cdot \left(C_{i,t} + \zeta_t \frac{\sigma_{R^*}}{N_{i,t}}\right)$$

while assuming a linear functional form for the regulatory cost function, $\Gamma(\cdot) = \zeta_t \frac{\sigma_{R^*} L_{i,t}}{N_{i,t}}$. Then rewriting the risk assessment in terms of deviations from realized defaults, $E\lambda_{i,s,t} = \lambda_{i,s,t} - \epsilon_{i,s,t}$, as

[22] Note that demand is homothetic across the size of total loans demanded, $L_{s,t}$. The CES demand structure is key in maintaining the tractability of the empirical measurement equation. Non-homothetic preferences over bank-specific loans may allow for the demand ratios for loans to vary across the total demand for loans, leading to a non-linear model of loan demand and potential identification issues measuring bank and regional fundamentals.

well as assuming a small default rate and marginal capital cost, $c_{i,t}$, implies the net profit margin:

$$\pi_{i,s,t} = r^*_{i,s,t} - c_{i,t} = \mu + \rho_{i,t} - \epsilon_{i,s,t}$$

The bank's profit margin is thus its observable net interest margin $(r^*_{i,s,t} - c_{i,t})$, which is equivalent to a linear combination of log markups $\mu = log(\frac{\sigma_{s,t}}{\sigma_{s,t}-t}) + s_{i,s,t}$, approximate residual unobservable costs $\rho_{it} = log(C_{i,t} + \frac{\zeta_t \sigma_{R^*}}{N_{it}}) - log(C_{i,t})$, and default assessment error $\epsilon_{i,s,t}$.

Total profits. The realized profits of a bank's regional loan portfolio in region s at time t are:

$$\begin{aligned}\Pi_{i,s,t} &= R^*_{i,s,t}\, L_{i,s,t} \\ &= \underbrace{(1-\lambda_{i,s,t})}_{\text{performing loans}} \cdot \underbrace{\eta_{i,s,t} R^{\sigma_{s,t}}_{s,t} L_{s,t}}_{\substack{\text{demand adjusted} \\ \text{aggregate conditions}}} \cdot \underbrace{\left[\frac{1}{1-E\lambda_{i,t}} \frac{\theta}{\theta-1}\left(C_{i,t} + \zeta_t \cdot \frac{\sigma_{R^*}}{N_{i,t}}\right)\right]^{1-\theta}}_{\text{expected return per loan}}\end{aligned}$$

which is a function of the realized vs expected loan losses, the aggregate lending conditions of the local market scaled by the bank-specific demand within the market, and the loan rate charged for each loan, while continuing to assume a linear functional form for the regulatory cost function, $\Gamma(\cdot) = \zeta_t \frac{\sigma_{R^*} L_{i,t}}{N_{i,t}}$. The banks total profits are then the sum of regional loan portfolio profits less fixed operating costs:

$$\Pi_{i,t} = \sum_{s \in S} \left\{ R^*_{i,s,t} L_{i,s,t} - f_{i,s,t} \right\} - (L_{i,t} - N_{i,t}) C_{i,t} - \Gamma(\sigma^R \frac{L_{i,t}}{N_{i,t}}) - f_{i,t}$$

recalling that the bank does not pay an operating cost $f_{i,s,t}$ if it does not issue loans in a region s in time t.

A.3.4 The sources of differences in bank lending volume and loan rates

Understanding the banks' regional market share Equation 12 begins with comparing the relative lending between any two banks in market s:

$$\frac{L_{i,s,t}}{L_{j,s,t}} = \frac{\eta_{i,s,t}\left(\frac{R_{s,t}}{R_{i,s,t}}\right)^{\sigma_{s,t}} L_{s,t}}{\eta_{j,s,t}\left(\frac{R_{s,t}}{R_{j,s,t}}\right)^{\sigma_{s,t}} L_{s,t}} = \frac{\eta_{i,s,t} R^{-\sigma_{s,t}}_{i,s,t}}{\eta_{j,s,t} R^{-\sigma_{s,t}}_{j,s,t}} \implies$$

$$\frac{s_{i,s,t}(1-s_{i,s,t})^{-\sigma_{s,t}}}{s_{j,s,t}(1-s_{j,s,t})^{-\sigma_{s,t}}} = \left(\frac{1/L_{s,t}}{1/L_{s,t}}\right)\frac{\eta_{j,s,t}}{\eta_{i,s,t}}\left[\frac{\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\phi_{j,t}}{\phi_{j,t}}\right)\right]\cdot\left(C_{j,t}+\zeta_t\frac{1}{N_{j,t}}\right)}{\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\phi_{i,t}}{\phi_{i,t}}\right)\right]\cdot\left(C_{i,t}+\zeta_t\frac{1}{N_{i,t}}\right)}\right]^{\sigma_{s,t}}$$

One can further take the log of both sides to show that the difference in lending is a function of log differences in their respective pricing factors and bank specific characteristic. In other words, the difference in market share, thus relative lending, between any two banks is a function of their relative appeal, borrower quality, ability, capital costs, and regulatory cost.

A similar analysis can be applied to comparing the loan rates charged by any two banks operating in the same region. In fact, that analysis is subsumed the comparison of lending quantities above, and the comparative loan rate quotient:

$$\frac{R_{j,s,t}}{R_{i,s,t}} = \left[\frac{\frac{1}{1-s_{j,st}}\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\phi_{j,t}}{\phi_{j,t}}\right)\right]\cdot\left(C_{j,t}+\zeta_t\frac{1}{N_{j,t}}\right)}{\frac{1}{1-s_{i,st}}\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\phi_{i,t}}{\phi_{i,t}}\right)\right]\cdot\left(C_{i,t}+\zeta_t\frac{1}{N_{i,t}}\right)}\right]$$

shows a similar result. The log-difference between any two banks' loan rates in the same region will be due to the comparative difference in their screening ability, capital cost, loan appeal (through differences in market share), and their pre-existing size (measured by net worth).

A.4 Proofs

In this section I prove the analytical characterization of loan rates, market share, and regional entry conditions.

A.4.1 Loan rate proof

Theorem 1 *A bank charges the region- and period- specific loan rate:*

$$R_{i,s,t} = \frac{1}{1-\mathbb{E}_t\lambda_{i,s,t}}\cdot\frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{1}{1-s_{i,s,t}}\cdot\left(C_{i,t}+\Gamma'(\sigma^R\frac{L_{i,t}}{N_{i,t}})\right)$$

where $\mathbb{E}_t\lambda_{i,s,t}$ is the expected loan default rate, $\sigma_{s,t}$ is the region-specific elasticity of intermediate loan substitutability, $s_{i,s,t}$ is the bank's regional loan market share, $C_{i,t}$ is the bank's capital cost, is the marginal regulatory cost based on the bank's leverage ratio $L_{i,t}/N_{i,t}$ and portfolio loan rate

return variance σ^R.

Proof. The bank's problem in series form follows:

$$\max_{\{R_{i,t}\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty}(1-m)^t\beta^t\Pi_{i,t} \quad \text{s.t.}$$

$$\Pi_{i,t} = \sum_{s\in S}\left\{R^*_{i,s,t}L_{i,s,t} - f_{i,s,t}\right\} - (L_{i,t} - N_{i,t})C_{i,t} - \Gamma(\sigma^R\frac{L_{i,t}}{N_{i,t}}) - f_{i,t} \qquad \text{(Profit function)}$$

$$R^*_{i,s,t} = (1-\lambda_{i,s,t})R_{i,s,t} \qquad \text{(Realized gross return)}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \qquad \text{(Net worth law of motion)}$$

The Lagrangian is:

$$\mathcal{L} = \sum_{t=0}^{\infty}(1-m)^t\beta^t\left[\sum_{s\in S}\left\{(1-\lambda_{i,s,t})R_{i,s,t}L_{i,s,t} - f_{i,s,t}\right\} - (L_{i,t} - N_{i,t})C_{i,t} - \Gamma(\sigma^R\frac{L_{i,t}}{N_{i,t}}) - f_{i,t}\right] \quad \text{s.t.}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \quad \text{with } N_{i,0} \text{ given}$$

Then using the bank-region-specific demand curve $L_{i,s,t} = \eta_{i,s,t}\left(\frac{R_{i,s,t}}{R_{s,t}}\right)^{\sigma_{s,t}}L_{s,t}$ to replace the loan rate $R_{i,s,t}$, and assuming the regulatory cost term takes a linear functional form: $\Gamma(L_{i,t}/N_{i,t}) = \zeta_t(L_{i,t}/N_{i,t})$ where $\zeta_t > 0$, the first order condition (FOC) with respect to loans issued in region s at time t is

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial L_{i,s,t}} = {} & L_{i,s,t}^{1/\sigma_{s,t}}(1-\lambda_{i,s,t})\eta_{i,s,t}R_{s,t}^{1-1/\sigma_{s,t}}L_{i,s,t}^{-1/\sigma_{s,t}}\left(\frac{\sigma_{s,t}-1}{\sigma_{s,t}}\right) - C_{i,t} - \zeta_t(\sigma^R/N_{i,t}) + \ldots \\ & \ldots + \left(\frac{L_{s,t}}{R_{s,t}}\right)^{1/\sigma_{s,t}}\frac{\partial R_{s,t}}{\partial L_{i,s,t}}R_{s,t}^{1-1/\sigma_{s,t}}\eta_{i,s,t}(1-\lambda_{i,s,t})\left(\frac{\sigma_{s,t}-1}{\sigma_{s,t}}\right) - \ldots \\ & \ldots - \left[\sum_{j=1}^{\infty}(1-m)^j\beta^j a_{i,j}\right] - \left[\sum_{j=1}^{\infty}(1-m)^j\beta^j b_{i,j}\right] = 0\end{aligned}$$

where $a_{i,j}$ is the marginal increase in net worth due to relaxing regulatory costs in period j from the marginal increase in retained earnings in period t, and $b_{i,j}$ is the marginal increase in net worth due to decreasing capital funding costs in period j from the marginal increase in retained earnings in period t. Note that the condition

$$[\sum_{j=1}^{\infty}\beta^j\eta_{i,j}] = [\sum_{j=1}^{\infty}\beta^j\zeta_{i,j}] = 0$$

must be true in the absence additional constraints. Assuming a non-zero Lagrange multiplier term implies a profit improving, thus net worth increasing, loan volume exists, but an unconstrained firm maximizing its lifetime stream of profits would then set its loan volume to that optimizing level. Formalizing a proof by contradiction using this logic is straightforward and is left to the reader. Thus the bank maximizes profits period-by-period and the FOC becomes a static equation:

$$\frac{\partial \mathcal{L}}{\partial L_i} = L_{i,s,t}^{1/\sigma_{s,t}}(1-\lambda_{i,s,t})\eta_{i,s,t}R_{s,t}^{1-1/\sigma_{s,t}}L_{i,s,t}^{-1/\sigma_{s,t}}\left(\frac{\sigma_{s,t}-1}{\sigma_{s,t}}\right) - C_{i,t} - \zeta_t(\sigma^R/N_{i,t}) + \ldots$$
$$\ldots + \left(\frac{L_{s,t}}{R_{s,t}}\right)^{1/\sigma_{s,t}}\frac{\partial R_{s,t}}{\partial L_{i,s,t}}R_{s,t}^{1-1/\sigma_{s,t}}\eta_{i,s,t}(1-\lambda_{i,s,t})\left(\frac{\sigma_{s,t}-1}{\sigma_{s,t}}\right) = 0.$$

Solving the FOC for $R_{i,t}$ is then a simple matter of algebra:

$$\frac{\partial \mathcal{L}}{\partial L_{i,s,t}} = 0 \implies \frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{C_{i,t}+\zeta_t(\sigma^R/N_{i,t})}{\eta_{i,s,t}(1-\lambda_{i,s,t})} = \frac{L_{s,t}}{L_{i,s,t}}^{1/\sigma_{s,t}}R_{s,t}^{1-1/\sigma_{s,t}} + L^{1/\sigma_{s,t}}R_{s,t}^{-1/\sigma_{s,t}}\frac{\partial R_{s,t}}{\partial L_{i,s,t}}L_{i,s,t}^{1-1/\sigma_{s,t}}$$

define $\chi_{i,s,t} = \frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{C_{i,t}+\zeta_t(\sigma^R/N_{i,t})}{\eta_{i,s,t}(1-\lambda_{i,s,t})}$ for convenience, then

$$\begin{aligned}\chi_{i,s,t}R^{1/\sigma_{s,t}} &= L_{s,t}^{1/\sigma_{s,t}}L_{i,s,t}^{-1/\sigma_{s,t}}R_{s,t} + L_{s,t}^{1/\sigma_{s,t}}\frac{\partial R_{s,t}}{\partial L_{i,s,t}}L_{i,s,t}L_{i,s,t}^{-1/\sigma_{s,t}} \\ &= s_i^{-1/\sigma_{s,t}} + s_i^{-1/\sigma_{s,t}}\frac{L_{i,s,t}}{R_{s,t}}\frac{\partial R_{s,t}}{\partial L_{i,s,t}}\end{aligned}$$

where $s_{i,s,t} = L_{i,s,t}/L_{s,t}$ is the bank's loan market share, and using Shepard's Lemma, $-\frac{\partial R}{\partial L_{i,s,t}} = s_{i,s,t}$, where $s_{i,s,t}$ is bank i's loan market share in region s and time t, we have:

$$\chi_{i,s,t}R^{-1/\sigma_{s,t}} = s_{i,s,t}^{-1/\sigma_{s,t}} - s_{i,s,t}s_{i,s,t}^{-1/\sigma_{s,t}} = s_{i,s,t}^{-1/\sigma_{s,t}}(1-s_{i,s,t})$$

$$\implies \frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{C_{i,t}+\zeta_t(\sigma^R/N_{i,t})}{\eta_{i,s,t}(1-\lambda_{i,s,t})}\frac{1}{1-s_{i,s,t}} = \left(\frac{L_{s,t}}{L_{i,s,t}}\right)^{1/\sigma_{s,t}}R_{s,t}^{1/\sigma_{s,t}-1}$$

$$\implies \frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{C_{i,t}+-\zeta_t(\sigma^R/N_{i,t})}{(1-\lambda_{i,s,t})}\frac{1}{1-s_{i,s,t}} = \eta_{i,s,t}L_{s,t}^{1/\sigma_{s,t}}L_{i,s,t}^{-1/\sigma_{s,t}}R_{s,t}^{1/\sigma_{s,t}-1} = L_{i,s,t}R_{i,s,t}$$

so the loan rate for a one unit loan is:

$$\frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{C_{i,t}+\Gamma'(\frac{L_{i,s,t}}{N_i})}{(1-\lambda_{i,s,t})}\frac{1}{1-s_i} = R_{i,s,t}$$

where $\Gamma'(\frac{L_{i,s,t}}{N_i})$ is the marginal regulatory cost with respect to bank lending $L_{i,s,t}$. ■

A.4.2 Market share

Theorem 2 *A bank's market share in region s at time t is determined as:*

$$\log(s_{i,s,t}) = \frac{\sigma_{s,t}\log\tilde{s}_{s,t}}{\sigma_{s,t}\log\tilde{s}_{s,t}+1}[(1/\sigma_{s,t})(\log(\eta_{i,t})-\tilde{\eta}_{s,t})+(\tilde{C}_{s,t}-\log(C_{i,t}))+\ldots$$
$$\ldots+(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t}))+(\mathbb{E}\tilde{\chi}^{\psi}_{s,t}-\mathbb{E}\log(\chi^{\psi}_{i,t}))]$$

where $s_{i,s,t}$ is the bank's regional loan market share, $\sigma_{s,t}$ is the region-specific elasticity of intermediate loan substitutability, $\chi^{\psi}_{i,t} = \frac{1-\psi_{i,t}}{\psi_{i,t}}$ is the bank's log-odds of screening ability, $C_{i,t}$ is the bank's capital cost, is the marginal regulatory cost based on the bank's leverage ratio $L_{i,t}/N_{i,t}$ and portfolio loan rate return variance σ^R, and tilde variables are the geometric means of bank-level values for banks operating in the region at time t.

Proof. Start by comparing the relative lending between any two banks in market s:

$$\frac{L_{i,s,t}}{L_{j,s,t}} = \frac{\eta_{i,s,t}\left(\frac{R_{s,t}}{R_{i,s,t}}\right)^{\sigma_{s,t}}L_{s,t}}{\eta_{j,s,t}\left(\frac{R_{s,t}}{R_{j,s,t}}\right)^{\sigma_{s,t}}L_{s,t}} = \frac{\eta_{i,s,t}R_{i,s,t}^{-\sigma_{s,t}}}{\eta_{j,s,t}R_{j,s,t}^{-\sigma_{s,t}}} \implies$$

$$\frac{s_{i,s,t}(1-s_{i,s,t})^{-\sigma_{s,t}}}{s_{j,s,t}(1-s_{j,s,t})^{-\sigma_{s,t}}} = \left(\frac{1/L_{s,t}}{1/L_{s,t}}\right)\frac{\eta_{j,s,t}}{\eta_{i,s,t}}\left[\frac{\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\phi_{j,t}}{\phi_{j,t}}\right)\right]\cdot\left(C_{j,t}+\zeta_t\frac{1}{N_{j,t}}\right)}{\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\phi_{i,t}}{\phi_{i,t}}\right)\right]\cdot\left(C_{i,t}+\zeta_t\frac{1}{N_{i,t}}\right)}\right]^{\sigma_{s,t}}$$

One can further take the log of both sides to show that the difference in lending is a function of log differences in their respective pricing factors and bank specific characteristic. In other words, the difference in market share, thus relative lending, between any two banks is a function of their relative appeal, borrower quality, ability, capital costs, and regulatory cost (ie risk aversion).

Next take the sum of the log-linearized relative market shares for bank i with respect to all banks:

$$\sum_{j\in\mathcal{B}_t}\log s_{i,s,t}-\sigma_{s,t}s_{i,s,t}-\log s_{i,s,t}+\sigma_{s,t}s_{j,s,t} = -J_{s,t}\sigma_{s,t}s_{i,s,t}+\frac{\log s_{i,s,t}}{\sum\log s_{j,s,t}}+\sigma_{s,t}$$

now define a tilde variable, for example, $\tilde{s}_t$ as the unweighted average of bank-level log values, so that, for example, $\tilde{s}_{s,t} = (1/J_{s,t})\sum\log s_{j,s,t}$, where $J_{s,t} = ||\mathcal{B}_{s,t}||$ is the cardinality of the set of

banks operating in a region at time t. We can then write both sides of the sum as follows:

$$s_{i,s,t}+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}}+1 = (1/\sigma_{s,t})(\log(\eta_{i,s,t})-\tilde{\eta}_{s,t})+(\tilde{C}_{s,t}-\log(C_{i,t}))+(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t}))+(\tilde{\chi}^{\psi}_{s,t}-\log(\chi^{\psi}_{i,t}))$$

We can use the properties of logs and definition of market share to rewrite the left hand side as follows:

$$\frac{L_{i,s,t}-L_{s,t}}{L_{s,t}}+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}} \approx \log(L_{i,s,t}/L_{s,t})+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}} = \ldots$$

$$\ldots = \log(s_{i,s,t})+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}} = \log(s_{i,s,t})\frac{1+\sigma_{s,t}\log \tilde{s}_{s,t}}{\sigma_{s,t}\log \tilde{s}_{s,t}}$$

and as a result, we have the following closed form equation for a bank's log market share:

$$\log(s_{i,s,t}) = \frac{\sigma_{s,t}\log \tilde{s}_{s,t}}{\sigma_{s,t}\log \tilde{s}_{s,t}+1}[(1/\sigma_{s,t})(\log(\eta_{i,t})-\tilde{\eta}_{s,t})+(\tilde{C}_{s,t}-\log(C_{i,t}))+\ldots$$
$$\ldots+(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t}))+(\mathbb{E}\tilde{\chi}^{\psi}_{s,t}-\mathbb{E}\log(\chi^{\psi}_{i,t}))]$$

where $\chi^{\psi}_{i,t} = \frac{1-\psi_{i,t}}{\psi_{i,t}}$. The bank's market share is an increasing function in its appeal relative to the average appeal, borrower quality relative to the average borrower quality, ability relative to the average ability, but decrease in capital and regulatory costs relative to average capital and regulatory costs. ■

A.4.3 Regional market entry

Theorem 3 *The expected profit at which a bank will be indifferent between entering a regional market or not then follows from the expected profit condition:*

$$\mathbb{E}log\bar{\Pi}_{i,s,t} = \mathbb{E}\left\{\log\frac{\sigma_{s,t}}{\sigma_{s,t}-1}+\exp\{\log s_{i,s,t}\}+\log s_{i,s,t}+w_{s,t}+\frac{1}{\alpha}y_{s,t}-\frac{1}{\alpha}a_{s,t}\right\} = \log f_{i,s,t}$$

where $log\bar{\Pi}_{i,s,t}$ is the cutoff expected profit at which a bank will enter a region. This condition is based on four primary forces, the banks expected competitiveness, local market frictions, depth of the regional market, and the distance from headquarters.

Proof. A bank will enter a regional market if $\mathbb{E}\Pi_{i,s,t} \geq 0$. Given oligopolistic structure of the regional loan markets, the condition in turn depends on whether the bank can make a large enough profit per loan to pay its fixed cost of operating in the given region. For a bank's headquarters region, then the fixed cost is f_t, and if the region is not the headquarters region then the fixed cost is $f_t + f(d_{i,s,t})$. Therefore, all else equal, as distance increase from the headquarters region, the

greater the cost of operating in a region and the less likely a bank will move into the new market.

The expected profit at which a bank will be indifferent between entering a regional market or not then follows from the condition (dropping expectation operator for now):

$$\bar{\Pi}_{i,s,t} = R^{*}_{i,s,t} L_{i,s,t} - c_{i,t} - \Gamma'_{i,t} - f_{i,s,t} = 0$$

$$\bar{\Pi}_{i,s,t} = (1 - \lambda_{i,s,t}) R_{i,s,t} \cdot s_{i,s,t} L_{s,t} - c_{i,t} - \Gamma'_{i,t} - f_{i,s,t} = 0$$

take the log of both sides of the condition:

$$log\bar{\Pi}_{is,t} = (\mathbb{E}\lambda_{i,s,t} - \lambda_{i,s,t}) + \log \frac{\sigma_{s,t}}{\sigma_{s,t} - 1} + \exp\{\log s_{i,s,t}\} + \log s_{i,s,t} + \log L_{s,t} = \log f_{i,s,t}$$

following from the firm's problem, log loans are: $l_{s,t} = w_{s,t} + \frac{1}{\alpha} y_{s,t} - \frac{1}{\alpha} a_{s,t}$. So then when we take the expectation of both sides we have the condition:

$$\mathbb{E} log\bar{\Pi}_{i,s,t} = \mathbb{E}\left\{ \log \frac{\sigma_{s,t}}{\sigma_{s,t} - 1} + \exp\{\log s_{i,s,t}\} + \log s_{i,s,t} + w_{s,t} + \frac{1}{\alpha} y_{s,t} - \frac{1}{\alpha} a_{s,t} \right\} = \log f_{i,s,t}$$

where $log\bar{\Pi}_{i,s,t}$ is the cutoff expected profit at which a bank will enter a region. This condition is based on thee primary forces, the banks expected competitiveness the depth of the regional market, and the trade distance. ■

B Data

This appendix provides additional details, discussion, and figures for the data used in the estimation of the bank, region fundamentals, market frictions, and regulatory costs.

B.1 Call Reports: bank balance sheet and income data

Bank-level data are from the Consolidated Report of Condition and Income, also known as Call Reports. A form of Call Report data was first collected in 1933 by the Federal Deposit Insurance Corporation (FDIC) as a part of the regulatory reform responding to the Great Depression. However, there was a substantial update to the required reporting frequency, the number of variables collected, and variable definitions in 1984, resulting in a bifurcation between the pre- and post-1984 Call Report sample history. Because of the incongruity between variable definitions across the two sample periods modern Call Report data effectively begins in 1984. I specifically use the Call Report files maintained by the Federal Reserve Bank of Chicago, although the data steward ship changed hands to the FFIEC after 2021.

The bank-level sample is representative of the U.S. banking sector. Figure 2 shows that the sum of deposits and loans reported in the micro-data (approximately) match the level of deposits and loans reported by the Federal Reserve Board at the national-level. Figure 2 likewise shows that the loan-weighted charge-offs ratio reported in the micro-data matches the national charge-off ratio reported by the Federal Reserve. The sample extends from 1984 through 2019 (where I end due to the COVID-19 pandemic), and includes over 11 thousand unique banks, although banks enter and exit the sample due to failures, mergers, and acquisitions, so that maximum number of banks any given period is no more than 7 thousand (see Figure 13 for further description of the number and entry/exit of banks).

Most variables of interest in my study are not directly observable from bank balance sheets and income statements. Table 5 reports the variable mnemonics used to construct the constructed variables. Net charge offs have been visualized in Figure 2, loan rates have been presented in Figure 3, so lastly, Figure 15 shows the quarterly median interest income and costs. The difference between these two loan characteristics is the observable net interest margin, which is used for estimating the regulatory costs and markups described in Section 4.3. Note that similar to the inferred loan rates, both interest income and costs have been in secular decline since the 1980s, but unlike loan rates, have a pro-cyclical component increasing before recessions.

B.2 Summary of Deposits: spatial distribution of banking activity

County-level deposit data are from the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits (SOD). The SOD is an annual survey that collects branch-level information on deposit balances for all FDIC-insured institutions as of June 30th each year. The data provide the geographic distribution of deposits across counties and states and include identifiers that allow deposits to be linked to bank holding companies and Call Reports. Because the survey is conducted annually and reflects deposits at a point in time, it is best suited for measuring cross-sectional exposure and geographic deposit shares rather than within-year flows. Consistent and comprehensive coverage begins in 1994 following improvements in reporting standardization and electronic submission procedures. For this reason, most analysis will begin in 1995 after including one year for the autoregressive lag in the state-space model. I will specifically use the SOD information on the branch state-level locations of each bank and the level of deposit holdings at each of those branches. These information are required to calculate regional loan market shares.

B.3 Calculating regional market shares

A bank's regional market share is calculated by the bank's loan share of all loans in a U.S. state. The algorithm for calculating a bank's regional market share is as follows: For banks that only have branch locations in one state, then the bank's entire loan portfolio is assumed to be made to borrowers in that state. For banks with branch locations across multiple states, the bank's loan portfolio is allocated across states based on the bank's deposit holdings by state, following Nguyen (2019). That is, if a bank has raised half of its deposits in Virginia and half in Maryland, then half of its loan portfolio is counted as its Virginia loan portfolio and the other half as its Maryland loan portfolio. These regional loan portfolios are then summed across all banks operating in a given region to determine the total region loan market size. Dividing the bank's regional loan portfolio by the total regional loan market size gives the bank's regional loan market.

Testing the spatial distribution of loans assumption. The key assumption motivating the the regional loan market calculation is that banks lend where they raise deposits. This is statutorily mandated by the Community Reinvestment Act of 1977. However, recent work also shows that banks sort into localities with excess deposits to lending ratios so they can use the additional deposits to fund loans in higher margin markets, see for example Oberfield et al. (2024). Therefore I test the validity of the spatial sorting assumption by calculating the within-bank correlation of a bank's county-level deposits and lending activity (either mortgage lending using HMDA data or small business lending using CRA data). Figure 17 shows a high correlation between where banks raise deposits and where they lend. The median bank has a correlation coefficient greater than 0.9

on average over the sample for either type of lending. I take this as sufficient evidence to allocate a bank's loan portfolio across states by using their relative deposit holdings, and in turn calculate its state-level loan market share.

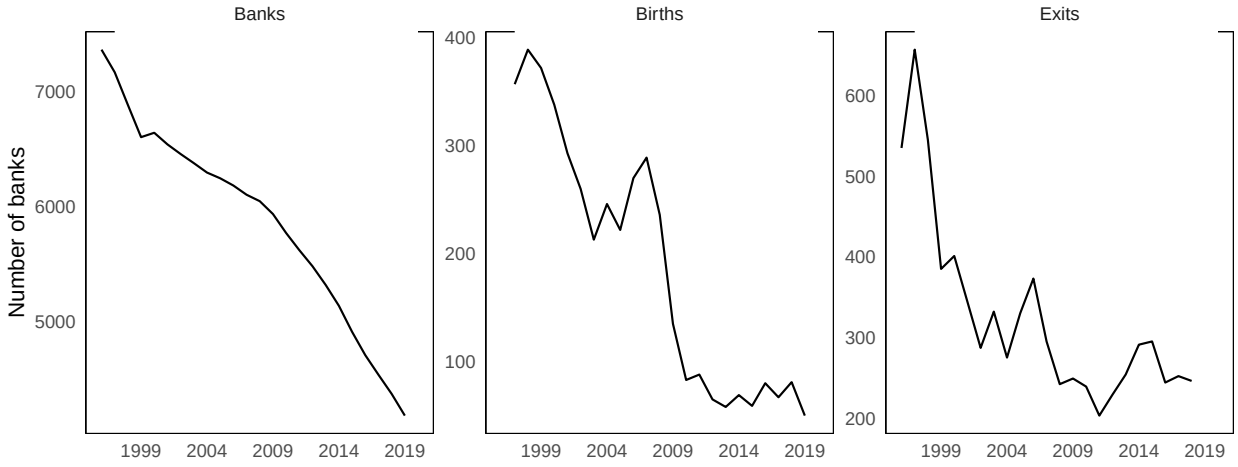
Figure 18 shows the average bank-state-level market share from 1995 through 2019, while Figure 19 shows the maximum bank-state-level market share across the sample per state. State-level market share is relatively low, with the maximum occurring in South Dakota and peaking near seven percent of the total state-level loan market share after the GFC. In comparison, Figure 16 shows that the national bank loan market is dominated by a small group of banks. The left panel shows the distribution of banks' national market shares. Notably, the mean market share is well above the 90th percentile of the distribution, pointing to extreme outliers. Moreover, these outliers are becoming more influential on the mean over time as the banking sector grows more concentrated. The right panel then charts the market share of the 50 largest banks (by loan portfolio). The four largest banks account for approximately 37 percent of loans in the U.S. by 2020.

B.4 Non-financial data

The key non-financial data used to estimate fundamentals and frictions are the state-level firm exit rate, which acts an observable proxy for the firm-level probability of receiving an exogenous exit shock, and the real median household income, which is a stand-in for household labor income. Both are collected from the U.S. Census Bureau, the latter from the Business Dynamism Statistics dataset and the latter from the report on Income and Poverty in the Unites States.

Figure 21 shows the mean, median, and interquartile range of either non-financial data series of interest. The data is a balanced panel of the 50 U.S. states. Firm exit rates are countercyclical, which their largest spike during the global financial crisis. Real median household income has been was generally stable from 1995 through 2010 and then slowly increased through 2019.

Figure 13: Banks, entry, and exit



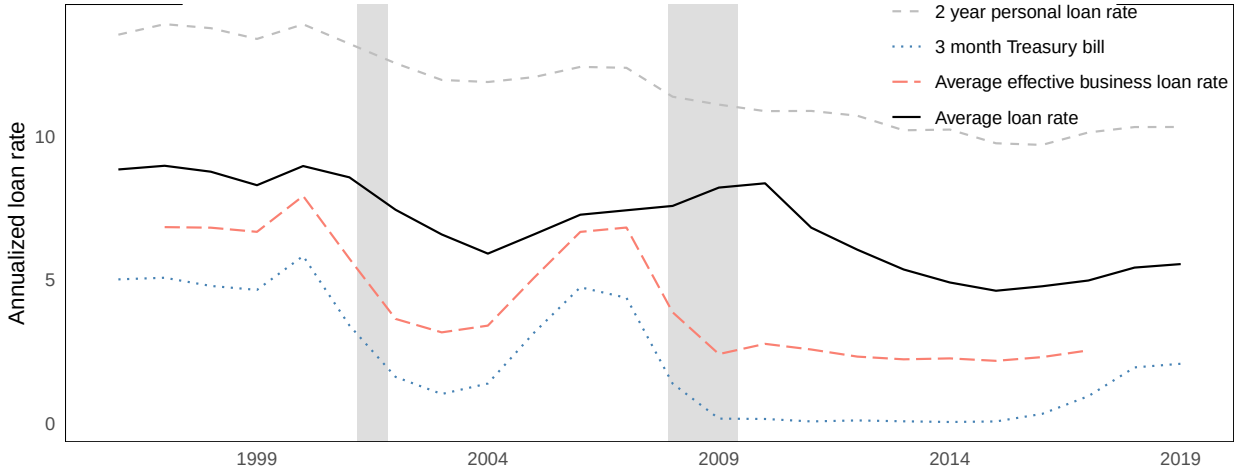
Notes: These plots present the number, entry, and exits of banks from the FFIEC Call Reports that operate a physical branch location. A bank is born when its unique RSSD ID is observed for the first time in the sample, while a bank is considered to have exited when its RSSD ID has been observed for the last time. Bank exits are not necessarily failures, but can also represent acquisitions and mergers. Data are annual from 1996 through 2019.

Table 5: Bank-level variable details

Variable	Variable	Formula	Call Report variables
Net worth	N		RCON3210
Loans	L		RCON2122, RCFD2122
Capital costs	C	total interest expense / total loans	RIAD4073, RCON2122, RCFD2122
Loan return	R^*	loan interest income / total loans	RIAD4010, RCON2122, RCFD2122
Charge-off ratio	λ	net charge offs / total loans	RIAD4635, RIAD4605
			RCON2122, RCFD2122

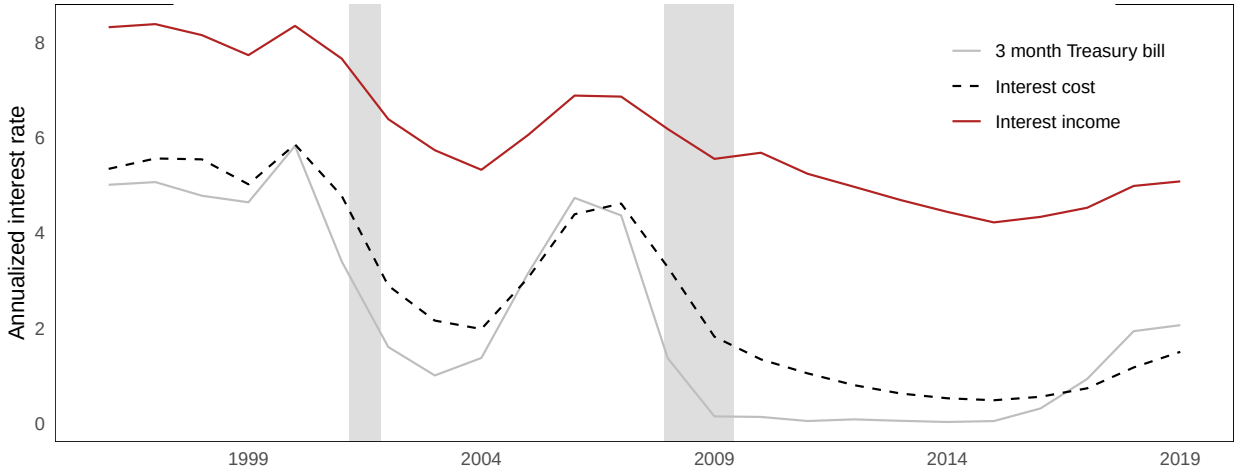
Notes: This table presents the Call Report filing variables used to create observable proxies for banks' capital cost premium, loan return, and charge off ratios. Variable codes are based on the Federal Reserve Bank of Chicago managed set of Call Report data.

Figure 14: Bank loan rates



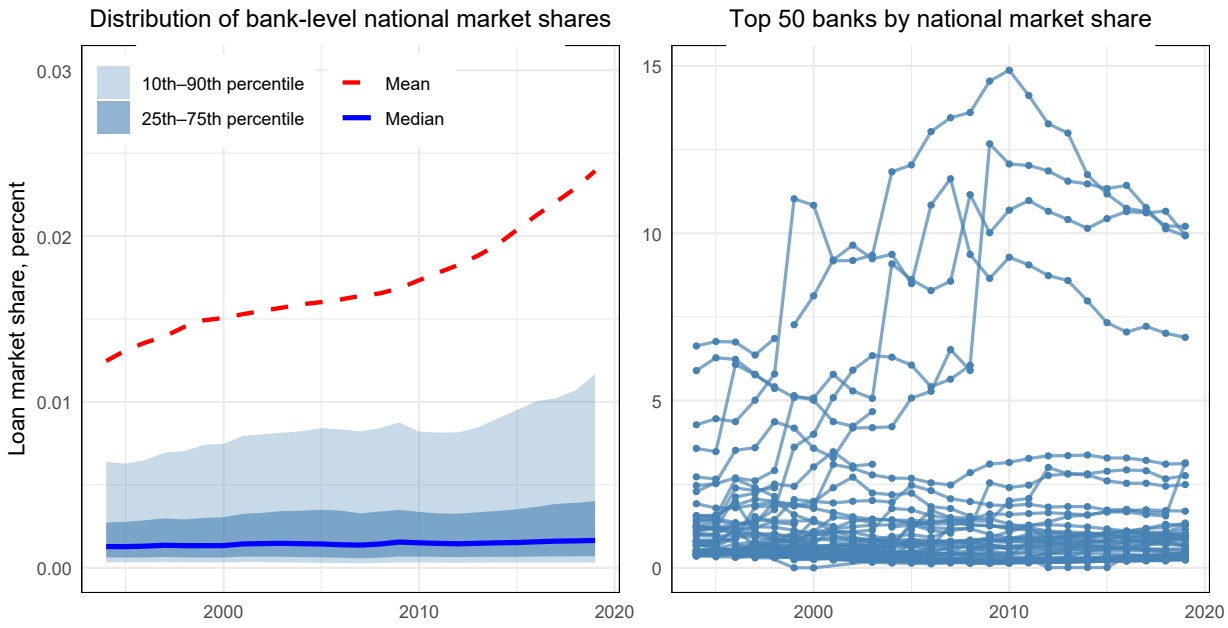
Notes: This plot compares the sample implied loan-weighted loan rate to published risk free and consumer loan rates. The bank-level sample loan rate is calculated as the reported quarterly loan income divided by total loans, scaled by the inverse one minus net charge-off ratio. The average loan rate is then the loan-weighted average of bank-level loan rates over the cross-section of banks in the given quarter. The risk-free rate is proxied by the US 3-month T-bill. The consumer loan rate is the Finance Rate on Personal Loans at Commercial Banks, 24 Month Loan, reported by the Federal Reserve Board's G.19 Consumer Credit. The average busies loan rate is weighted-average effective loan rate for all commercial and industry Loans reported in the Federal Reserve Board's E.2 Survey of Terms of Business Lending. Gray shaded regions mark NBER dated recessions. Data are annual from 1996 through 2019.

Figure 15: Loan income and costs



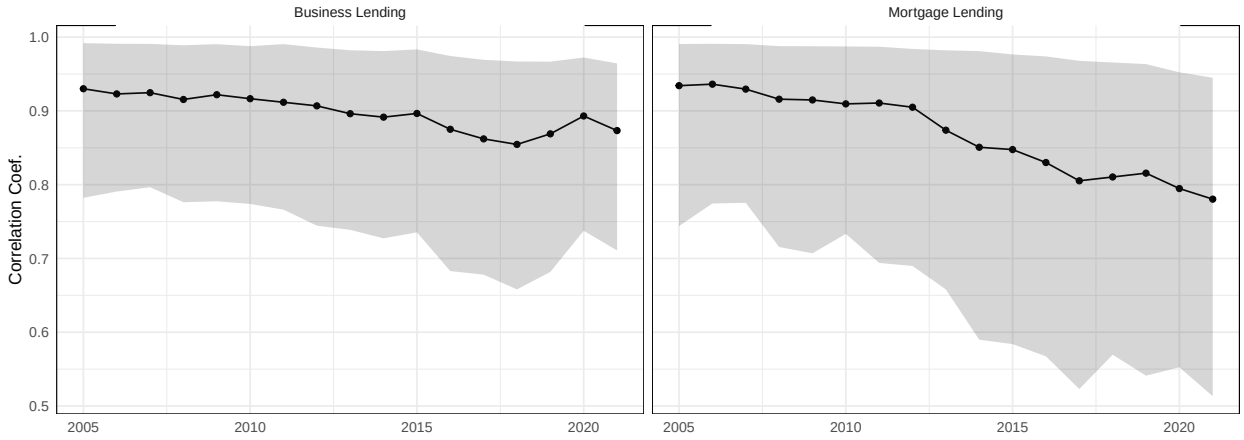
Notes: This plot presents loan interest income, costs, and the 3-month Treasury rate for comparison. Bank loan interest income and costs are loan-weighted. Gray shaded regions mark NBER dated recessions. Data are annual from 1996 through 2019.

Figure 16: Loan market shares over time



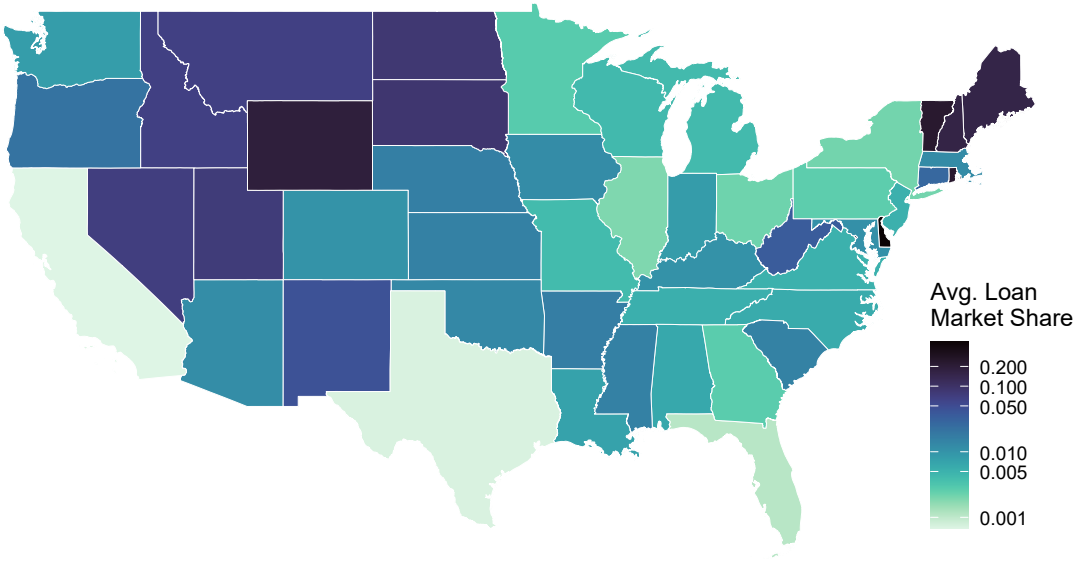
Notes: This plot shows the distribution of banks' loan market shares over time. The left panel shows the mean, median, 10-90 percentile range, and inter-quartile range of bank-level national market shares. The right panel shows the individual national market share of the 50 largest banks by loan portfolio size. Data are annual from 1996 through 2019.

Figure 17: Correlation between county lending and deposits shares within banks



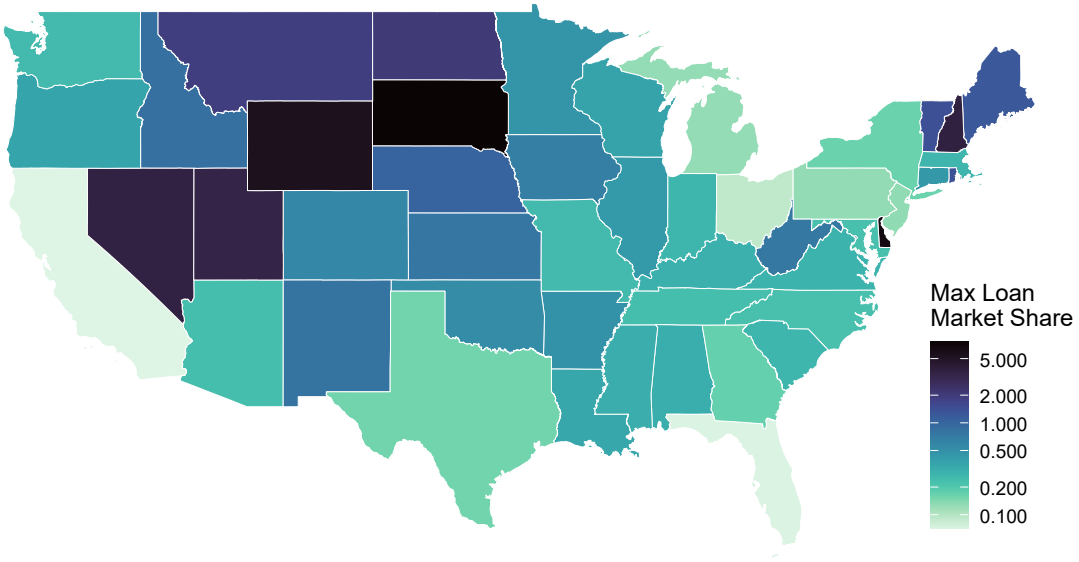
Notes: This plot shows the within-bank correlation between branch deposit holding and lending originations. The left panel shows the annual correlation between a bank's county-level small business lending (gathered from CRA filings) and branch deposit holdings (gathered from the FDIC's Summary of Deposits report). The right panel shows the annual correlation between a bank's county-level mortgage lending origination and branch deposit holdings. The gray bands mark the inter-quartile range of bank-level correlation coefficients, while the black line marks the median bank-level correlation. Data are annual from 2005 through 2021.

Figure 18: Average bank market share by state



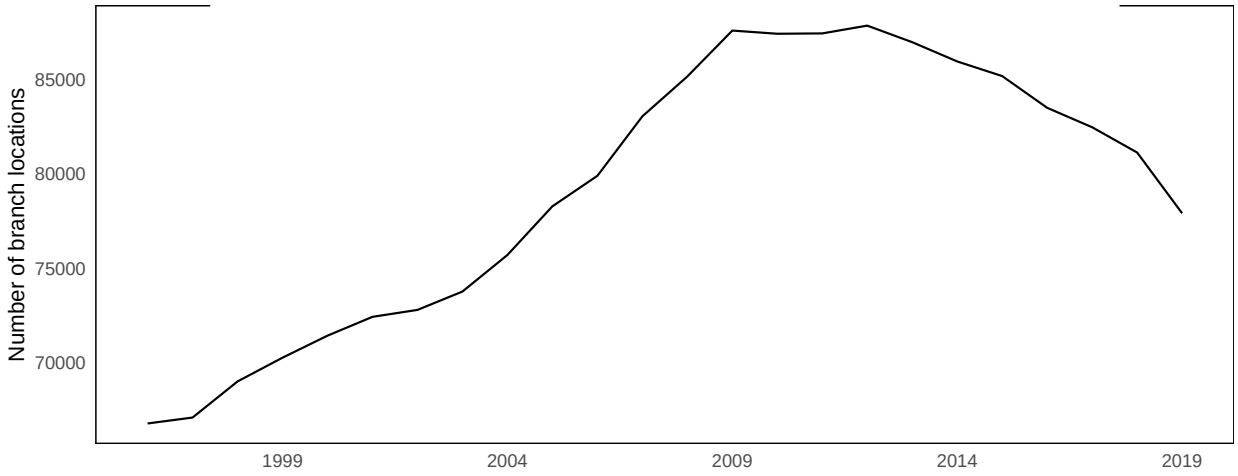
Notes: This plot shows the average loan market share of banks by U.S. state. Market share averages are pooled across time by state. Data are annual from 1995 through 2019.

Figure 19: Maximum bank market share by state



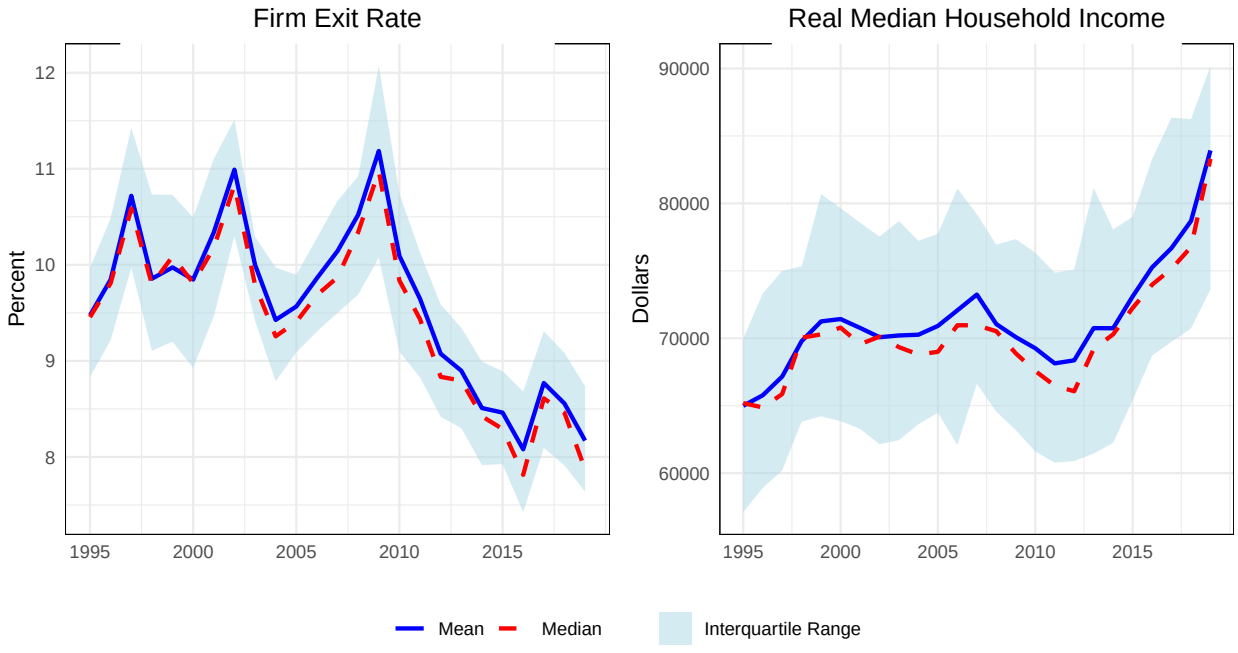
Notes: This plot shows the maximum loan market share of any bank in a given U.S. state. Data are annual from 1995 through 2019.

Figure 20: Number of FDIC Insured Bank Branches in Sample



Notes: This plot presents the number of branch locations operated by banks in the estimation sample. Branch locations are identified by BHC or bank owner RSSID in the FDIC's Summary of Deposits. Data are annual from 1996 through 2019.

Figure 21: State-level Non-financial Data



Notes: This plot summarizes the distribution of the state-level firm exit rates and real median household income. The solid blue line is the unweighted mean, the red dashed line the median. The blue bands mark the inter-quartile range. Data is annual from 1995 through 2019.

C Details on measuring fundamentals and market frictions

This appendix discusses the details of state-space model, constructing the bank-specific borrower quality proxy, estimating non-financial productivity wedges and shocks, the joint estimation algorithm, and the estimation accuracy of the algorithm.

C.1 State-space model details

This appendix provides a detailed description of the dynamic state-space model used to estimate bank-level fundamentals, regional borrower quality, and bank–region loan appeal.

State vector and transition equation. Let $i \in \mathcal{B}$ index banks, $s \in \mathcal{S}$ index regions, and $t = 1, \ldots, T$ index time. The latent state vector is stacked as

$$X_t = \left[\log \chi^{\psi}_{1,t} \cdots \quad \log \chi^{\psi}_{B,t} \quad g_{1,t} \cdots \quad g_{B,t} \quad \log \chi^{q}_{1,t} \cdots \quad \log \chi^{q}_{S,t} \quad \eta_{1,1,t} \cdots \eta_{B,S,t}\right]'$$

where:

- $\chi^{\psi}_{i,t} = \frac{1-\psi_{i,t}}{\psi_{i,t}}$ is the log-odds transformation of screening ability.
- $g_{i,t}$ is the bank capital cost premium.
- $\chi^{q}_{s,t} = \frac{1-q_{s,t}}{q_{s,t}}$ is the log-odds transformation of borrower quality.
- $\eta_{i,s,t}$ is bank–region loan appeal.

The law of motion is linear and Gaussian:

$$X_t = \Gamma X_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q).$$

The matrix Γ is comfortable block diagonal, with AR(1) persistence parameters for each latent component and their constant. The shock covariance matrix Q is diagonal, implying conditional independence across banks and regions given past states.

Measurement Equation and observables. Let the stacked observable vector be

$$Y_t = \left[\log \tilde{G}_{1,t} \cdots \quad \log \tilde{G}_{B,t} \quad \log \lambda_{1,t} \cdots \quad \log \lambda_{B,t} \quad \log E_{1,t} \cdots \quad \log E_{S,t} \quad \log s_{1,1,t} \cdots \quad \log s_{B,S,t}\right]'.$$

where

- $\tilde{G}_{i,t}$ is the observable approximation of a bank's capital cost premium

- $\lambda_{i,t}$ is a bank's loan portfolio default rate
- $E_{1,t} = \frac{1-q_{s,t}}{q_{s,t}}$ is the observable log-odds firm exit rate
- $s_{i,s,t}$ is a bank's regional loan market share

The measurement equation is

$$Y_t = \gamma + H_t X_t + \tilde{H}_t X_t + D Z_t + \nu_t, \quad \nu_t \sim \mathcal{N}(0, R).$$

The matrix H_t maps bank- and region-specific fundamentals into observables implied by equilibrium conditions. The matrix $\tilde{H}_t$ constructs region-level geometric averages of bank fundamentals that enter the market share equation. The measurement error covariance R is diagonal. The vector Z_t includes exogenous terms, bank-level regulatory costs and the nominal federal funds rate, to the measurement equation. The vector Z_t is a stacked analogously to X_t, with bank-specific regulatory costs in order and then the nominal federal funds rate repeated for each bank.

To make the measurement matrix concrete, consider the model with only one bank and one region so that that the system would follow as:

$$\mathbf{X}_{i,s,t} = \begin{bmatrix} \log G_{i,t} \\ \log \eta_{i,s,t} \\ \log \chi^{\psi}_{i,t} \\ \log \chi^{q}_{s,t} \end{bmatrix}, \quad \mathbf{Y}_{i,s,t} = \begin{bmatrix} \log \tilde{G}_{i,t} \\ \log \lambda_{i,t} \\ \log E_{s,t} \\ \log s_{i,s,t} \end{bmatrix} = \gamma_t + \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ \bar{\sigma}_{s,t} & 0 & \frac{\bar{\sigma}_{s,t}}{\sigma_{s,t}} & \bar{\rho}^{\psi}_{s,t} & 0 \end{bmatrix}}_{\equiv \mathbf{H}_t} \mathbf{X}_{i,s,t} + \tilde{H}_t X_t + B_t Z_t + \nu_t$$

$$\bar{\sigma}_{s,t} \equiv \frac{\sigma_{s,t} \log \tilde{s}_{s,t}}{\sigma_{s,t} \log \tilde{s}_{s,t} + 1} \qquad \rho^{\psi}_{s,t} \equiv \rho^{\psi} \bar{\sigma}_{s,t}$$

where both s and i are equal to one. The model with multiple banks and regions then expands this simplified matrix to be a blocked matrix with scalar ones becoming identity matrices and parameters being sorted accordingly. The blocks of the full measurement matrix are determined by the stacking of the latent state vector X_t.

The regional matrix $\tilde{H}_t$ collects the fundamentals of all banks operating in a region at time t and produces the unweighted mean of their log fundamentals. This results in a stacked vector of region-specific geometric means of bank fundamentals, which are necessary for computing the bank-region-specific market share. This additionally introduces the possibility of regional

spillovers, and resembles the spatial autocorrelation term present in spatial vector-autoregression and panel models.

Aggregate-Idiosyncratic Decomposition. The system admits an orthogonal decomposition that separates economy-wide common variation from bank- and region-specific deviations. This decomposition achieves computational efficiency while preserving the full covariance structure. The following decomposition has been used for these reasons to estimate modern regional NK DSGE models, including Beraja et al. (2019) and Jones et al. (2022). The following will describe the decomposition.

For any latent state $x_{i,t}$, define:

$$\bar{x}_t \equiv \frac{1}{N}\sum_{i=1}^{N} x_{i,t}, \quad \tilde{x}_{i,t} \equiv x_{i,t} - \bar{x}_t$$

The linearity of the state and measurement equations ensures this decomposition preserves the Gaussian structure:

$$\begin{aligned}
\text{Aggregate system:} \quad & \bar{X}_t = \bar{\Gamma}\bar{X}_{t-1} + \bar{\varepsilon}_t, && \bar{\varepsilon}_t \sim \mathcal{N}(0, \bar{Q}) \\
& \bar{Y}_t = \bar{H}_t\bar{X}_t + \tilde{\bar{H}}_t\bar{X}_t + \bar{B}_t\bar{z}_t + \bar{\nu}_t, && \bar{\nu}_t \sim \mathcal{N}(0, \bar{R}) \\
\text{Idiosyncratic system:} \quad & \tilde{X}_t = \tilde{\Gamma}\tilde{X}_{t-1} + \tilde{\varepsilon}_t, && \tilde{\varepsilon}_{i,t} \sim \mathcal{N}(0, \tilde{Q}) \\
& \tilde{Y}_t = \tilde{H}_t\tilde{X}_t + \tilde{\tilde{H}}_t\tilde{X}_t + \tilde{B}_t\tilde{Z}_t + \tilde{\nu}_{i,t}, && \tilde{\nu}_t \sim \mathcal{N}(0, \tilde{R})
\end{aligned}$$

where matrices are now divided by mapping idiosyncratic or aggregate laws of motion.

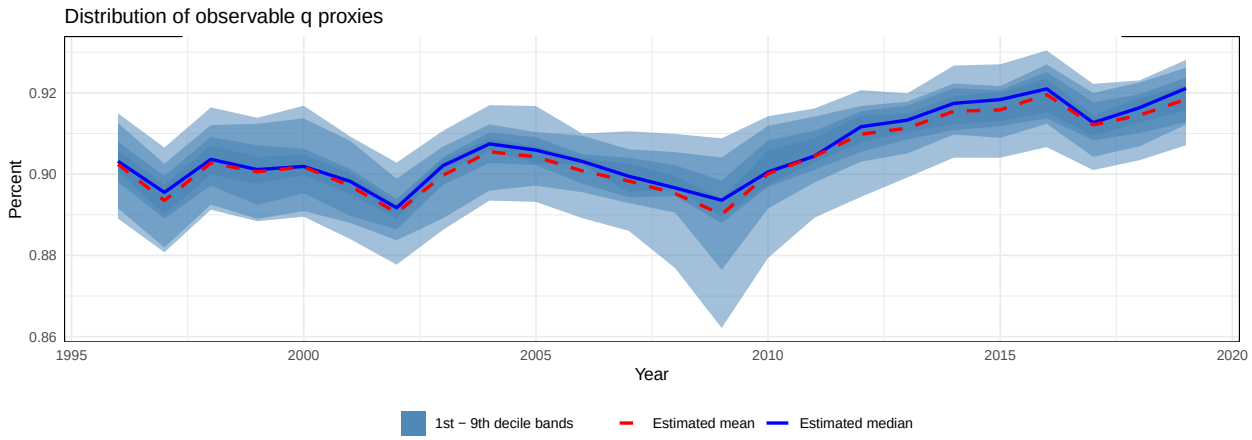
By construction, $\mathbb{E}[\bar{x}_t\tilde{x}_{i,t}] = 0$ for all i, t, and the likelihood function factors:

$$\mathcal{L}(Y_1, \ldots, Y_T \mid \theta) = \mathcal{L}^{agg}(\bar{Y}_1, \ldots, \bar{Y}_T \mid \theta) \times \prod_{i=1}^{N} \mathcal{L}^{idio}(\tilde{Y}_{i,1}, \ldots, \tilde{Y}_{i,T} \mid \theta)$$

This factorization permits separate estimation of the two subsystems, dramatically reducing computational burden when N is large, as is the case in our setting.

Resulting estimates. The resulting estimation yields a set of aggregate and idiosyncratic shocks and laws of motions for bank-level fundamentals (screening ability and capital cost premiums), and region-level fundamentals (firm exit rates and non-financial productivity). However, bank-region loan appeal is only constructed at the idiosyncratic level, because its normalization keeps

Figure 22: Observable proxy for local borrower quality



Notes: This plot shows the observable proxies for the bank-location specific market log-odds borrower quality. The solid blue line marks the median and the dashed red line the unweighted mean. The blue banks mark the first through ninth deciles in each period. Data are annual from 1996 through 2019.

the aggregate level of appeal constant. I will later exploit this bifurcation in structural exercises considering the macro and regional effects of shocks to bank fundamentals.

C.2 Bank-specific borrower quality

This section discusses the procedure to estimate bank-specific proxies for borrower quality. The first step is to create an observable proxy for firm exits in a tight geographic region around the bank's geographic footprint. The second step is to then estimate the bank-specific elasticity of loan losses to local firm exits.

C.2.1 Local firm exits

I create a proxy for the firm survival rate in a tight geographic region around a bank's branch footprints. That is, I assume banks lend to borrowers residing or operating close to the bank's physical branch locations and in turn create a proxy for the borrower quality of firms operating in that narrowly defined banking market. I calculate the narrowly defined $\bar{q}_{i,t}$ as the (deposit-weighted) average county-level firm survival rate for counties with active bank branches. Formally, the proxy is calculated as:

$$\bar{q}_{i,t} = \sum_{c \in C_{i,t}} \left(\frac{d_{i,c,t}}{\sum_{j \in C_{i,t}} d_{i,j,t}} \right) q_{c,t}$$

where $C_{i,t}$ is the set of counties a bank is operating in at time t, $d_{i,c,t}$ is the number of deposits a bank holds in branches located in county c, and $q_{c,t}$ is the firm survival rate of county c. This first steps ameliorates concerns that banks are exposed to credit markets that differ in their regional exit rate shocks compared to the broader state-wide economy.

Figure 22 shows the distribution of borrower quality proxy follows closely the state-level distribution of log-odds borrower quality proxies shown in Figure 29. This shows that while adopting a proxy for borrower quality based on bank-specific geographic footprints, the aggregate dynamics of the borrower quality represented in the sample are unaffected by this transformation.

C.2.2 IV for bank-specific elasticity of loan losses to local firm exits

I use an instrumental variable approach and estimate the the following model:

$$\log \lambda_{i,t} = \alpha_i \left[\log \frac{\widehat{1-\bar{q}_{i,t}}}{\bar{q}_{i,t}} \right] + \epsilon_{i,t}, \qquad \epsilon_{i,t} \equiv \log \frac{1-\psi_{i,t}}{\psi_{i,t}} + \eta_{i,t}$$

$$\log \frac{1-\hat{q}_{i,t}}{\hat{q}_{i,t}} = a + b x_t + \nu_{i,t} \qquad \nu_{i,t} \perp \eta_{i,t} \text{ and } \nu_{i,t} \perp \psi_{i,t}$$

where the elasticity of the default rate to local borrower quality, α_i, is the statistic of interest, $\hat{q}_{i,t}$ is the observable proxy for borrower quality $q_{i,t}$ estimated in the previous step, $\widehat{\frac{1-\bar{q}_{i,t}}{\bar{q}_{i,t}}}$ is the predicted borrower quality term, x_t is the instrument for borrower quality shocks, and $\epsilon_{i,t}$ is the residual that contains the unobservable log odds screening ability and measurement error $\eta_{i,t}$. The instrument must satisfy the standard relevance condition, as well as a two-faced exclusion restriction:

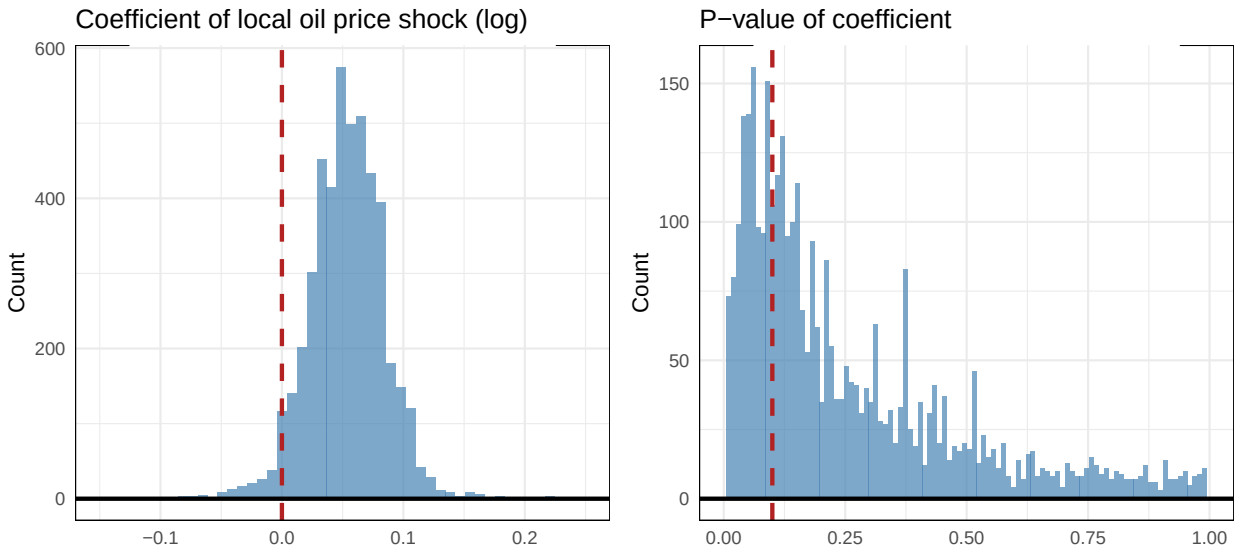
$$\mathbb{E}[x_t \eta_{i,t}] = 0, \quad \mathbb{E}[x_t \psi_{i,t}] = 0$$

That is, the instrument must be exogenous (statistically independent) to both the measurement error $\eta_{i,t}$ and the bank-specific productivity $\psi_{i,t}$. Estimating the model will result in a consistent estimate of α_i and isolate the unobservable bank productivity in the residual loan default rate.

I use Känzig (2021) oil news shocks as the instrument x_t. Känzig (2021) identifies oil supply news shocks by exploiting high-frequency changes in oil futures prices around OPEC announcements. By focusing on asset price fluctuations during a tight window around the macroeconomic news announcement, he isolates variation in oil prices to supply shocks. The following sections scrutinize the relevance and exclusion restrictions.

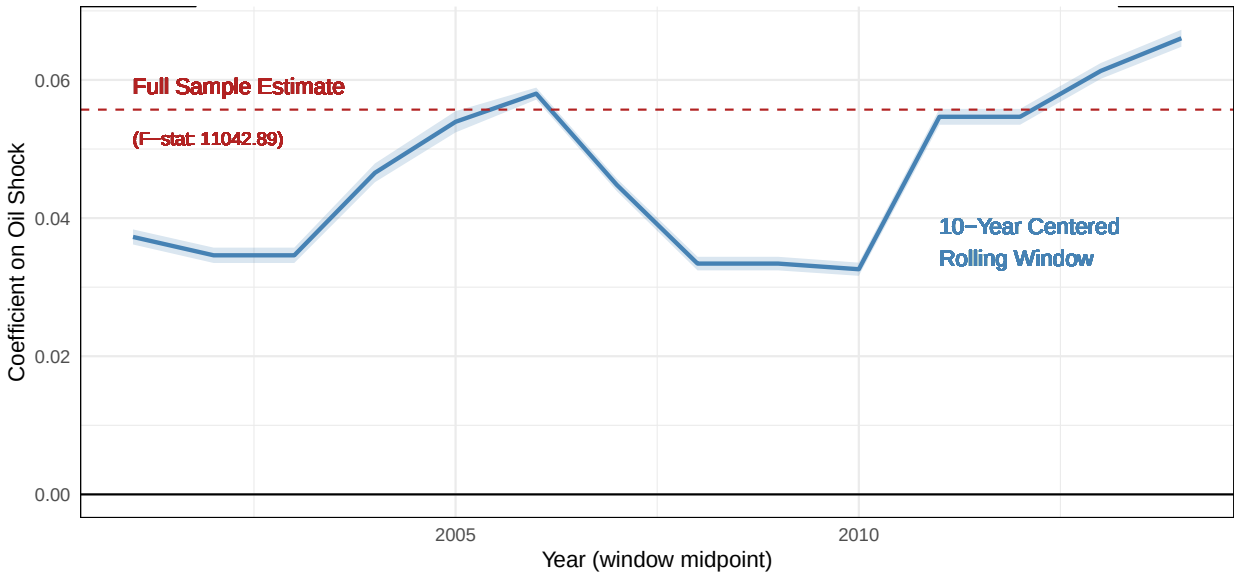
Relevance criterion

Figure 23: Distribution of bank-level first stage regressions



Notes: These plot shows the distribution of bank-level first stage elasticities of borrower quality to oil price shocks and their probability values of being statistically different than zero. The left panel shows the elasticities estimated by bank-level univariate models with a constant. The red dashed line marks zero, where positive values are the theoretically expected results. The left panel shows the p-values of the elasticities and the red line marks the 10 percent confidence level.

Figure 24: Pooled first-stage regression



Notes: This plot shows the first stage elasticity of borrower quality to oil price shocks estimated with the pooled sample or with 10-year rolling windows. Each regression includes bank fixed effects. The rolling window estimated elasticity is marked its 99-percent confidence interval. Standard errors are clustered at the bank-level.

The IV must satisfy the standard relevance condition. The Känzig (2021) oil price new shocks satisfy the relevance condition both on theoretical and empirical grounds. First, oil price shocks are effectively a marginal cost shock for non-financial firms that induce exits, holding all else equal, in standard Hopenhayn (1992) style models of firm dynamics. While my model of the economy does not explicitly model these non-financial entry/exit forces and ultimately represents regional production as a representative firm, the regional firm exit shocks can be considered a reduced-form stand in for the individual firm-level entry and exit problems that are abstracted by the representative agent framework.

Second, the oil price news shocks are empirically relevant. Figure 23 shows the distribution of bank-level first-stage coefficients and corresponding probability values that the coefficient is different than zero. The coefficient maps the log oil price news shock to the log odds of the observable proxy for local firm exits, meaning they can be interpreted as elasticities. The mean coefficient is 0.051 and the median 0.054, indicating that a one percent increasing oil price shock will increase the log-odds of firm exits by approximately five basis points on average. Only 4.12 percent of bank-level elasticities falling below zero. That is, approximately 96 percent of banks have the "correctly" signed first-stage, consistent with the theoretical framework. In a just-identified IV, the correctly signed coefficient is a sufficient first stage condition to then rely on inference tests of the second stage estimates, following Angrist and Kolesár (2024). However, only, 31.3 percent of bank-level regressions report coefficient p-values below the standard 0.1 threshold.

Since the low p-values may reflect noise from a small sample problem (banks can only have up to 24 observations), I additionally estimate an analogous pooled model that includes bank-level fixed effects and clustered errors at the bank-level. Figure 24 shows the full sample estimated coefficient is approximately 0.55, in line with the average and mean of the bank-level regression distribution. The pooled sample has a robust F-statistic of 11042.89, well exceeding the standard relevance thresholds (see Stock and Yogo (2002)). The sign and significance of the pooled sample elasticity are also stable over time. Figure 24 likewise shows the centered 10-year rolling window estimated coefficient of interest, which remains centered around 0.049 and is always positive, consistent with the theoretically implied effect of oil price shocks on the log-odds of firm exits.

Exclusion restriction

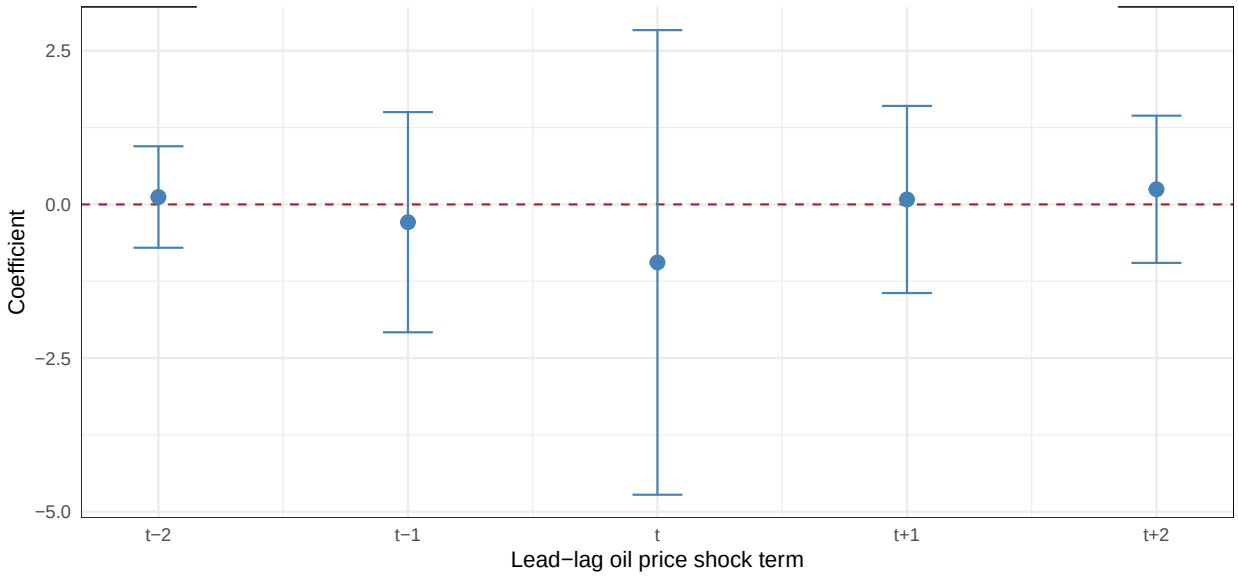
The oil news shocks must additionally satisfy the exclusion restriction —bank's screening technology is not impacted by oil price shocks. This condition holds in the context of the theoretical

Table 6: Testing the effects of oil price shocks on loan losses

Specification	Coefficient	Std. Error	p-value
only oil shocks	-0.371	0.168	0.0478
adding borrower quality...	-0.164	0.112	0.170
adding capital cost and net worth...	-0.163	0.112	0.170
adding loan market shares...	-0.161	0.111	0.172

Notes: This table shows the estimated coefficient, standard error, and implied p-values for regressions mapping bank-level loan losses to oil price news shocks. The first reports the univariate regression with only oil price shocks, while the following rows successively add controls. Each regression has bank- and year-level fixed effects. Standard errors are clustered at the year-level. Each regression is estimated on a sample of 73.5 thousand bank-year observations.

Figure 25: Lead-Lag Structure: Oil Shocks and Loan Losses



Notes: This plot shows the estimated coefficients and 95 percent confidence intervals for a regression mapping bank-level loan losses to leads and lags oil price news shocks. The x-axis marks the coefficients by the number of period leads or lags on the oil price shock regressor. The regression also include the observable proxy for borrower quality. The regression has bank- and year-level fixed effects. Standard errors are clustered at the year-level. The regression is estimated on a sample of 73.5 thousand bank-year observations.

framework guiding the empirical estimation. There is modeled channel for oil price shocks to influence the exogenous shocks to the bank-level screening ability. However, it may fail if we relax the assumption of banks' operating with full-information rational expectations. For example, if banks exhibit bounded attention, large oil price shocks may divert resources away from screening

and toward processing aggregate information, violating the exclusion condition.

I empirically test the exclusion restriction with two exercises. I first test that oil price shocks are uncorrelated with loan losses while controlling for borrower quality, showing that if the shocks have an effect on loan losses, it is through the borrower quality channel as assumed. Second, I test that loan losses do not anticipate future or lag oil price new shocks, disqualifying the possibility that the oil price shocks violate the exclusion restriction by working through non-borrower quality channels with a lead or delay that would keep them hidden from the first test.

Independence from screening ability. Table 6 reports a progressive controls test. I regress bank loan losses on oil shocks, sequentially adding controls for predicted borrower quality, capital costs, net worth, and market share. In Specification (1), without controls, oil shocks predict loan losses and are statistically significant at the five percent level. However, once I control for the observable borrower quality proxy in Specification (2), the oil shock coefficient shrinks considerably and becomes statistically insignificant at any standard level. Adding capital costs and market share controls in Specifications (3) and (4) does not meaningfully change this result. The attenuation of the oil shock coefficient to zero once borrower quality is controlled for supports the exclusion restriction: oil shocks affect loan losses only through borrower quality, not through screening ability.

Pre-trends and lead-lag structure. A violation of the exclusion restriction might manifest as oil shocks predicting future loan losses, even after controlling for current borrower quality. Figure 25 report the results of a lead-lag test. I include two leads and two lags of oil shocks in the regression, controlling for the observable borrower quality proxy. The lead coefficients are small and statistically insignificant ($p > 0.75$ for both $t + 1$ and $t + 2$), indicating no anticipation effects or omitted variable bias. The contemporaneous and lag effects are also insignificant once borrower quality is controlled for.

Collectively, these tests provide support for the exclusion restriction. Oil supply news shocks predict loan losses only through their effect on borrower quality, with no direct effect on screening ability.

C.3 The complete joint estimation algorithm

I jointly estimate the latent bank- and region-level fundamentals, time-varying loan market frictions, and regulatory costs using the following full-information EM algorithm.

1. **Residualize loan default rates.** Use bank-specific localized firm exit rates and oil price shocks as instruments to estimate the borrower quality component of observed loan losses. Then residualize the observable portfolio-level loan default rate to the borrower quality proxy. Use the residualized loan default rate it in place of the bank-region-specific loan default rate in the observables vector and zero out its loading on the regional borrower quality latent state in the measurement equation.

2. **E-step (Kalman Smoother).** For the aggregate and idiosyncratic system, separately use the Kalman filter and Rauch–Tung–Striebel (RTS) smoother to recover smoothed latent states $\hat{x}_{i,t}$, given the current measurement equation parameters.

3. **M-step.**

   - *Update State Transition Parameters.* Estimate the autoregressive coefficients $\rho^{\psi}, \rho^{q}, \rho^{g}, \rho^{\eta}$ by regressing smoothed latent states on their own lags at the bank- and region-level for both idiosyncratic and aggregate systems.
   - *Estimate Structural Loan Rate Equation.* After combining the idiosyncratic and aggregate states:
     - Construct expected loan default risk $\mathbb{E}[\lambda_{i,s,t+1}]$ from estimated bank ability and borrower quality states and their updated laws of motion to then construct bank-level portfolio average loan rates.
     - Estimate the regulatory cost term via a weighted regression of residualized loan rates on the risk-adjusted inverse bank net worth.
     - Invert the banks' first-order conditions to recover the elasticity of substitution $\sigma_{s,t}$
   - *Update Measurement Equation Coefficients.* Use the estimated $\sigma_{s,t}$ and average market shares to construct:

$$\bar{\sigma}_{s,t} = \frac{\sigma_{s,t} \cdot \log \tilde{s}_t}{\sigma_{s,t} \cdot \log \tilde{s}_t + 1}, \quad \bar{\rho}_t^{\psi} = \bar{\sigma}_{s,t} \cdot \rho^{\psi}, \quad \bar{\rho}_t^{q} = \bar{\sigma}_{s,t} \cdot \rho^{q}$$

     These coefficients define the time-varying measurement matrix for the next E-step.

   - *Update Exogenous States*. Use the estimated bank-level regulatory costs to update the vector of exogenous states in the measurement equation, Z_t.

4. **Convergence Check.** Terminate the algorithm if the maximum absolute change in all parameters falls below a pre-specified tolerance level.

5. **Repeat.** Iterate steps 2–4 until convergence.

6. **Non-financial fundamentals.** Using the estimated latent states, market frictions, and regulatory costs, estimate the region-specific non-financial productivity level, shocks, and laws of motion.

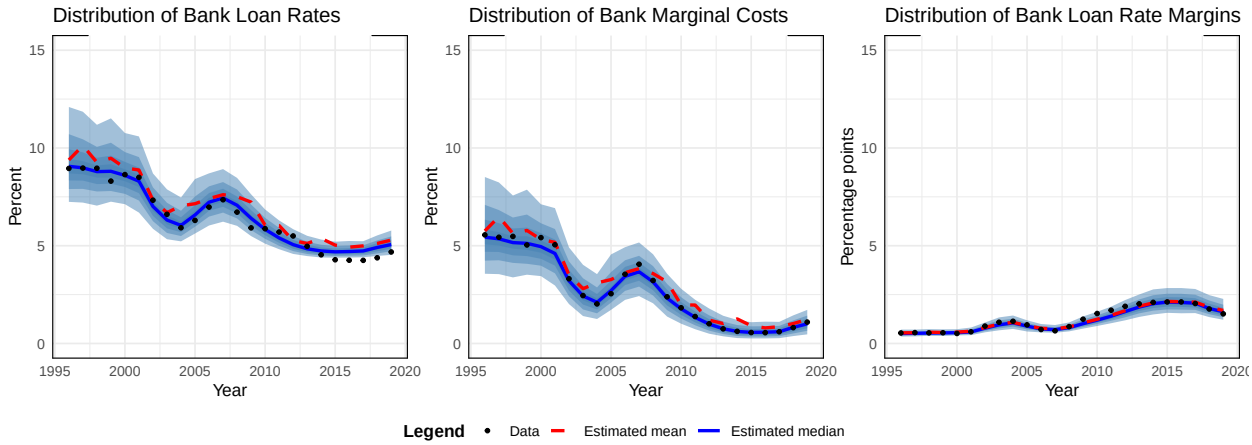
This procedure efficiently combines the state-space estimation of the bank-level fundamentals and structural parameter estimation of the regional market frictions and regulatory costs into a into a single full-information approach. The result is a set of consistently estimated fundamentals $\{\psi_{i,t},\ q_{s,t},\ \eta_{i,s,t},\ c_{i,t}\}$, market frictions $\{\sigma_{s,t}\}$, and regulatory cost function $\Gamma(\cdot)$.

C.4 Estimation accuracy

I lastly characterize the accuracy of my estimation algorithm to recover bank-level loan rates. Figure 26 shows the distribution of the empirically estimated bank-level (loan-weighted) loan rates. The blue bands mark the first through ninth deciles of the distribution, while the red dashed line marks the estimated mean. The figure further decomposes loan rates into their marginal costs and net interest margin components. Across all three panels the average estimated loan rate or its component part well matches its observable counterpart. This is evidence that the empirical estimation is not only successfully reproducing observable loan rates, but is also correctly estimating its marginal costs and markup components.

I then additionally consider the accuracy of the estimation procedure by comparing the mean and inter-quartile range of the unweighted and loan-weighed year-by-year loan rate distributions. Figure 27 shows these comparisons, and makes it clear that the mean of the distribution is well predicted across both unweighted and loan weighted treatments. In comparison the spread of loan rates is not well matched, with the estimation consistently producing too narrow of a loan rate distribution.

Figure 26: Distribution of estimated loan rate



Notes: This plot describes the estimated loan rates, marginal capital cost, and net interest margins, comparing them to observable data. The blue banks mark the first through ninth deciles, while the solid blue line indicates the median. The dashed red line is the mean, and the black dots are the average observable loan rate, cost, or spread. Data is from 1996 through 2019.

Figure 27: Comparing estimated and observable loan rates

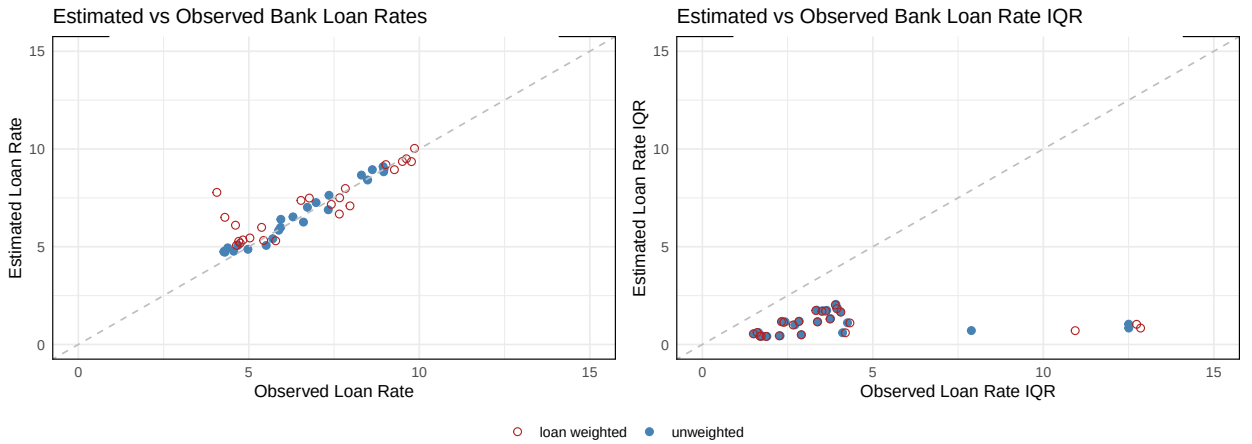


Figure 28: Notes: This plot compares the mean and inter-quartile ranges of estimated and observable loan rates. The solid blue dots marks moments from the unweighted distribution while the red clear dots mark moments from the loan weighted distribution. The dashed gray line marks the forty-five degree line; when dots are on the forty-five degree line, then the model has perfectly reproduced the data.

D Estimated non-financial fundamentals

This appendix describes the estimated borrower quality and non-financial productivity.

Estimating region-specific non-financial productivity wedges. After estimating the bank fundamentals, regulatory costs, and market frictions, I need to estimate one more object, the regional non-financial productivity shifter, $A_{s,t}$. The goal of this estimation is not to precisely measure local industry TFP, but rather to measure a reduced-form wedge that characterizes fluctuations in regional activity for reasons other than shocks originating in banks.

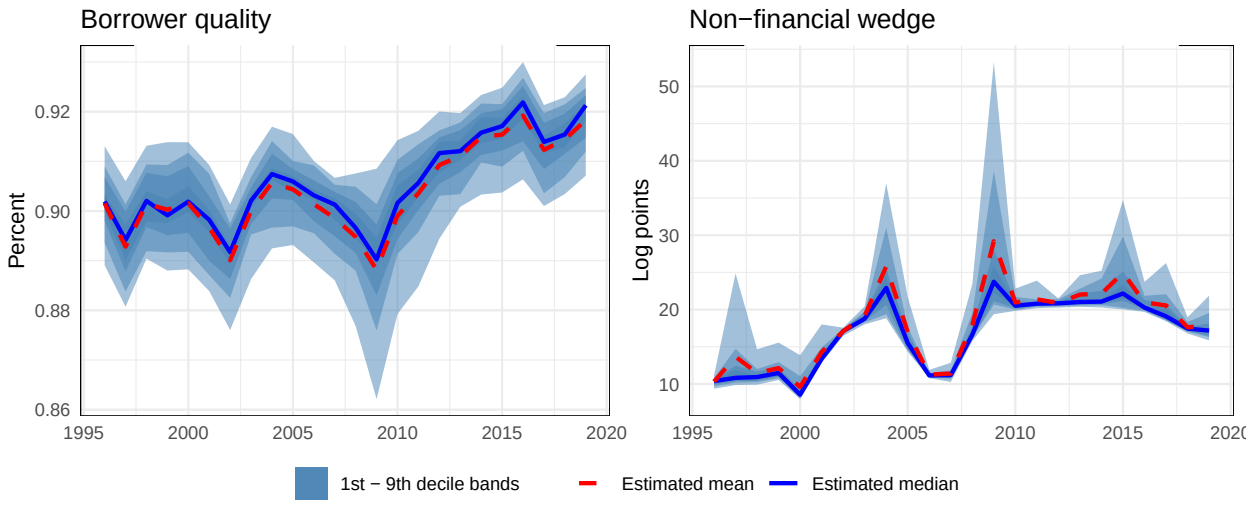
I construct the regional productivity shifter starting from the model-implied structural wage equation,

$$W_{s,t} = \left[A_{s,t}^{\varrho} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\upsilon} \right]^{(\upsilon+1)^{-1}}, \qquad \upsilon = \frac{\alpha\varrho + 1}{1-\alpha},$$

where α denotes the labor share of output, governing the curvature of decreasing returns to scale, and ϱ is the elasticity of labor supply. Solving this expression for $A_{s,t}$ delivers a structural measure of local productivity that can be expressed as a function of observable real income, regional loan rates, and calibrated deep parameters. Empirically, I identify $A_{s,t}$ with the residual variation in local real income after netting out the effects of local financial conditions, summarized by regional loan rates. This residual captures regional heterogeneity in real economic fundamentals and serves as the empirical proxy for the local productivity shifter.

Estimated region-level fundamentals. Figure 29 shows the distribution of regional borrower quality (one minus the regional firm exit rate) and the log non-financial productivity wedge. The distribution of borrower quality mirrors the distribution of observable firm exit rates shown in Figure 21. Borrower quality is pro-cyclical, was stable from 1995 through the mid-2000s, but began a secular rise after the GFC. In comparison, the non-financial wedge has been largely increasing since 1995, with cyclical downturns in 2001 and the GFC. The non-financial productivity distribution becomes notably dispersed during periods of rapid growth, namely the mid-1990s and the recovery directly after the 2001 recession. However its widest dispersion occurs during the 2009 recession. There is more consistent dispersion in borrower quality, and Figure 30 shows the average borrower quality by state.

Figure 29: Regional fundamentals and frictions



Notes: This plot shows the estimated distribution of region-specific borrower quality and non-financial productivity. The solid blue line marks the median and the dashed red line the unweighted mean. The blue banks mark the first through ninth deciles in each period. Data are annual from 1996 through 2019.

Figure 30: Average borrower quality by state

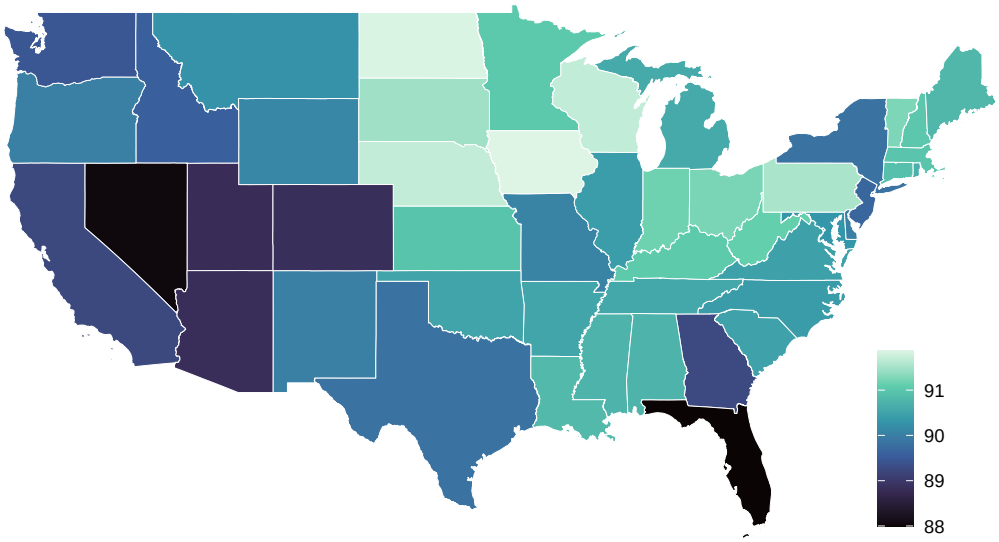


Figure 31: Notes: This plot shows the sample average borrower quality by state. Data are annual from 1995 through 2019.

E Quantitative calibration and simulations

This appendix provides a detailed description of the quantitative model calibration, solution methods, and simulation procedures used to analyze the macroeconomic and distributional effects of bank fundamentals and loan market frictions.

E.1 Model calibration

The quantitative model combines empirically estimated laws of motion for bank and regional fundamentals with standard parameter values from the macroeconomic literature. This subsection details the calibration strategy for each component.

E.1.1 Estimated Fundamentals

Table 7: Estimated Fundamental Laws of Motion

	ρ_g	ρ_ψ	ρ_q	ρ_a	ρ_η	Var(c)	Var(ψ)	Var(q)	Var(a)	Var(η)
Micro-level parameters										
Micro	−0.347	0.821	0.983	0.300	0.937	15.6	0.403	0.0604	5.09	9138
Macro-level parameters										
Macro	0.927	1.020	1.000	0.311	—	0.00542	0.159	0.0114	2.12	—

Notes: The table reports estimated autoregressive coefficients and unconditional variances for micro- and macro-level fundamentals. Macro-level fundamentals exhibit greater persistence, while micro-level fundamentals display substantially higher volatility.

The laws of motion for bank-level fundamentals, regional fundamentals, and market frictions are calibrated using the empirical estimates from Section 4. This approach ensures that the quantitative model's stochastic environment reflects the data-generating process observed in U.S. banking sector from 1994 through 2019. Table 7 reports the estimated parameters defining the persistence and volatility of the fundamentals and frictions' laws of motion. The following will outline the estimated fundamentals, frictions, and their functional forms in more detail.

Bank-level fundamentals. Each bank i is characterized by three time-varying fundamentals: screening ability $\psi_{i,t}$, capital cost premium $g_{i,t}$, and bank-region loan appeal $\eta_{i,s,t}$. These fundamentals follow AR(1) processes estimated via the Kalman smoother in the E-step of the EM

algorithm:

$$\log \chi^{\psi}_{i,t} = \rho^{\psi} \log \chi^{\psi}_{i,t-1} + \varepsilon^{\psi}_{i,t}, \quad \chi^{\psi}_{i} = \frac{1-\psi_i}{\psi_i}, \tag{29}$$

$$g_{i,t} = \rho^{g} g_{i,t-1} + \varepsilon^{g}_{i,t}, \tag{30}$$

$$\log \eta_{i,s,t} = \rho^{\eta} \log \eta_{i,s,t-1} + \varepsilon^{\eta}_{i,s,t}, \tag{31}$$

where $\varepsilon^{\psi}_{i,t}$, $\varepsilon^{g}_{i,t}$, and $\varepsilon^{\eta}_{i,s,t}$ are i.i.d. Gaussian innovations with variances recovered from the state-space model. In the simulation exercises I directly shock $\log \chi^{\psi}_{i,t}$ and transform the variable to levels of $\psi_{i,t}$ when needed. The bank-region loan appeal parameters reflect the unnormalized appeal. In the simulation the appeals with be normalized by region each period.

The estimated persistence parameters $\{\rho^{\psi}, \rho^{g}, \rho^{\eta}\}$ and shock variances $\{\sigma^2_{\psi}, \sigma^2_{g}, \sigma^2_{\eta}\}$ are taken directly from the M-step of the empirical estimation. Banks are initialized with draws from the unconditional distribution of fundamentals, ensuring that the initial cross-sectional dispersion matches the data.

Regional fundamentals. Each region s is characterized by borrower quality $q_{s,t}$ and non-financial productivity $A_{s,t}$, which also follow estimated AR(1) processes:

$$\log \chi^{q}_{s,t} = \rho^{q} \log \chi^{q}_{s,t-1} + \varepsilon^{q}_{s,t}, \quad \chi^{q}_{s} = \frac{1-q_s}{q_s}, \tag{32}$$

$$a_{s,t} = \rho^{a} a_{s,t-1} + \varepsilon^{a}_{s,t}, \quad a_s = \log A_s, \tag{33}$$

where regional productivity $A_{s,t}$ is constructed as the structural residual from the wage equation, netting out the effects of regional loan rates. The persistence and variance parameters are estimated from state-level data on firm exit rates and real median household income.

E.1.2 Estimated frictions, regulatory and distance costs

Market frictions. Regional loan market frictions are captured by the time-varying elasticity of substitution across banks, $\sigma_{s,t}$. The elasticity evolves according to the AR(1) process:

$$\log \sigma_{s,t} = \mu^{\sigma} + \rho^{\sigma} \log \sigma_{s,t-1} + \varepsilon^{\sigma}_{s,t}, \tag{34}$$

which is recovered from the structural net interest income equation via the M-step. The $\mu^{\sigma} = 1.107$, $\rho^{\sigma} = 0.898$, and $Var(\sigma_{s,t}) = 1.394e-05$.

Regulatory costs. Bank i's regulatory cost in at time t is:

$$\Gamma_{i,t} = \zeta_{1t} \log(\sigma^R) + \zeta_{2t} \log(1/N_{i,t}), \tag{35}$$

where $N_{i,t}$ is the bank's net worth, σ^R is the portfolio risk, and ζ_{1t}, ζ_{2t} are the time-varying aggregate regulatory cost coefficients. However I keep the regulatory cost coefficients fixed for simulations to isolate changes in banks' decision making to the effects of frictions, fundamentals, and net worth fluctuations.

The estimated regulatory cost function coefficients are $\zeta_2 = 0.0689$ and $\zeta_1 = -8.985$, and the portfolio return risk is $\sigma^R = 0.4820696$. This implies the portfolio risk term will contribute a constant 2.847 percentage points to loan rates.

Distance costs. I will set the fixed cost $f = 0$ for banks operating in their headquarters region. The cost of operating in region s' for a bank headquartered in region s is:

$$c(s, s') = 2.5 \times \log(d_{s,s'}) + 1, \tag{36}$$

where $d_{s,s'}$ is the geographic distance between the centroids of regions s and s'. The elasticity of 2.5 is calibrated so that as the log-distance doubles the distance cost doubles, matching the distance cost estimated in Oberfield et al. (2024).

E.1.3 Deep structural parameters

The deep parameters governing preferences, technology, and nominal rigidities are calibrated to values commonly employed in the New Keynesian DSGE literature. Table 8 summarizes these parameters. The curvature of the production function $\alpha = 0.33$ follows standard calibrations for the capital share of output in the real business cycle literature. I use the standard capital share because the emphasis of the study is on the role of capital lending. However, using the standard labor share value $\alpha = 0.66$ leads to qualitatively similar results. The inverse Frisch elasticity $\varrho = 1$ follows Nakamura and Steinsson (2014) and while larger than typically used in microeconomic studies, is meant to capture the extensive margin of labor supply and is standard in the macroeconomic literature. The elasticity of intertemporal substitution $\epsilon = 1.50$, the Rotemberg adjustment cost parameter $\theta = 78$, and the discount rate is $\beta = 0.99$, implying a modest one percent steady state risk-free interest rate.

The monetary policy rule parameters follow standard Taylor (1993) specifications, with $\phi_\pi = 1.5$

Table 8: Deep parameters and model dimensions

Parameter	Symbol	Value	Description
Model dimensions			
Number of regions	n_s	50	Total number of regions in the economy
Banks per region	n_b	500	Total Number of banks operating in the economy
Simulation periods		100	Number of simulated time periods
Deep parameters			
Discount rate	β	0.99	Time discounting rate of banks and households
Input share	α	0.33	Curvature of the firm's production function
Inverse Frisch elasticity	ϱ	1.00	Determines responsiveness of labor supply to real wages
Rotemberg adjustment cost parameter	θ	78.0	Cost of adjusting prices; governs nominal rigidity
Elasticity of intertemporal substitution (EIS)	ϵ	1.5	Responsiveness of consumption growth to real interest rate
Monetary policy rule			
Inflation response coefficient	ϕ_π	1.5	Taylor-rule coefficient on inflation gap
Output response coefficient	ϕ_y	0.5	Taylor-rule coefficient on output gap
Interest rate smoothing	ρ_r	0.8	Persistence of nominal policy rate
Monetary policy shock persistence	ρ_i	0.85	AR(1) persistence of monetary policy shock
Monetary policy shock variance	σ_i^2	0.025	Variance of innovations to monetary policy shock

and $\phi_y = 0.5$ satisfying the Taylor principle and ensuring determinacy of the equilibrium. The interest rate smoothing parameter $\rho_r = 0.8$ captures the observed persistence in the federal funds rate. The monetary policy shock follows an AR(1) process with persistence $\rho_i = 0.85$ and innovation variance $\sigma_i^2 = 0.025$.

E.2 Simulation and solution details

This section describes the steps to solving and simulating the economy. Note that given the structure of the economy, most region-level outcomes can be solved analytically after solving for the banks' entry/exit decisions with a bounded rationality rule à la Krusell and Smith (1998) and Weintraub et al. (2008).

Simualtion. I simulate economies with 500 banks and 50 regions for 100 periods. For each of these simulations I take the following steps:

- Initialization:
  - Create S ex ante identical regions and generate an $S \times S$ matrix determining the relative distance of regions (for trade cost calculations). Distances are drawn from a log-normal random distribution with mean 1, standard deviation 0.1.

- Create B ex ante identical banks. Uniformly distribute banks across the regions.

- Each Period:

  - Banks decide where to operate based on expected profits

    * Banks forecast each regions banking sector average fundamentals, market friction, its own appeal, and total loan demand to calculate its expected market share and profits if it enters. Banks decide to enter a region if its forecasted profits satisfy condition 13.

      This follows a Krusell-Smith bounded rationality approach to make the problem computationally tractable. The standard oligopolistic entry problem is potentially a high dimension one, depending on the full distribution of the banking sector and each bank's best response function. However, I assume a "large" region inspired by the Oblivious Equilibrium approach proposed by Weintraub et al. (2008), so that banks can meaningful forecast the region's banking sector average fundamentals in a reduced-form approach 'a la Krusell and Smith (1998).

  - Banks draw fundamentals $G_{i,t}$, $\eta_{i,s,t}$ (conditional on entering region s), and forecast $\mathbb{E}q_{s,t}$, $\mathbb{E}\psi_{i,t}$. Regions draw $\sigma_{s,t}$ and $A_{s,t}$.

  - Banks decide on $\{L_{i,s,t}, R_{i,s,t}\}$. Firms and households solve $Y_{s,t}$, $M_{s,t}$, $W_{s,t}$, given fundamentals and $\{L_{i,s,t}, R_{i,s,t}\}$.

    * Loan rates are solved analytically using Equation 11. With loan rates and fundamental shocks realized, regional planned output, employment, loan demand, and market clearing wages can all be solved analytically with Equations 26, 24, 25, and 23 respectively.

  - Production occurs and firm exit rate shocks are realized. Loan losses are realized and bank net worth are updated. The Central Bank determines the policy rate for the start of the next period.

  - Banks are hit with exit shocks, new banks enter. Update banking sector distribution.

The regional entry forecasting rule. The banks use a set of seemingly unrelated AR(1) models to forecast regional banking sector average fundamentals, their own appeal, and regional frictions before deciding to enter a region in a given period. These forecasting rules are estimated following the standard Krusell-Smith algorithm. I instantiate the forecasting rules as random walks, then (1) simulate the economy and (2) use the simulated paths to update the forecasting rules, repeating steps (1) and (2) until the forecasting rules have converged. Because the forecasting rules are at the bank, region, or bank-region level, the estimation preserves the panel structure of the simulated

data, meaning the number of observations per simulation is quite large, making the algorithm computationally efficient in terms of the number of simulations required to converge.

Solution method and computational considerations. After solving the bank's regional entry problem with a Krusell-Smith bounded rationality forecasting approach, all other economic decisions can be solved analytically given the structure of the economy. This was done intentionally to keep the computational problem of simulation the economy tractable enough to be feasible with a large panel of banks and regions.